

Global Equity Derivatives & Strategy

Year Ahead 2025: The roaring 2020s?

Risks in both tails as policy + bubble reminiscent of 1920s

2025's risk will be misjudging the scope of a tech-bubble amplified by possibly the most radical US policy shift in 100yrs. 80s deregulation/tax cuts & the 90s internet bubble drove record booms, while laissez-faire, speculation & innovation fueled the roaring 20s. The ensuing '87, '00 & '29 crashes taught us: 'Bigger the boom, bigger the bust'.

Vol says AI can still run, but it's too late to avoid a bust

We still believe AI may see a multi-year bubble analogous to the 90s (see [Vol suggests AI isn't a bubble...yet](#)). Vol & prices rising together are the tell of a bubble, although they say we are closer to '96/'97 than '98/'99. From today's valuations though, bear markets have always ensued. We're in too deep; own NDX vol to ride bubble through the pop.

Fragility + bond vigilantes are biggest left tail risks

August's VIX shock reminded us markets remain fragile amidst crowding & failing trading liquidity. With stocks setting fragility records, a market-wide event is overdue. Bond vigilantes 'calling time' on policy makers given historic US debt & potential profligacy may be biggest left tail risk. Bond/equity correlation helps create a cheap hedge.

Why 2017 vol lows are unlikely under Trump 2.0

Won't the 2017 (Trump 1.0) vol crush repeat? We say no as bubble + policy risks, macro uncertainty & equity P/Es are different today. Stocks are also more volatile, fragile & faster to re-price news. SPX gamma would need to 8x to sustain 2017-style vol alone.

Ride boom without needing to time bust: From A to SkeU

Fundamental investors will struggle chasing irrational valuations, risking under-performance. Most who do will fail to time the top. Hence, efficient optionality is key. Our systematic A-trade would've tracked S&P rallies with 1.4x Sharpe & *crucially* insulated investors in selloffs. Our convex yet carry-neutral SPX option strategy, SkeU, benefits from fragility & boosted equity Sharpe by 25% as part of a defensive overlay.

Europe cheap + unloved; we like upside hedges, carry & RV

EU vol may be less painful to short than the market, as there are fewer crowds to exit in a shock & stocks may squeeze higher in case of a positive surprise. We like SX5E calls, cyclical vs defensive RV, carry from dislocated TRFs, FTSE divs, return & vol dispersion.

Hedge Taiwan, largest AI risk; own Nuclear, next AI leg

While unlikely, Taiwan worst-case scenarios, with a clear Nvidia link, cannot be ignored; trade >22x max payout tail hedges. Benefit from AI-induced power demand with more exposure to Asia nuclear technology leaders offering better performance and lower vol than Western-focused alternative.

Trading ideas and investment strategies discussed herein may give rise to significant risk and are not suitable for all investors. Investors should have experience in relevant markets and the financial resources to absorb any losses arising from applying these ideas or strategies.

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Refer to important disclosures on page 55 to 57. Analyst Certification on page 54. 12754783

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[See Team Page for List of Analysts](#)

See the end of the report for a list of all trades opened in this piece.

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2025 Outlook

Risks in both tails

As we enter 2025, the risk is not appreciating what a historic time this is and therefore how much risk lies in the tails. We are far enough into the AI boom that equities will likely either accelerate towards a more bubble-like state or unwind their already significant gains. The US could see the most radical public policy shift in a century with Trump's sweeping win, and markets remain historically fragile. Macro uncertainty remains elevated post-Covid and the 100yr high US debt levels amplify the risks.

History suggests bubbles and large public policy shifts can drive wide swings in asset prices, and 2025 could witness *both* simultaneously (something not seen perhaps since the 1920s). This should support equity volatility no matter the market direction, as risk historically rises when valuations are this stretched. While asset bubbles have proven resilient to macro risk, the potential for bond vigilantes to "call time" on profligate policy and push back should not be ignored. VIX's record shock this summer is a reminder of markets' structurally higher fragility, and a broad-market fragility event is likely overdue.

The potential mistake will be underappreciating the risks and remaining pensive, hoping for more clarity which never comes. Getting out on the front foot, recognizing we are in a period of rising risk and rising return is key. Few will successfully time the market top or the next shock, but volatility collapsing post the US election and strong cross-asset dislocations are helping to create cheap optionality to hedge either tail.

Our outlook on a page:

AI becomes more bubble-like before it pops: 90's internet bubble suggests still early days for AI trade with higher returns, higher volatility & larger drawdowns in coming yrs.

Trump's sweep amplifies risks: proposals for largest de-regulation since '80's Reagan, tariffs since '30's Smoot-Hawley & historic deportation amplifies boom/bust risks.

Market-wide fragility shock overdue: VIX's historic Aug-24 spike, record single stock fragility & continued failing trading liquidity suggest a broad fragility shock is overdue.

Volatility flat to up: regardless of market direction, equity vol should be supported as vol rises in bubbles. Though some see 2017 post-Trump vol crush, this time is different.

Bond vigilantes + macro uncertainty biggest left tail risk: Fed may not cut as markets want given macro uncertainty; bond vigilantes may return with record US debt.

But Fed put strike rising supports LTD: Fed put strike is rising as inflation falls, supporting dip-buying in the left tail; don't underestimate the power of free Fed puts.

Europe is still cheap & unloved: Positioning heavily favors US. Don't get caught short an EU surprise or animal spirits lifting all boats. Short EU vol has less tail risk than US.

Asia has pockets of attractive upside, but also critical risks: Taiwan biggest AI threat; Asia-focused Nuclear a key next AI trade; China may bounce in '25 but with risks

Risks to the view:

Markets muted on falling vol: AI rally stalls, markets rotate to value while Trump's policies are all bark and no bite. Belief in option supply induces 2017-style vol crush pushing VIX below 10 as fragility falls with Tech positioning. Low risk/low returns ensue.

2025 outlook doesn't play out until 2026: Pace of long tech super-cycle is hard to time. Big-cap tech rally stalls in '25 & rotates to smaller AI beneficiaries. Major indices muted as big caps too heavy-weight. Bubble resumes in earnest nearer '26 as use-cases from big investments become more apparent.



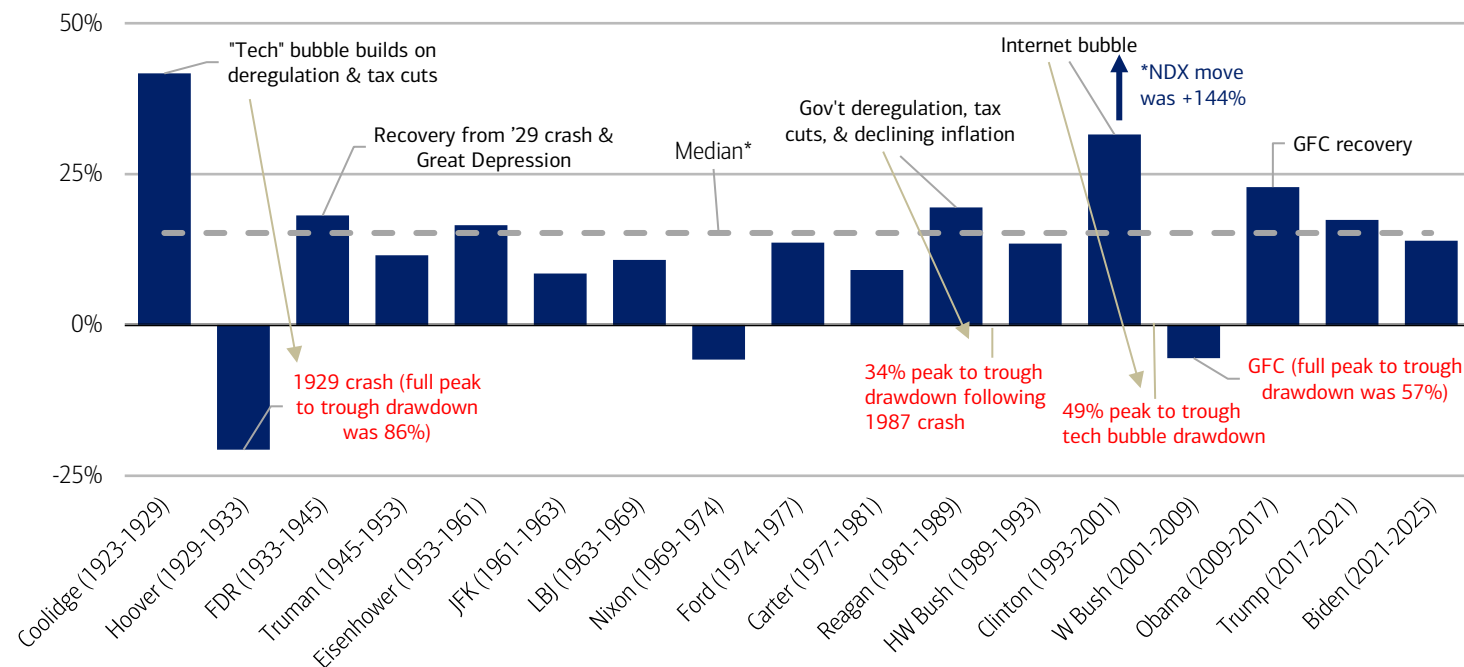
The roaring 2020s?

Bubbles and policy experiments collide. Not since the 1920s has the US seen today's mix of asset bubble potential combined with public policy experimentation. History suggests either of these on their own could drive large booms/busts in asset prices (Exhibit 1), and today we potentially have both. Trump's win and Republicans' sweep of both houses of Congress have given him a mandate for change, and his cabinet picks so far demonstrate that desire. While the policy that comes to fruition is yet unknown, based on campaign promises, it could potentially be the largest government deregulation since Ronald Reagan's administration ('81 – '89), the largest use of Tariffs since the Smoot-Hawley act of 1930, and the largest ever plans for the deportation of illegal immigrants in US history.

While the analog to the 1920s can't be taken too seriously given the extreme lack of financial regulation at the time, it was an era of laissez-faire economics, tax cuts, and deregulation, combined with a technology boom (automobiles, radios, electricity & consumer appliances). Not likely since then have we seen asset bubbles and policy experimentation operate simultaneously to the extent we could see from here.

Exhibit 1: Not since the 1920s have we seen the potential for a major tech boom to coincide with significant deregulation & tax cuts. Historically, both have driven large booms in equities; however, big busts have always followed.

Largest gains or losses in US equities during historical presidential terms normalized by their time in office (plotted as positive if biggest move is a gain, negative if biggest move is a loss).



Source: BofA Global Research. Data from 1-Jan-1921 to 21-Nov-2024. Prior to the S&P 500 data beginning on 30-Dec-27 we use Dow Jones Industrial Average returns. Largest presidential term equity move is calculated as the max absolute return during the presidential term starting from inauguration day, normalized by the length of their term. We plot returns as positive if the max move is a gain and negative if the max move is a loss. * Median is of positive bars only.

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Big booms are followed by big busts. Exhibit 1 shows that big equity booms that coincided with periods of deregulation and tax-cuts (Reagan, Coolidge) or big tech booms (Clinton, Coolidge) were followed by big busts ('29 crash, '87 crash, early 2000s internet-bubble unwind). In fact, if we look at the largest US equity draw-ups preceding bear markets since 1921 in US equities, they tend to be followed by the largest drawdowns in the year following the peak (Exhibit 2).

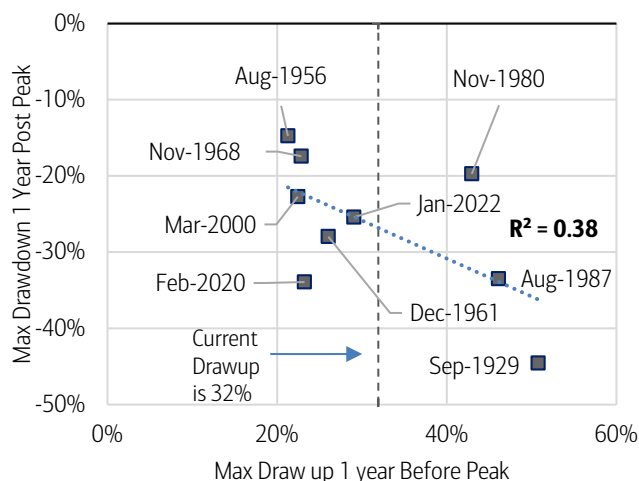
Biggest busts tend to be crashes, while rest are grinds. Exhibit 3 shows that the largest booms prior to the peak ('29 and '87) were followed by the fastest crashes (like the Covid shock in 2020, which was not driven as much by a prior boom but rather an exogenous crisis). Most other large sell-offs historically, including that of the 2000's



internet bubble (at least in S&P 500 terms), was more of a grind lower. Though interestingly as Exhibit 4 & Exhibit 5 show, in Nasdaq terms the 2000's tech bubble bust was convex. When it comes to hedging against these downside risks, planning for either eventuality is important, as we highlight in the trade sections below.

Exhibit 2: Bigger the boom, the bigger the bust: largest draw-ups prior to bear markets tend to be followed by larger drawdowns

Largest draw ups in US equities leading into peaks vs max drawdowns 1 year post the peak.

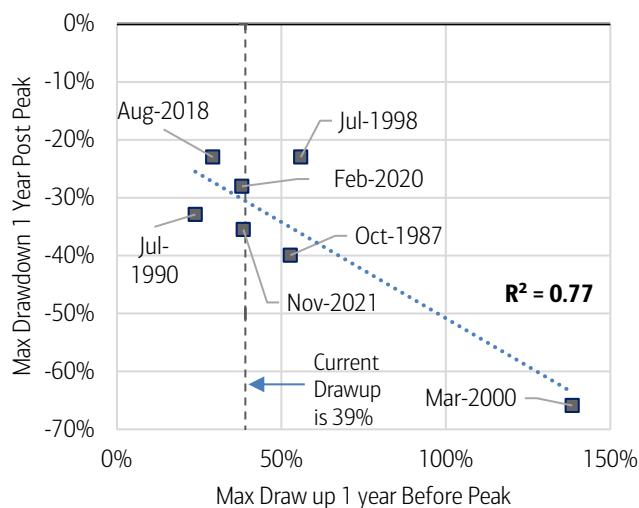


Source: BofA Global Research. Data from 1-Jan-1921 to 21-Nov-2024. Prior to the S&P 500 data beginning on 30-Dec-27 we use Dow Jones Industrial Average returns. Peaks are defined as the highs before a drawdown greater than 20% and peak dates are noted in the labels. A draw up is the return from the low in the year prior to the peak while a drawdown is the return from the peak to the low within the following one year.

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Exhibit 4: While the 2000s tech bubble wasn't the largest boom in S&P, it was in Nasdaq; it was also the biggest bust

Largest draw ups in the NASDAQ-100 leading into peaks vs max drawdowns 1 year post the peak

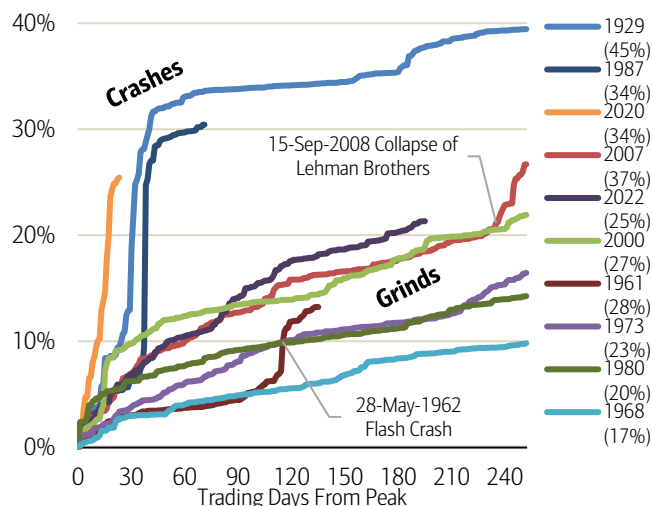


Source: BofA Global Research. Data from 4-Feb-1985 to 22-Nov-2024. Peaks are defined as the highs before a drawdown greater than 20% and peak dates are noted in the labels. A draw up is the return from the low in the year prior to the peak while a drawdown is the return from the peak to the low within the following one year.

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Exhibit 3: Biggest booms ('29,'87) were followed by crashes like Covid ('20) while most others (including '00 and '22) resulted in "grinds" lower in the S&P 500

Accrued Variance (vol pts) of the 10 largest S&P 500 Drawdowns during their first year (or market bottom if sooner)

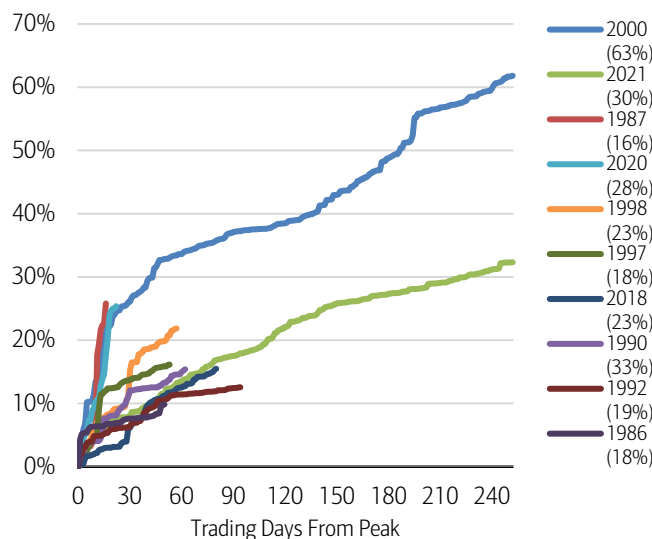


Source: BofA Global Research. Data from 1927 to 2024. The max drawdowns 1 year from peaks are noted in parenthesis in the legend. 252 trading days of data shown unless the trough was reached before then.

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Exhibit 5: While the 2000 tech bubble bust was a grind lower in S&P, in Nasdaq it started off as more of a crash similar to '87 & '20

Accrued Variance (vol pts) of the 10 largest NASDAQ-100 Drawdowns during their first year (or market bottom if sooner)



Source: BofA Global Research. Data from 1985 to 2024. The max drawdowns 1 year from peaks are noted in parenthesis in the legend. 252 trading days of data shown unless the trough was reached before then.

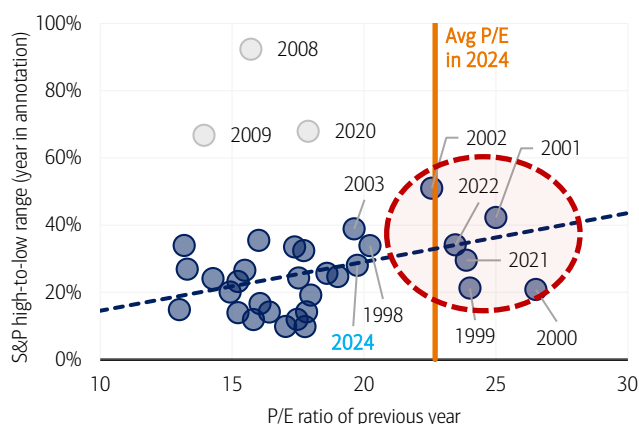
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AI rally has gone too far to neither boom nor bust. With just 2 years since ChatGPT kicked off the latest AI boom, the Nasdaq 100 is up 85% and Magnificent 7 stocks¹ have risen ~200%. As we argued in June (see [Volatility suggests AI isn't a bubble...yet](#)), the likelihood the AI boom turns into a much larger asset bubble before it's over seems high. And given it has progressed this far, with P/E ratios already historically stretched, chances are that the boom either continues or it busts leading to a reversal of already substantial gains. Exhibit 6 shows that the terminal distribution of equity market returns widens with rising valuations. Starting from today's forward P/E ratio of 25, history suggests one should expect relatively wide swings.

Internet bubble analog argues higher risk, higher return. The best (though not perfect) analog to today's AI boom is the internet bubble of the late 90s. That lasted over 5 years, witnessed an 11x increase in the Nasdaq-100 and saw broadly increasing risk and return as it progressed (Exhibit 7). Since ChatGPT was launched in Nov-22, the Nasdaq-100 has nearly doubled and Mag 7 stocks are 3x (though NVDA is 10x). However, looking at both volatility and valuations of tech stocks broadly suggests we are closer today to 1996/1997 in that analog than 1998/1999 (Exhibit 11 & Exhibit 15).

Exhibit 6: Historically, as valuations expand, so have swings in equity markets, as witnessed by the positive relationship between P/E ratios & high-to-low ranges in the S&P in the following 1yr period

S&P 500 high-to-low range vs P/E ratio of prior year

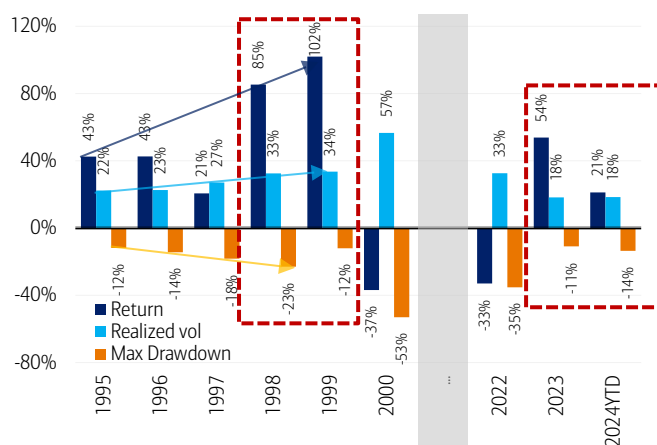


Source: BofA Global Research, Bloomberg. Data from 1-Jan-90 to 15-Nov-24. P/E ratio based on NTM expected EPS, and is calculated as a calendar year average. Average P/E in 2024 is YTD.

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Exhibit 7: The Tech bubble period saw the Nasdaq increase 11x, with rising vol & drawdown sizes over the 5 year span. By contrast, the AI era's gains and fluctuations have been relatively muted so far

Nasdaq 100 returns. Realized vol & max drawdown for the tech bubble (1995-2000) & the AI wave (2022-)



Source: BofA Global Research. Data from 1-Jan-95 to 15-Nov-24. Metrics are calculated for each calendar year.

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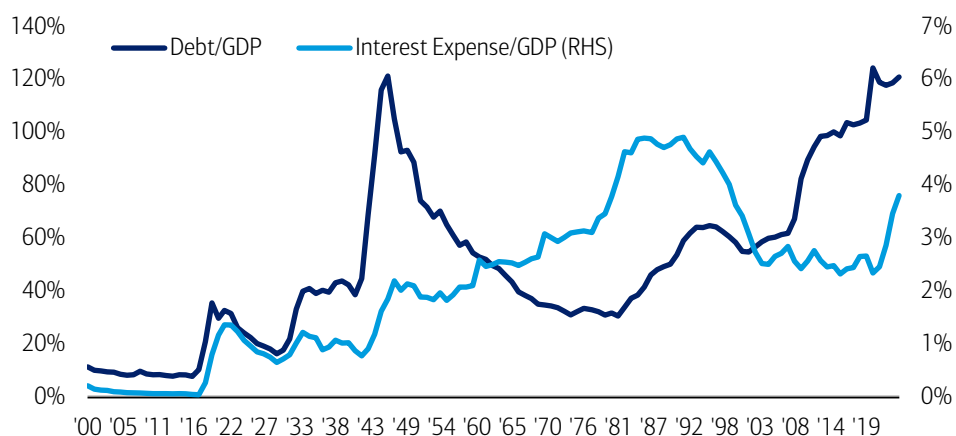
But could macro + fragility pop the euphoria? US Debt/GDP is near a record 120% today versus just 30% in the 1920s (Exhibit 8), and it even exceeds the WWII peak. With macro uncertainty and policy risking higher-for-longer interest rates, and potential policy shifts risking higher budget deficits (with higher debt servicing costs amplifying deficits), the risk is bond markets push back. This perhaps is the largest visible left tail risk today, which would be compounded by already elevated equity valuations given the AI boom to-date. As noted, busts almost inevitably follow booms.

¹ "Magnificent 7" stocks are defined as the group of Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL), Meta (META), Microsoft (MSFT), Nvidia (NVDA), & Tesla (TSLA).



Exhibit 8: US debt levels are reaching uncharted territory while servicing costs are sharply rising and at risk from higher yields and profligate policy

US Debt/GDP and Interest Expense/GDP ratios



Source: BofA Global Research, Federal Reserve, US Office of Management and Budget, Congressional Budget Office, & US Census Bureau.
Data from 1900 to 2024.

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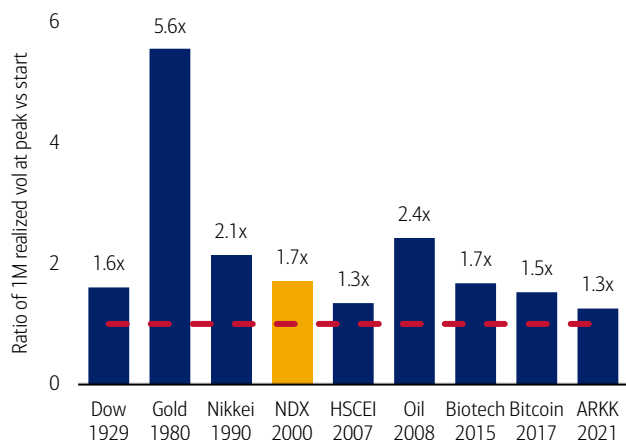
Volatility suggests AI likely to become a bigger bubble

Tech revolutions result in bubbles and bubbles result in volatility

In June-24 (see [Volatility suggests AI isn't a bubble...yet](#)) we noted that, in periods when Technology developments hold the promise of large productivity gains, this often results in asset bubbles, as markets compete to pull future potential forward. This happened during the first industrial revolution in 1830-60's Britain within railroad stocks, the Radio & Auto boom in the 1920s US and most recently in the 1990s internet bubble. Asset bubbles are also consistently marked by rising volatility along with rising prices (Exhibit 9 & Exhibit 10).

Exhibit 9: Looking at 9 notable asset bubbles in the last 100 years, volatility has consistently risen as the bubble developed

Rise in 1m realized vol from start to peak for historical bubbles

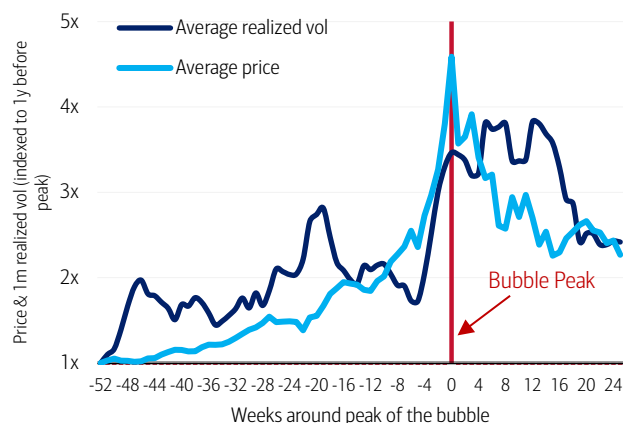


Source: BofA Global Research. Data from Sep-1927 to 09-Feb-22.

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Exhibit 10: Historically, asset bubbles have seen a rise in volatility concurrent with a rise in asset prices, with volatility continuing to rise after the bubble pops

Price and average 1m realized vol in historical bubbles (vs 1y before peak)



Source: BofA Global Research. Data from Sep-1927 to 09-Feb-22. Historical asset bubbles cover Dow Jones in Sep-29, Gold in Jan-80, Nasdaq in Mar-00, HSCEI in Oct-07, oil in Jul-08, biotech in Jun-15, Bitcoin in Dec-17 and ARKK in Feb-21.

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Vol & valuation suggest more room to run

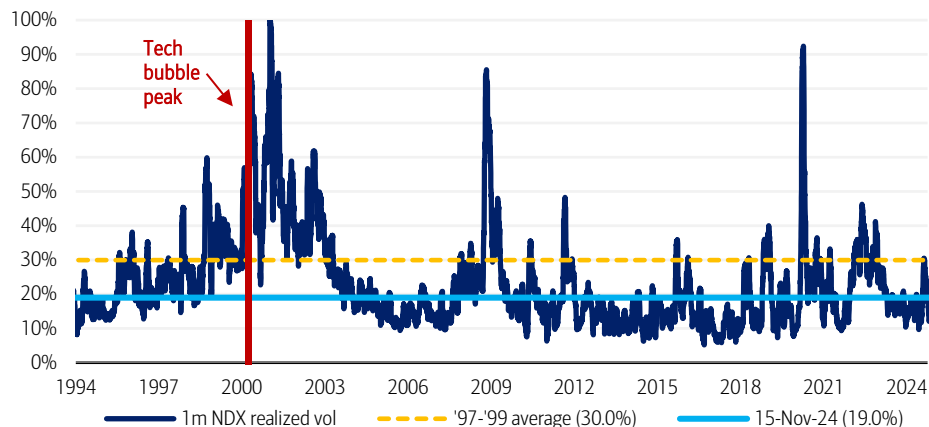
Updating the work from our Jun-24 piece suggests we are yet to see the kind of rise in volatility we saw in the latter stages of the 90's internet bubble – the most reasonable (but not perfect) analog to today's AI boom as Exhibit 11 to Exhibit 13 show.

Furthermore, tech valuations remain far from the peak levels recorded during the '90s internet bubble (Exhibit 14 & Exhibit 15). An impressive feature of the 90s internet bubble was also how resilient it was to macro risk. That period witnessed 2 major macro crises ('97 Asian crisis and '98 Long Term Capital Management failure), a bear market in Nasdaq in '98 and over 100bps of Fed rate hikes (Exhibit 16). If we get further Fed cuts in 2025 as markets expect (without an economic collapse), this could further amplify upside risks.



Exhibit 11: Nasdaq vol has risen materially from the lows of earlier this year; however, levels still sit below those seen during the 1997-99 period (particularly the peaks), indicating that there may be room for even higher vol as the AI boom progresses

Nasdaq 1m realized vol

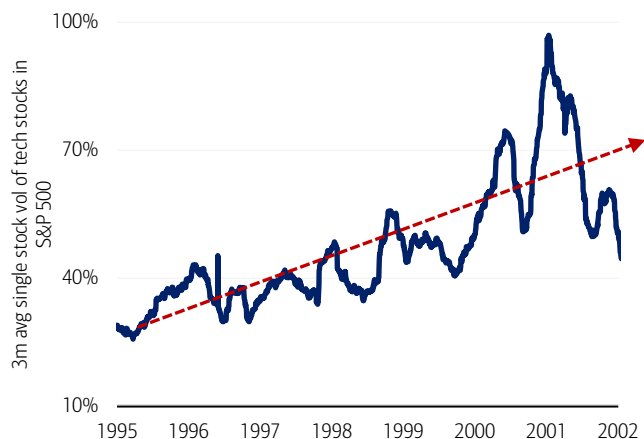


Source: BofA Global Research. Data from 1-Jan-94 to 15-Nov-24.

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Exhibit 12: Single stock volatility within the tech sector rose during the Tech Bubble

Single stock 1m realized vol for S&P 500 tech stocks in the Tech Bubble era

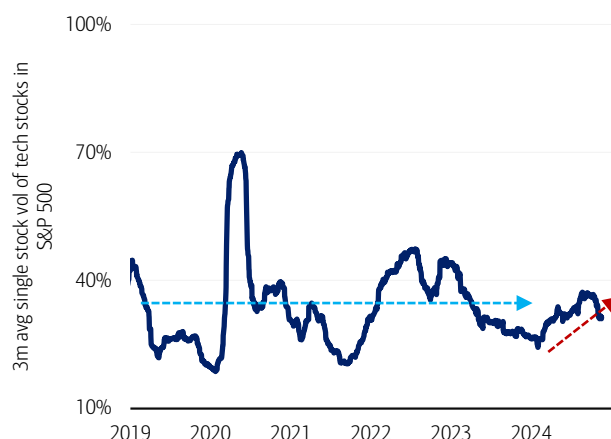


Source: BofA Global Research. Data from 1-Jan-95 to 31-Dec-01. "Tech" is defined by the Information Technology GICS classification along with primarily tech-driven stocks in other classes (e.g. TSLA, AMZN, etc.). Single stock vol is calculated as the weighted average of the volatilities of S&P 500 constituents (based on prevailing index weights).

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Exhibit 13: Single stock tech vol in the AI era has been trending higher but remains far from levels seen during the Tech Bubble

Single stock 1m realized vol for S&P 500 tech stocks over past 5 years

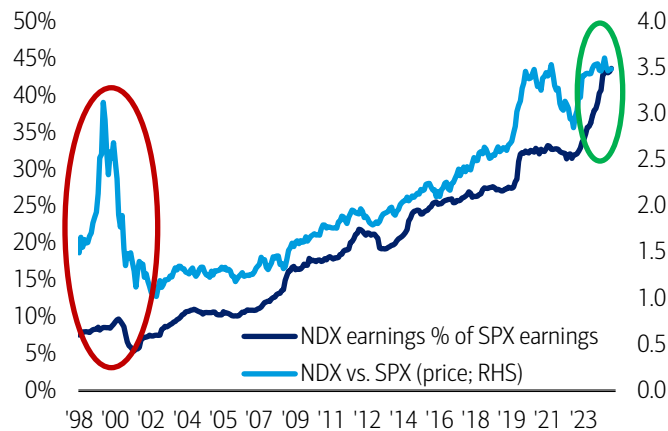


Source: BofA Global Research. Data from 1-Jan-19 to 15-Nov-24. "Tech" is defined by the Information Technology GICS classification along with primarily tech-driven stocks in other classes (e.g. TSLA, AMZN, etc.). Single stock vol is calculated as the weighted average of the volatilities of S&P 500 constituents (based on prevailing index weights).

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Exhibit 14: During the 90's Tech Bubble, stock prices clearly decoupled from fundamentals, a phenomenon yet to be seen recently

Nasdaq 100 vs S&P 500 – earnings & performance

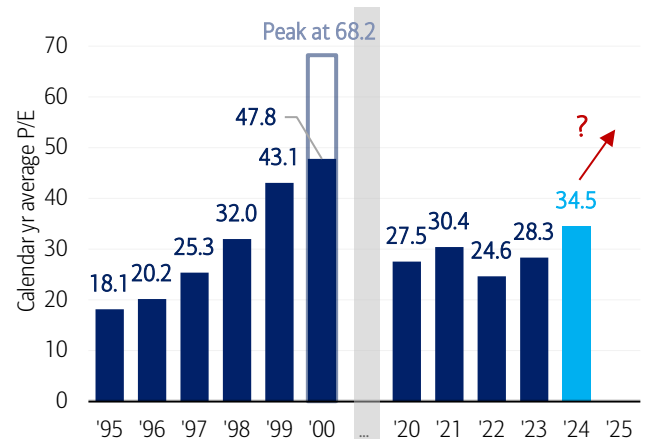


Source: BofA US Equity & Quant Strategy, FactSet. Data from Jan-98 to Oct-24.

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Exhibit 15: While valuations of tech stocks have risen materially since 2022, they remain below those seen during the Tech Bubble, suggesting there may be further room for expansion

S&P 500 Tech sector P/E ratio for the tech bubble (1995-2000) & the AI wave (2022-)

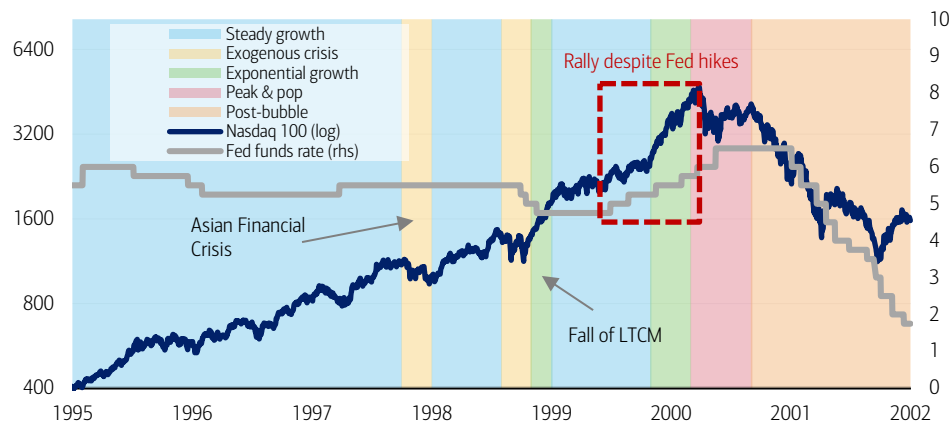


Source: BofA Global Research, Bloomberg. Data from 1-Jan-95 to 15-Nov-24. P/E ratio based on NTM expected EPS, and is calculated as a calendar year average.

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Exhibit 16: Between 1995 & early 2000, the Nasdaq grew more than 11x while withstanding the shocks of the Asian Financial Crisis and the fall of LTCM, as well as 100bps of Fed rate hikes

Nasdaq 100 prices (log scale) over the tech bubble era



Source: BofA Global Research, Bloomberg. Data from 1-Jan-95 to 31-Dec-01.

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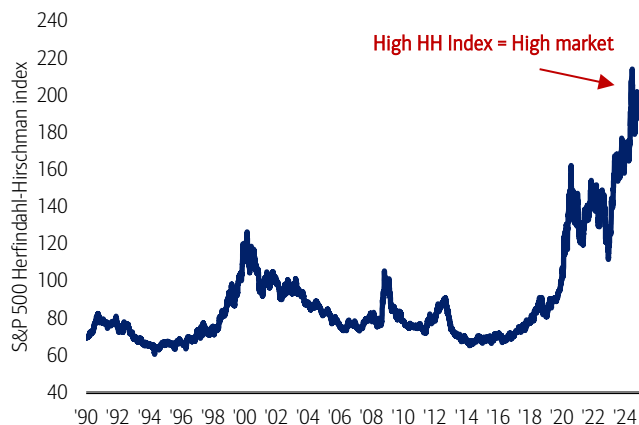
90s analog not perfect either

As we noted in June, there are also key differences between today's AI boom and the 90s internet bubble, most notably the extreme concentration and size of the leaders today (Exhibit 17 & Exhibit 18). It may be harder for the largest beneficiaries in the AI boom such as NVDA to reach the valuations recorded in the 90s, given the extreme size it would imply (NVDA at CSCO's peak tech bubble valuation would be valued at ~\$11tn and >45% of today's US GDP). Hence, it may require a broadening of bubble beneficiaries to smaller companies to generate similar valuations. Nonetheless, history suggests there may be little reason for the current AI momentum to run out of steam yet either. The bottom line is to expect higher risk & higher returns before it's over.



Exhibit 17: The concentration of market cap within the S&P 500 is far higher today than that witnessed even at the peak of the 90s Tech Bubble

Herfindahl-Hirschman Index for the S&P 500

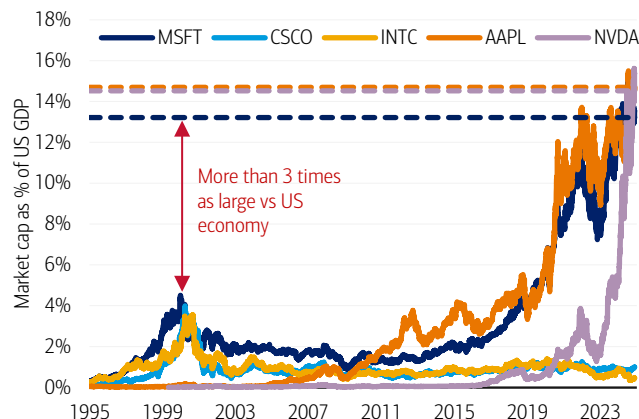


Source: BofA Global Research. Data from 31-Jan-90 to 15-Nov-24. Herfindahl-Hirschman index is a market share concentration measure, equal to sum of squares of single stock weightings.

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Exhibit 18: Compared to the Tech Bubble, today's tech megacaps are almost 3x larger with respect to the entire US economy, which may act as a hurdle in the pace at which an AI bubble forms

Market cap as % of US GDP (Top 3 tech stocks in 90's Tech Bubble & today)



Source: BofA Global Research. Data from 1-Jan-95 to 15-Nov-24.

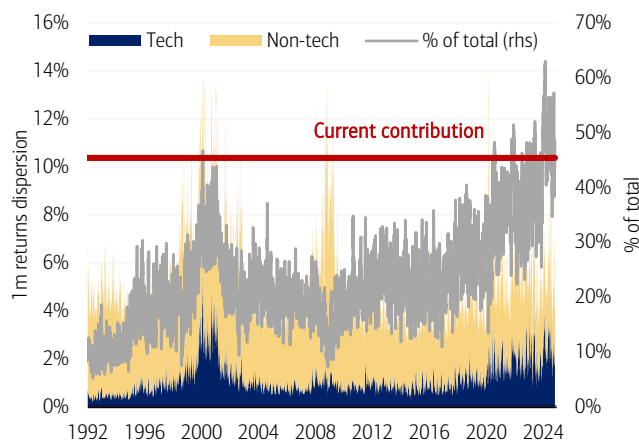
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Concentration remains a significant risk for active managers

Record concentration in US equities is resulting in historic concentration of active risk. If stock volatility rises further (and concentration grows) as asset bubbles inflate, this will only amplify the challenge of managing stock portfolios against a market-cap weighted benchmark. As Exhibit 19 shows, within the S&P 500, tech stocks are already contributing record return dispersion (a.k.a. active risk), even exceeding that in the 90's tech bubble, however, this is primarily due to the size of tech and not its underlying dispersion (Exhibit 20). Hence this risk will likely grow further as the bubble progresses. On top of this, single stocks are becoming more fragile (Exhibit 22 & Exhibit 23) making it even more necessary to manage active risks into key catalysts such as earnings. Thankfully, options are increasingly underpricing this risk (see Exhibit 30 & our [22-Oct-24 GEVI](#) report).

Exhibit 19: Today, tech's contribution to returns dispersion (active equity risk) in the S&P is close to all-time highs and near its (extraordinary) contribution during the tech bubble peak...

Tech contribution to SPX single stock 1m returns dispersion (cap-weight)

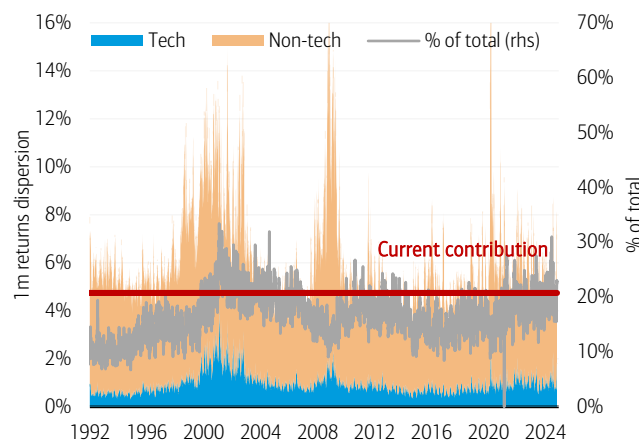


Source: BofA Global Research. Data from 1-Jan-92 to 15-Nov-24. "Tech" is defined by the Information Technology class of the Global Industry Classification Standard along with companies in other classifications that are primarily internet/tech driven (e.g. Tesla, Amazon, Uber etc.).

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Exhibit 20: ...but much of this is driven by increasing market cap concentration, since contribution on an equally weighted basis has only risen moderately

Tech contribution to SPX single stock 1m returns dispersion (equal-weight)



Source: BofA Global Research. Data from 1-Jan-92 to 15-Nov-24. "Tech" is defined by the Information Technology class of the Global Industry Classification Standard along with companies in other classifications that are primarily internet/tech driven (e.g. Tesla, Amazon, Uber etc.).

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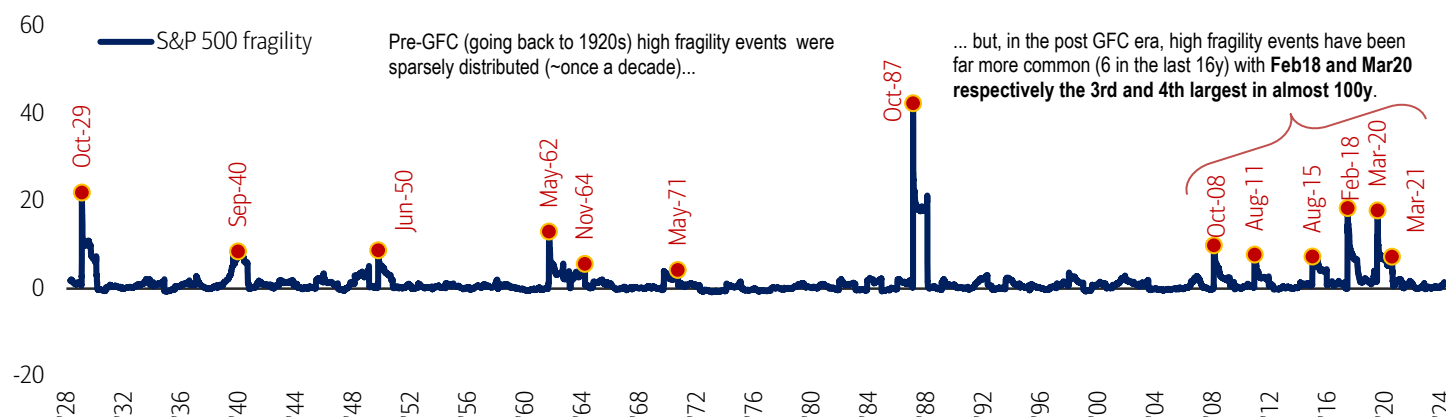
Fragility remains a risk

VIX shock a reminder the underpinnings of fragility persist

Markets have become structurally more fragile in the last several decades. This owes to a feedback loop of investors crowding into a limited set of momentum trades and then facing a dearth of trading liquidity upon exit (when liquidity is needed most). We first put forward the underlying thesis for this structural shift towards asset distributions becoming more leptokurtic (longer periods of calm followed by fatter tails) in our 2016 outlook, [Fragility is the new volatility](#). Since then, the S&P 500 has recorded 2 of its 4 largest fragility shocks in nearly 100yrs (Feb-18 and Mar-20) and has become ~5x more fragile since the GFC (Exhibit 21). This is also a cross-asset phenomenon, with increased fragility in rates, credit, FX and commodity markets. The most recent example of extreme fragility emanated from the Aug 5, 2024 VIX shock, the biggest fragility event in its 34yr history (see [VIX 65: Its' a liquidity problem](#)). This is a stark reminder that the underpinnings of fragility persist and that positioning is an important driver (see VIX imbalances growing: what it means & how to trade it section).

Exhibit 21: The S&P 500 has become ~5x more fragile since the GFC

S&P 500 fragility



Source: BofA Global Research. Data from 30-Dec-1927 to 19-Nov-2024. Fragility is measured as the rolling 1 year kurtosis of weekly log returns.

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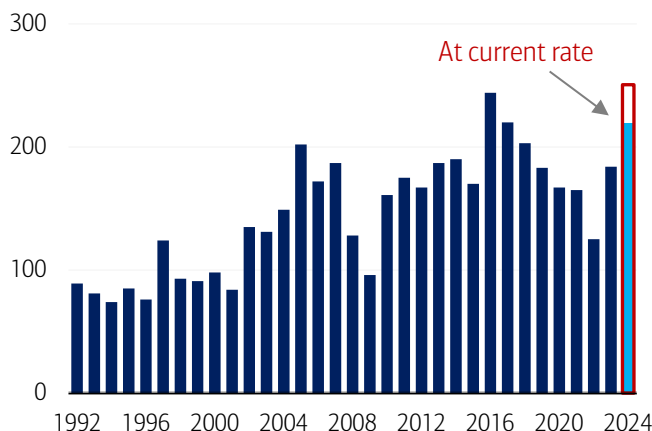
Single stock fragility at record levels and may grow as tech bubble inflates

More recently, fragility has also been rising among US single stocks. Indeed, it is now near 30yr+ extremes in both frequency and magnitude (Exhibit 22 & Exhibit 23). Much of this fragility has been taking place around earnings announcements. Among S&P 500 stocks, the average absolute reaction to earnings is at post-GFC highs (Exhibit 24), as is the share of annual realized volatility coming from those earning reactions (Exhibit 25).



Exhibit 22: So far in 2024 we are on-pace to see the most number of fragility events among US tech stocks in history

Number of fragility events (sigma move magnitude > 3) for S&P 500 Tech stocks

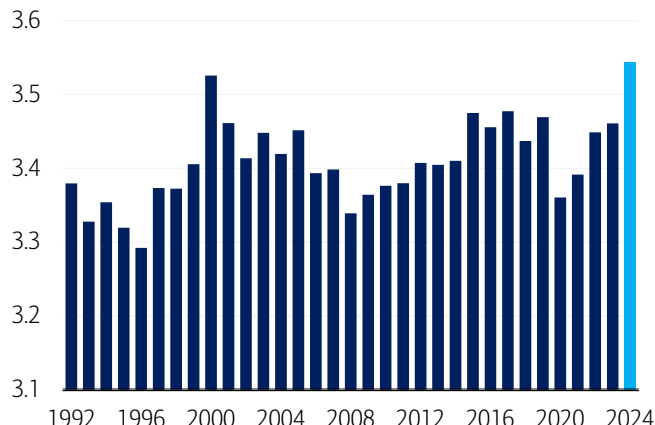


Source: BofA Global Research. Data from 1-Jan-92 to 15-Nov-24. "Sigma move" is measured by dividing a stock's 1d return by its 21d trailing volatility. It is a measure of how large a shock is vs its risk projected by recent volatility and is one measure of price "fragility". "Tech" is defined by the Information Technology GICS class along with stocks in other classes that are primarily tech-driven (e.g. Tesla, Amazon, etc.). Components of index kept fixed from the start of every calendar year

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Exhibit 23: US stocks are also setting records for the size of their fragility events, exceeding even that at the peak of the internet bubble

Average magnitude of fragility events (when sigma move magnitude is > 3) for the largest 50 stocks in the S&P 500



Source: BofA Global Research. Data from 1-Jan-92 to 15-Nov-24. "Sigma move" is measured by dividing a stock's 1d return by its 21d trailing volatility. It is a measure of how large a shock is vs its risk projected by recent volatility and is one measure of price "fragility". Components of index kept fixed from the start of every calendar year.

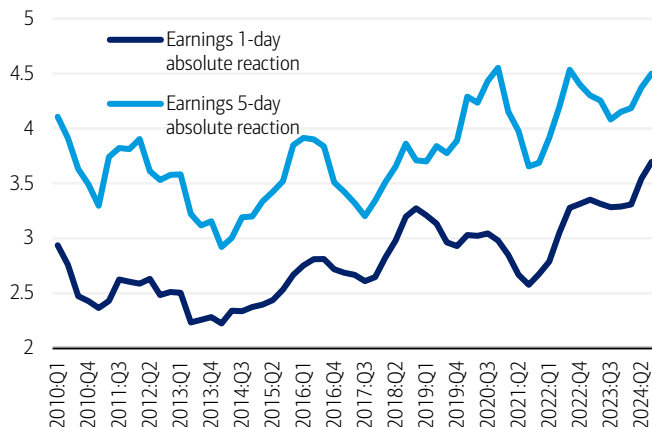
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Earnings reactions are not just larger, but also faster

This hyperreactivity to earnings results is not due to more volatile fundamentals, as neither earnings volatility nor the size of beats or misses are elevated relative to history (Exhibit 26). At least part of the reason may be that price adjustments to earnings results are happening *faster*. As seen in Exhibit 27, a larger percentage than ever of the 5-day earnings reaction has been taking place during the first day. And of that first day's move, the overnight session had never contributed as much as it has in recent quarters (for stocks reporting after the close). Interestingly, this result is partly driven by stocks increasingly overshooting on the downside (more negative returns on the 1st day than over the first 5 days; Exhibit 28). Given that the same is not true on the upside, we can likely rule out liquidity- or positioning-driven overshoots and rather point to buy-the-dip forces driving stocks higher after an initial selloff.

Exhibit 24: Among S&P 500 stocks, the average absolute reaction to earnings is at post-GFC highs

Median absolute 1-day & 5-day reactions to earnings results (S&P 500 stocks; 4 quarter moving average)



Source: BofA Global Research. Median of all S&P 500 stocks each quarter. Reaction in trading days.

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Exhibit 25: So is the share of annual realized volatility coming from those earning reactions

Median 12m realized vol / realized vol exc. earnings reactions (S&P 500 stocks; 12m MA)

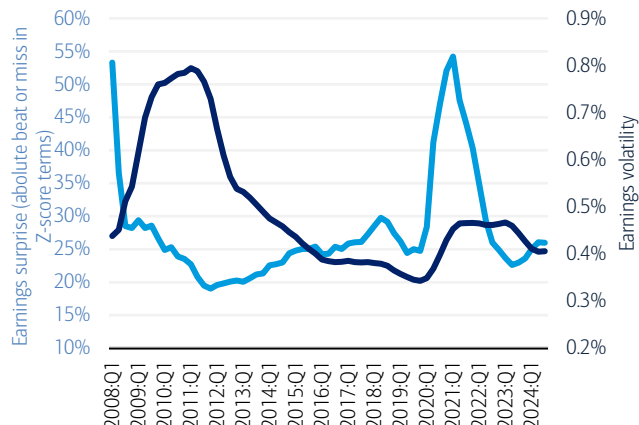


Source: BofA Global Research. The denominator excludes the 1-day reaction to earnings.

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Exhibit 26: This hyperreactivity of stocks to earnings is not due to either higher earnings volatility or the size of beats or misses

Median earnings surprise and 12-quarter earnings volatility (S&P 500 stocks; 4 quarter moving average)

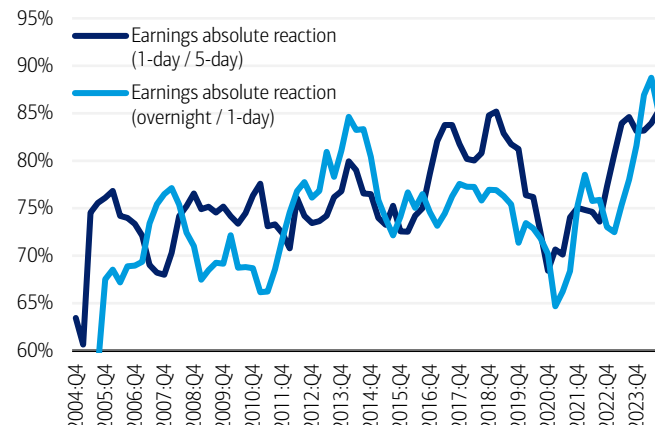


Source: BofA Global Research. Surprise Z-score = (actual EPS – consensus EPS)/12-quarter standard deviation of EPS

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Exhibit 27: At least part of the reason may be that price adjustments to earnings results are happening faster

Median absolute 1-day vs 5-day earnings reactions and overnight vs 1-day reactions (S&P 500 stocks; 4 quarter moving average)



Source: BofA Global Research. Median of all S&P 500 stocks each quarter. Reaction in trading days.

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Faster reactions support realized vol

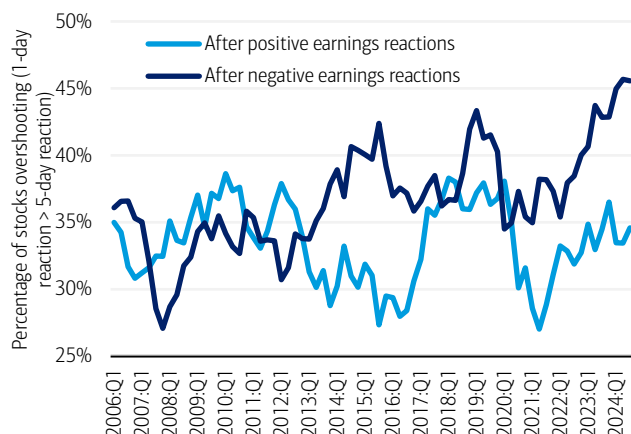
One notable consequence of this faster reactivity is its positive contribution to single stock realized vol. As the hypothetical example in Exhibit 29 shows, when returns are concentrated in fewer days, the mechanical result is higher volatility, even if the stock ends up at the same level over time.

Option prices haven't fully adjusted to higher single stock fragility

The rise in single stock fragility and earnings reactivity hasn't happened overnight. Yet option prices haven't fully adjusted to this dynamic. Both around earnings (Exhibit 31) and more generally (Exhibit 30), the performance of long single-stock option positions has been steadily improving post-Covid.

Exhibit 28: More and more stocks are seeing post-earnings dips bought, while post-earnings rallies are not being sold more than usual

Percent of stocks rising/falling more in 1st day post-earnings than by day 5



Source: BofA Global Research. Percent of S&P 500 stocks each quarter giving up some gains or recovering some losses between days 1 and 5 post-earnings.

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Exhibit 29: Faster price adjustments mechanically lead to higher volatility

Hypothetical stock price paths and realized volatility



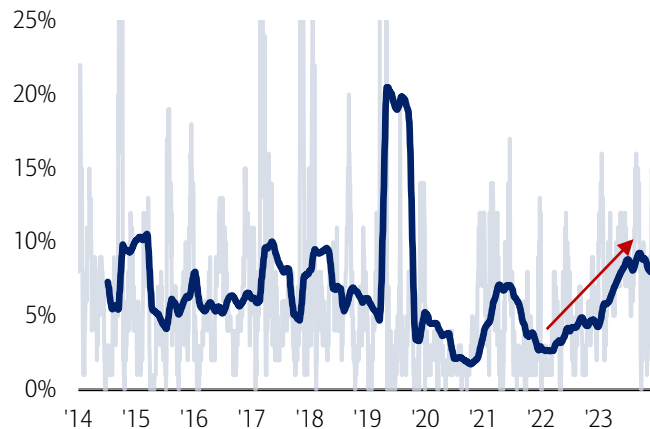
Source: BofA Global Research.

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Exhibit 30: Despite the increased incidence of single stock fragility, options markets don't seem to have reacted commensurately

Percent of S&P Top 50 stocks with 1-day moves larger than 1m 25-delta option* breakeven return (6m moving average)

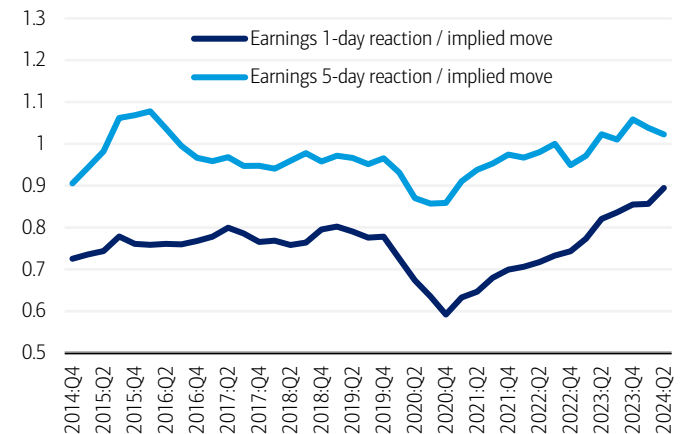


Source: BofA Global Research. Data from 12-Nov-14 to 15-Nov-24. *Average of call & put premium, uses theoretical breakeven return based on indicative mid prices. We compare this option breakeven to the largest 1-day return over the following 1m.

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Exhibit 31: Option implied moves around earnings are not keeping up with the rise in realized moves post-Covid

Median earnings reaction divided by implied move 5 days prior (S&P 500 stocks, 4 quarter moving average)



Source: BofA Global Research. Implied moves from Bloomberg.

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Are we due a broad market fragility shock?

As noted, we have seen an approximate 5-fold increase in the frequency of fragility shocks in the S&P 500 compared to the 80 years prior. Whereas from 1928 – 2008, we recorded a major fragility event approximately every 10 years, this frequency has increased to every 2-4 years since. Given the last major shock was March-2020, history would suggest an index-wide event may be overdue. As crowding and illiquidity are the primary drivers fragility, a continuation in AI momentum, possibly fueled by populist policy, should only further increase the risks of instability in 2025.

Return of the biggest right tailwind: the Fed put

Fed put strike rising with falling inflation

The biggest visible left tail risk post-Covid that never came to fruition was the loss of the high strike Fed put for the first time in its 30+year history. However, with US inflation having fallen from near double digits at its post-Covid peak to within striking distance of the Fed's 2% target (Exhibit 32), this risk has dissipated materially. It may not be entirely restored to its pre-Covid power due to still high economic uncertainty (and new inflation risks in 2025 could prolong this), however, dip-buyers have nicely carried us across the deepest part of the chasm (Exhibit 34).

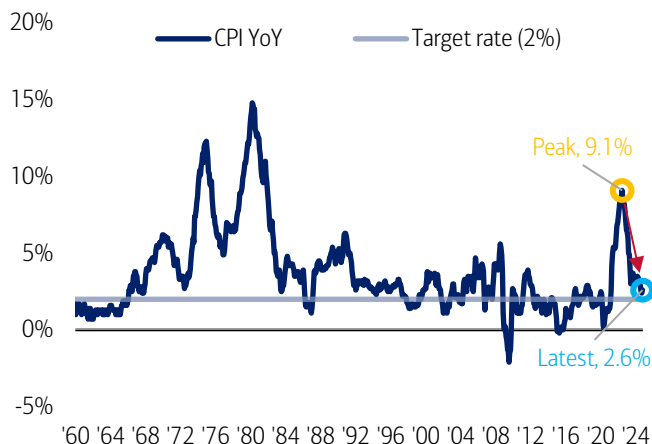
High strike Fed put perhaps single most important driver of equities ever

This is important as the high strike Fed put arguably has been among the most important drivers of US equities in the last 100 years. In the 37 years since Alan Greenspan first introduced the Fed put during the 1987 crash, the P/E ratio of US equities has nearly doubled compared to the 87 years prior (Exhibit 33). By construction, if you have a free put (in unlimited amounts), assets are simply worth more.

And supports dip-buying, crowding, and fragility

This has incentivized investors to buy equity dips, a trend that has also strengthened since 1987 (Exhibit 35) and has helped to reduce equity risk, as evidenced by the S&P recording consistently longer periods without 5% pullbacks (Exhibit 36). Dip-buying also supports momentum, crowding and ultimately higher fragility, which has materially increased in frequency in recent years as noted prior (Exhibit 21). Thus while the left tail protection of the Fed put does not come for entirely for free (it comes at a cost of greater asset instability), it is nonetheless a potentially important additional right tailwind from here.

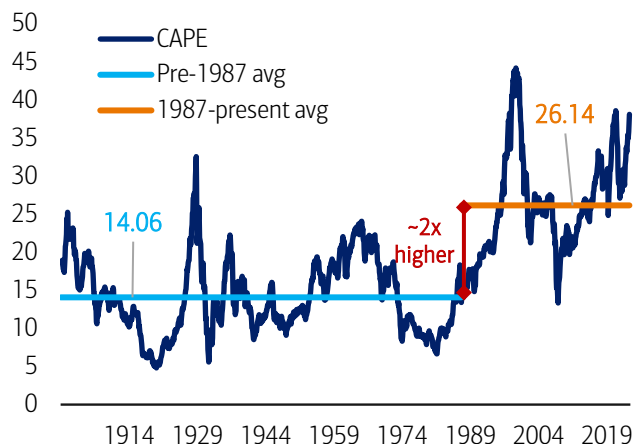
Exhibit 32: Inflation has come off rapidly from its >9% peak in 2022 and has been moving towards the Fed's 2% target
US CPI YoY since 1960



Source: BofA Global Research, Bloomberg. Data from Jan-1960 to Oct-24.

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Exhibit 33: Valuations in US equities have nearly doubled since 1987 when measured using cyclically-adjusted P/E ratios
US stock market cyclically adjusted P/E (CAPE) ratio



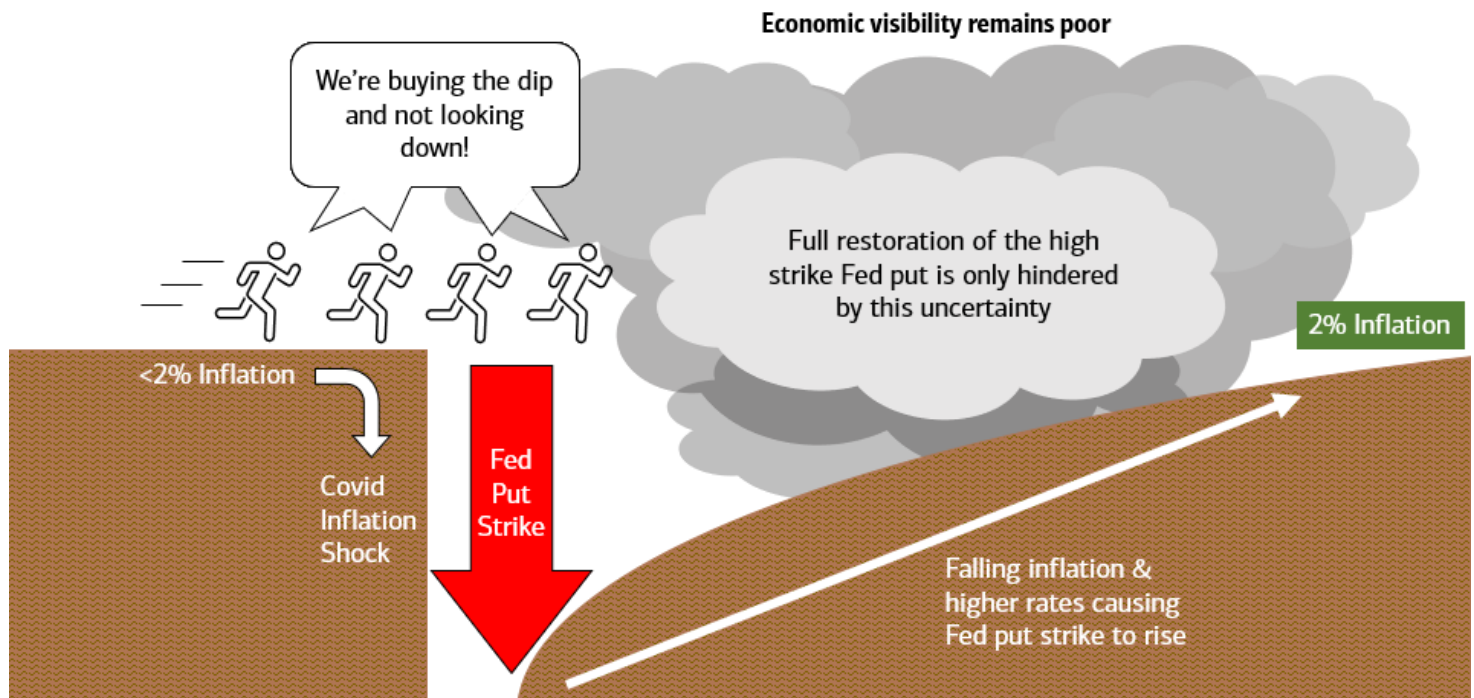
Source: BofA Global Research, Robert Shiller. Data from Jan-1900 to Nov-24. CAPE ratio is based on the average inflation-adjusted earnings from the previous 10 years.

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Exhibit 34: As inflation has fallen back closer to 2%, dip-buyers have carried across the chasm, back to a world of higher strike Fed puts

Stylized illustration of the evolution of the Fed put strike over time

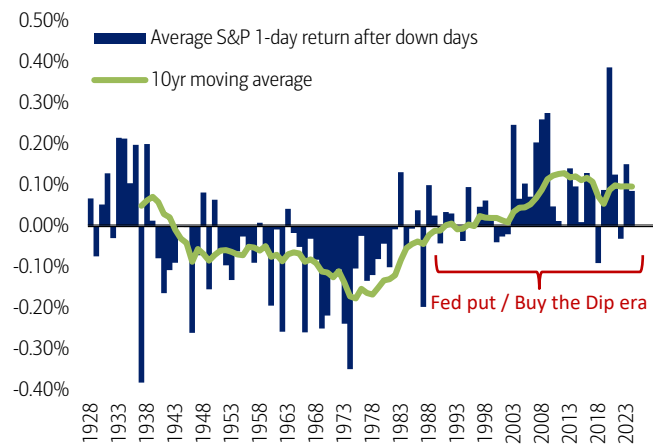


Source: BofA Global Research.

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Exhibit 35: Dip-buying has become a structurally more attractive strategy post 1987

S&P 1-day returns after down days (calendar-year average)

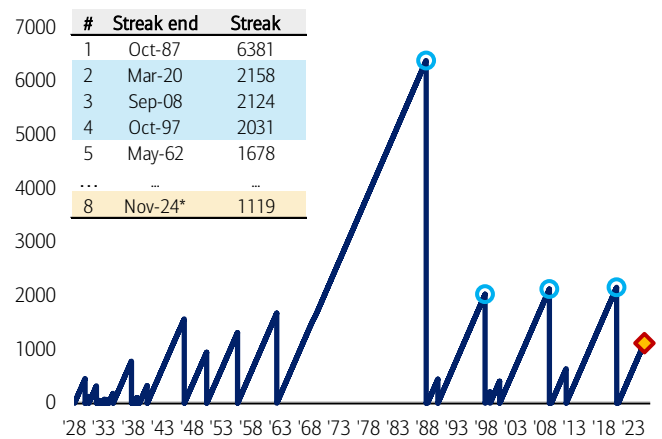


Source: BofA Global Research. Data from Jan-1928 to 20-Nov-24.

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Exhibit 36: Buy-the-dip conditioning has meant that 3 of the 5 longest periods without a 5% S&P pullback have occurred post-1987

No. of consecutive days without an S&P 5% down day



Source: BofA Global Research. Data from Jan-1928 to 20-Nov-24. *Nov-24 streak is ongoing.

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Why a 2017-style vol crush is unlikely in 2025

A hot topic post-election has been whether we will see a repeat of the ultra-low vol environment of 2017, when the S&P generated a meagre 6.8% realized volatility (Exhibit 37) and the VIX closed below 10 a staggering 52 times in Trump's first year in office.

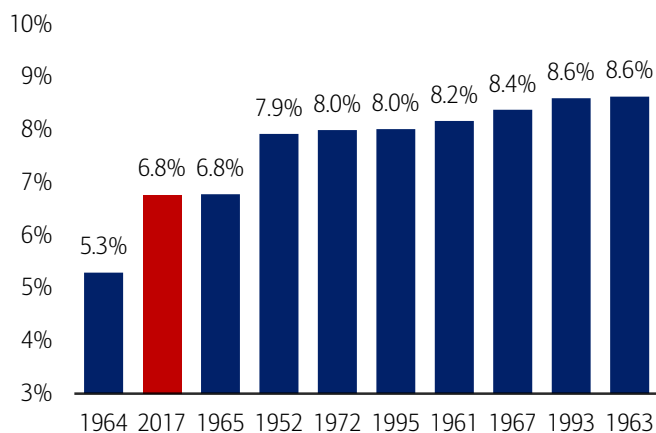
We have argued that "this time is different" due to today's mix of asset bubble potential + public policy experimentation, 100yr-high US debt levels, and still-elevated macro uncertainty post-Covid. But we expect many investors to be cynical that such *top-down* factors alone can keep US equity vol sufficiently supported next year. What does a more *bottom-up* approach suggest about the potential for a 2017-style vol crush in 2025?

Today's stretched valuations inconsistent with 2017's ultra-low vol

Higher equity market valuations tend to correlate with higher equity volatility over the subsequent year (Exhibit 38). 2024's elevated multiples (average forward P/E of 23 vs 18 in 2016) suggest S&P realized vol should at minimum be supported at 2024 levels of 12-13% (all points to the right of 2024 are at least as high) and show what a further outlier a repeat of 2017's 6.8% realized vol would be from today's initial conditions.

Exhibit 37: 2017 generated the 2nd-lowest calendar year realized vol since 1928 (6.8%) and saw the VIX close beneath ten 52 times; will Trump 2.0 bring about a repeat?

Calendar years with lowest S&P 500 realized volatility since 1928

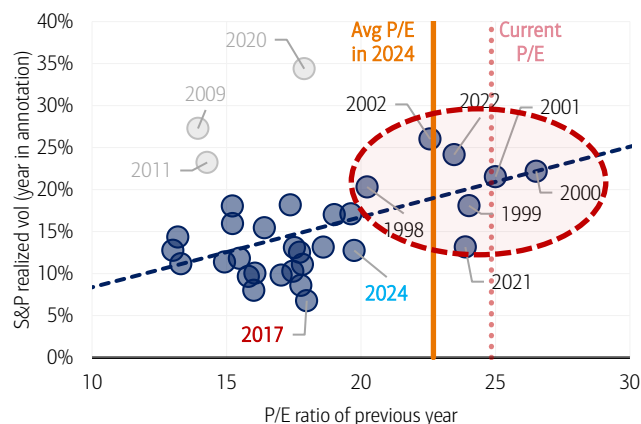


Source: BofA Global Research. Data from Jan-1928 to 20-Nov-24.

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Exhibit 38: Based on average P/E so far in 2024, we have never seen S&P volatility realize lower than what has been realized so far in 2024, and current P/E ratios suggest higher volatility still

S&P 500 calendar year realized volatility vs P/E ratio of prior year



Source: BofA Global Research, Bloomberg. Data from 1-Jan-90 to 15-Nov-24. P/E ratio based on NTM expected EPS and is calculated as a calendar year average. Average P/E in 2024 is YTD.

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Single stocks only becoming more volatile & more fragile

Similar to 2017, stock correlation has already hit and sustained historic lows in 2024 (see Exhibit 39 & Exhibit 41). Hence, a 2017 redux would primarily have to come from a collapse in single stock realized vol, which has been ~50% higher in 2024 than in 2017 (26% vs 17%; Exhibit 40) and is currently in the 67th %ile since 1991 (Exhibit 39).

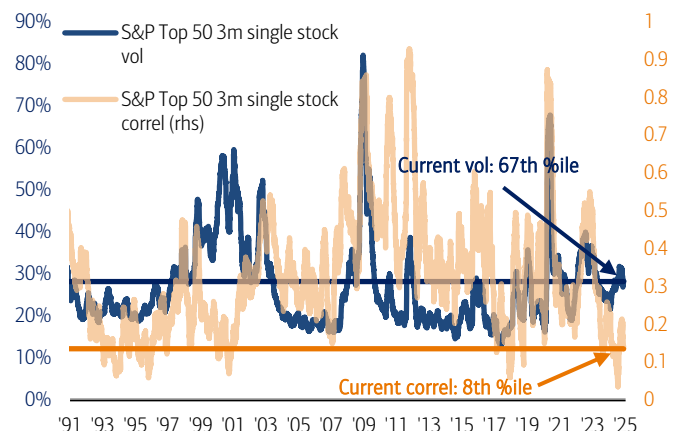
We see this as unlikely, however, for several reasons:

- S&P stock volatility has been on a sustained uptrend since hitting all-time lows in 2017 (Exhibit 39), in part as the reaction of stocks to earnings has been ratcheting higher (Exhibit 24).
- Earnings reactions have also been getting faster (e.g., concentrated in a shorter time window; Exhibit 27), not just larger, and faster price adjustments mechanically drive greater variance accrual, hence volatility (Exhibit 29).
- The magnitude of fragility shocks in the largest S&P stocks reached 30yr+ extremes in 2024 (Exhibit 23), with few signs of dissipating if the AI boom continues.



Exhibit 39: Correlation has been pinned at historic lows in 2024, hence a 2017-style vol crush would have to be driven by a collapse in single stock realized volatility

3m weighted avg. realized vol & correlation for S&P Top 50 stocks



Source: BofA Global Research. Data from 1-Jan-91 to 15-Nov-24. Average single stock vol weighted by prevailing index weights.

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Exhibit 40: However, single stocks have only become more volatile & more fragile in recent years, with stock vol running ~50% higher in 2024 than in 2017 (26% vs 17%)

1m weighted avg. realized vol for S&P 500 stocks in 2017 vs 2024 YTD

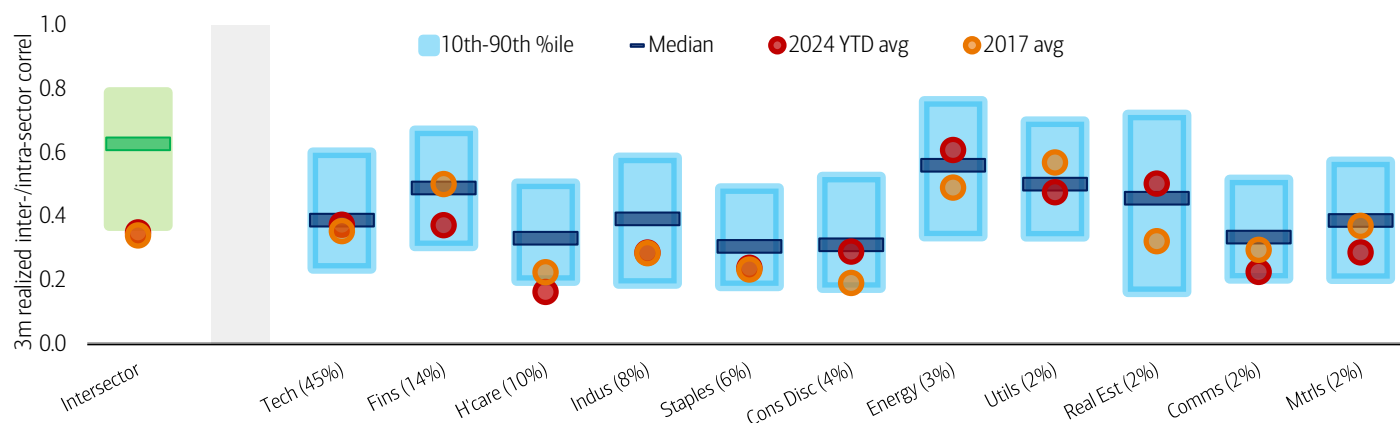


Source: BofA Global Research. Data from 1-Jan-17 to 29-Dec-17 for 2017 and 1-Jan-24 to 15-Nov-24 for 2024. Average single stock vol weighted by prevailing index weights.

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Exhibit 41: Inter-sector correlation in 2024 sat in its bottom decile since 1990 and close to 2017 levels; intra-sector correlations in heavyweights like Tech, Financials, and Healthcare (~70% of S&P in 2024) were at or below 2017 levels

Historical distribution of 3m inter-sector and intra-sector correlations for the S&P 500



Source: BofA Global Research. Data from 1-Jun-90 to 15-Nov-24. S&P sector weights in parentheses are as of 15-Nov-24. Sectors are defined as per the Global Industry Classification Standard along with re-classification of certain companies (primarily those not in tech that are largely internet/tech driven, e.g., Tesla, Amazon, Uber etc.). Intra-sector correlations are ranked by decreasing current sector weight in the S&P.

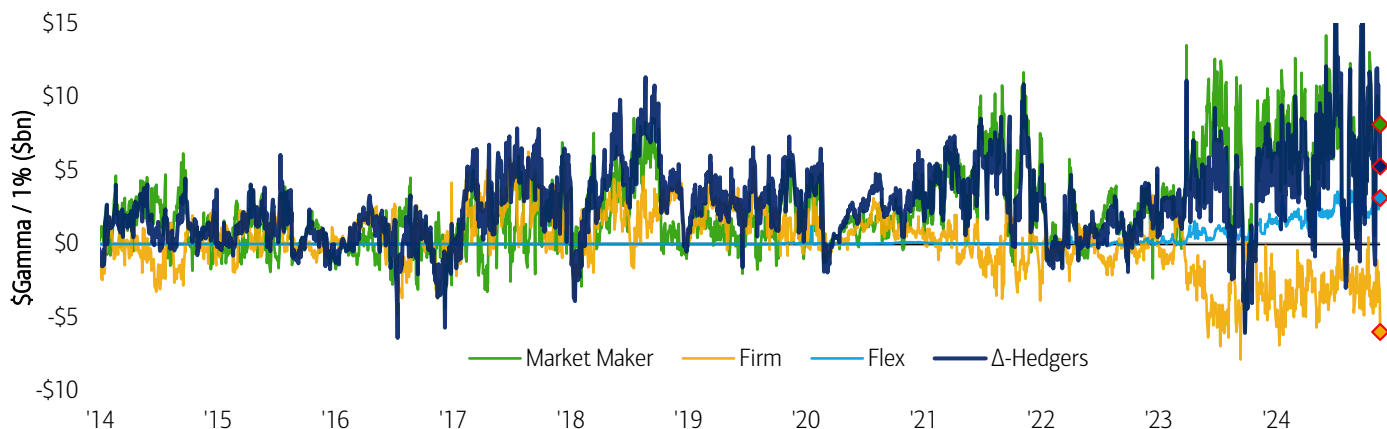
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What about all the option selling though?

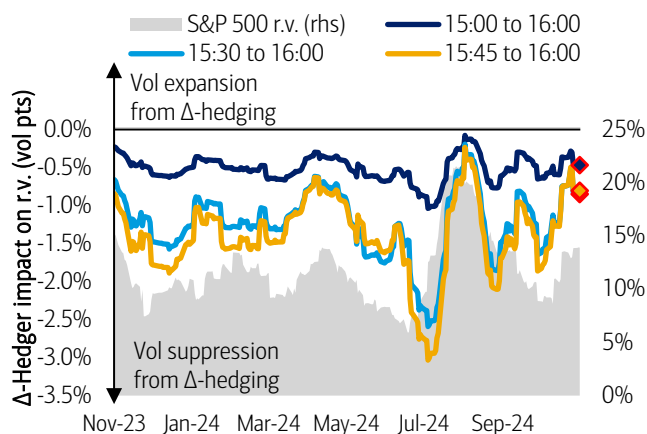
The growth of option selling funds has continued unabated in 2024, taking SPX long gamma to all-time highs (Exhibit 42). However, in terms of its influence on the level of S&P realized vol, we view gamma as the icing on the cake rather than the cake itself. Indeed, we estimate that over the past year, option delta-hedging suppressed S&P 1m realized vol by only 1.4 vol points (or 11% of the vol level; Exhibit 43).²

The scenario analysis in Exhibit 44 suggests that for S&P realized vol to fall from ~13% in 2024 to 2017 levels <7% (a 6pt decline or 46% of the level) – due to the impact of vol supply alone – SPX gamma would need to reach and sustain ~\$50bn. This is over 8x the average level of gamma seen in 2024 (\$5.8bn, itself already a record) and 2x larger than the max level ever recorded on a single day (Exhibit 42).

² Assuming that the delta-hedge is executed in the last 15 minutes of the trading day.

Exhibit 42: The growth of option selling funds continued unabated in 2024, driving SPX long gamma to all-time highs (averaging \$5.8bn YTD in '24)EOD SPX gamma for Δ -Hedgers (= Market-Maker + Firm + Flex)

Source: BofA Global Research, CBOE. Data as of 20-Nov-24. The gamma is derived from exchange data which classifies trades by market participant type (i.e., market-maker, firm, broker-dealer, customer, pro-customer). As shown above, "Market-Maker" and "Firm" represent the gamma from regular options for market-makers and firms respectively, while "Flex" is the gamma from flex options for market-makers only. BofA GLOBAL RESEARCH

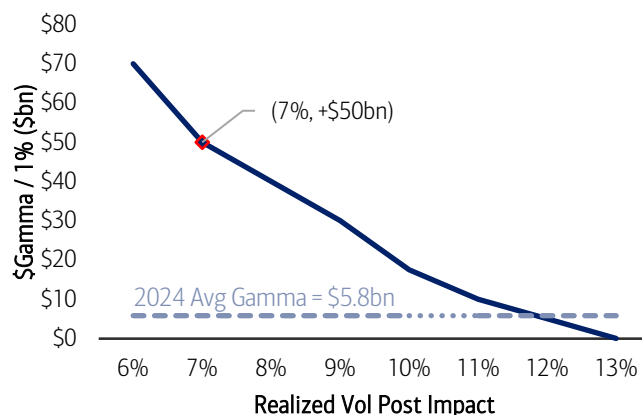
Exhibit 43: Despite SPX gamma reaching highs over the past year, our estimates suggest it suppressed realized vol by only 1.4 vol points (or 11% of the vol level)Estimated impact of SPX gamma on S&P e-mini 1-month realized volatility, with Δ -hedging confined to different time windows near the close

Source: BofA Global Research, CBOE. Data as of 20-Nov-24. Impact figures are quoted as market observed realized vol - estimated vol without gamma.

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Exhibit 44: For S&P realized vol to fall from ~13% in 2024 to 2017 levels <7% due to the impact of vol supply alone, SPX gamma would need to reach and sustain an extraordinary ~\$50bn next year

Simulated post-gamma-impact realized vol (x-axis) for different gamma levels (y-axis) assuming a pre-gamma-impact 10d realized vol of 13%



Source: BofA Global Research. Data as of 22-Nov-24. Delta-hedging is confined to the last 15-minutes of the trading session.

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What if we're still wrong?

Put differently, given the above, what would it take to get a 2017-style vol crush in 2025 that pushes the VIX below 10 on a sustained basis?

- Stretched valuations prove to be more bark than bite, perhaps as the AI rally stalls and investors quietly rotate into "value" and other laggards (keeping correlation and vol subdued).
- Single stocks reverse multiple structural shifts over the last several years, becoming less fragile, less hyperreactive to earnings, and slower to re-price post-events.
- Vol supply goes exponential as yield-hungry investors gorge on option selling funds. The perception that it's futile to fight the "guardrails" put up by dominant delta-hedging flows becomes a reality.



VIX imbalances growing: what it means & how to trade it

The August 5th vol shock was a stark reminder that markets are fragile and liquidity may rapidly vanish when it is needed most (see our [VIX 65: It's a liquidity problem](#) report). Further, the event saw the VIX decouple from equity moves to an extreme degree not seen since Feb18 highlighting: (i) the potential for volatility markets to be a source of true idiosyncratic risk distinct from the underlying and (ii) the clear need for better positioning intelligence to monitor emerging flow imbalances within the VIX market.

Here we arm ourselves with the tools needed to *monitor* and *monetize* VIX positioning.

See our [12-Nov-24 GEVI](#) for more details on our approach to quantifying VIX option imbalances & our [weekly/daily](#) Systematic Flows reports for related work on SPX options.

VIX option imbalances set new records in '24: Across many positioning measures, the VIX market developed material imbalances over the course of 2023 & 2024: (1) VIX delta imbalances reached levels hardly seen since the 2017/18 era. (2) The 3m average VIX gamma set new records in '24. (3) At the extremes of '24, the dealer-rebalance for VIX options in a +10pt vol shock equated to ~100-200k VIX futures to buy or ~50-100% of typical VIX futures volumes.³ See Exhibit 46 through Exhibit 49.

Each of these positioning measures, while related to one another, provides unique insight into the state of risk within the VIX market. E.g., one might conclude that VIX futures may be more reactive if dealers are short VIX gamma in size. However, extreme gamma is not always required to see fireworks in the VIX market. For example, 05-Aug was more likely the result of extreme *illiquidity* rather than VIX gamma. Going into August 5th VIX positioning was clearly stretched as evidenced by the elevated VIX delta imbalance, however, you might have missed this solely focusing on VIX gamma which was unremarkable at the time.

Aug24 was not a clearing event for the VIX complex: A distinctive feature of large vol shocks historically, such as Feb18 & Mar20, is for VIX option imbalances (e.g. delta, gamma, etc.) to quickly collapse towards 0, consistent with end-users monetizing hedges and/or shorting vol at elevated levels. This “reset” in positioning may be one reason that consecutive vol shocks frequently have multi-year gaps between events (i.e. it takes time for leverage to rebuild within the system). However, Aug24 appears not to have been a clearing event for the VIX options complex. Imbalances declined in August but did not fully reset and are still material. As such, in our view, it should *not* be taken as a given that the next stress event in the VIX complex is necessarily several years away.

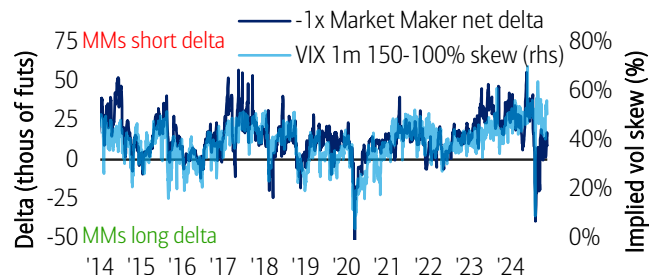
Positioning is closely linked with VIX option prices: Do markets actually care about positioning? Intuitively the answer should be yes, as prices are a function of supply & demand, but evidence to this effect is often elusive.

However, in the case of the VIX options market we have found a strong empirical relationship with over a decade of history between the net VIX option delta of market-makers and VIX call skew (Exhibit 45). Namely, as market-makers become shorter VIX option delta, VIX skew tends to be steeper.

While it is challenging to prove causality, this relationship would be consistent with market-makers (who are often considered the key price setters within markets) steepening VIX call skew as they build up an inventory of net short VIX calls.

Exhibit 45: Empirically, VIX call skew tends to be steeper when market-makers are shorter VIX option delta

Market-Maker VIX option net delta (scaled by -1x) vs VIX 1m 150-100% skew



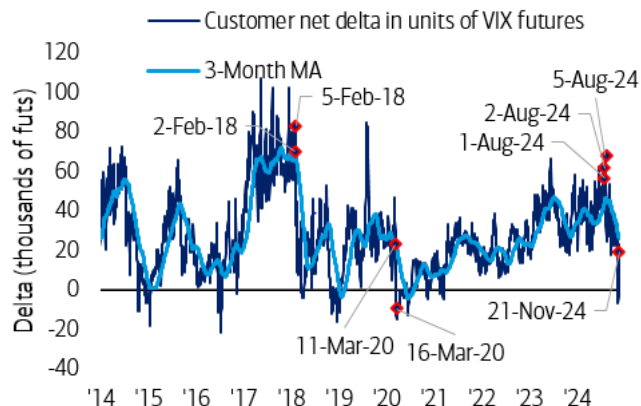
Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. MM = Market-Maker.

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³ As a point of clarification, the dealer delta-rebalance estimates strictly capture flows originating from VIX options, i.e., the futures rebalance associated with VIX ETNs/ETFs is not included. Consequently, the total rebalance footprint across all VIX products (not just options) may have been larger historically than the VIX option delta rebalance figures we see today. Aside from VIX futures one can also source VIX delta exposure by trading a combo. This additional channel of delta-one liquidity is not included in the normalization factor of the delta-rebalance / futures volume estimates.

Exhibit 46: Over the course of 2023/24, the VIX options market developed significant imbalances in the direction of end-users becoming increasingly long VIX delta & gamma (reverse for dealers)

Customer VIX option beta adjusted net *delta*

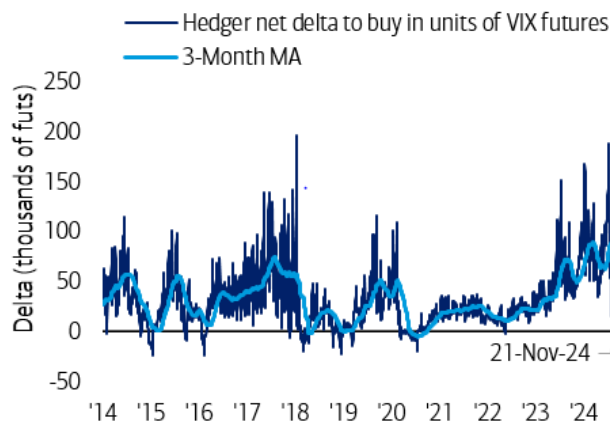


Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. The Black-76 delta is first computed for each option expiration separately. The deltas are then scaled using the beta of each expiry with respect to the 1m constant maturity future and aggregated together.

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Exhibit 48: The hedging needs of dealers in a +10pt vol shock show a demand for ~100-200k VIX futures at the extremes of '24 ...

Hedger theoretical VIX option net delta to buy after a +10pt shift in the 1m constant maturity future

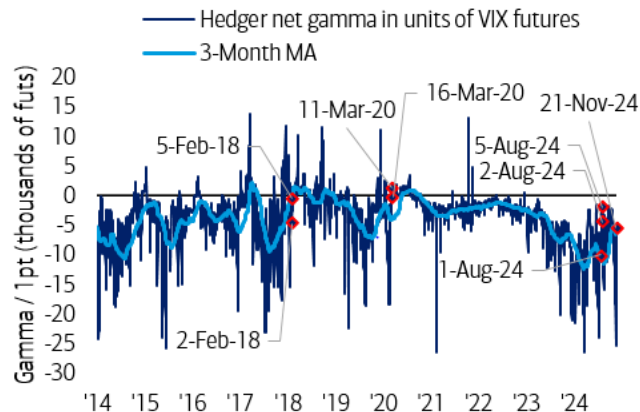


Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. Each listed future is shifted according to its beta to the 1m constant maturity future. Next the change in the Black-76 delta is computed using the new future level (holding implied vols & option positions constant). Then the delta-changes for each expiry are scaled by their respective betas and aggregated together.

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Exhibit 47: Feb18 & Mar20 were clearing events for the VIX complex (e.g., delta/gamma/dealer-rebalance all fell to 0), however, Aug24 was not; VIX imbalances declined in Aug but are still material

Hedger (= reverse of customer) VIX option net *gamma*

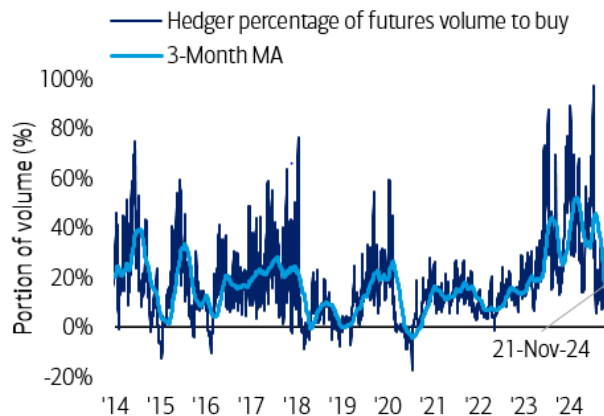


Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. The Black-76 gamma is first computed for each option expiration separately. The gammas are then scaled using the beta of each expiry with respect to the 1m constant maturity future and aggregated together.

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Exhibit 49: ... which translates to ~50-100% of typical VIX futures volumes to buy, easily setting new all-time-highs

Hedger theoretical percentage of VIX futures volume to buy after a +10pt shift in the 1m constant maturity future (= delta-to-buy / avg. fut volume)



Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. The percentage of volume to buy is computed as the ratio between (A) the estimated VIX option delta to buy after an x-point shift in the 1m constant maturity VIX future and (B) the 1m average beta adjusted VIX futures volume.

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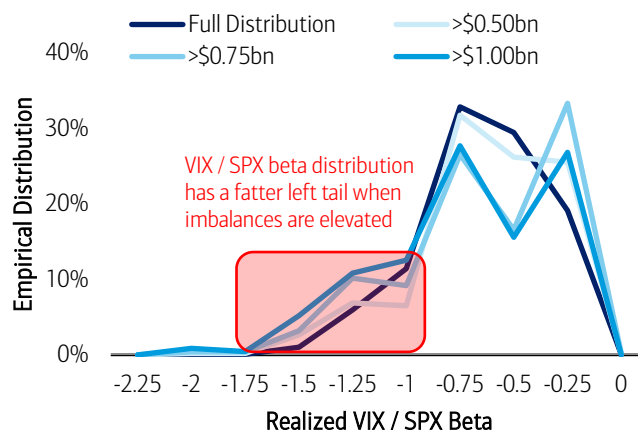


Monetizing VIX option positioning: Here we discuss two schemes to help monetize the VIX positioning intelligence discussed in this piece:

1. **Realized VIX / SPX Beta:** We compare VIX option PnL for customers in a +10pt vol shock (a measure of how long customers are in VIX options) with the subsequent realized VIX / SPX beta. We observe that extreme negative betas are more frequent when positioning is elevated (Exhibit 50). Hence, VIX positioning shows promise to be used alongside trades which seek to monetize the VIX / SPX beta.
2. **VIX Calls vs SPX Puts:** To complement systematic long VIX call vs short SPX put tail hedges, we use the following two quantities as a signal: (A) the VIX vol shock PnL measure mentioned above and (B) the ratio of the cost of VIX calls to SPX puts. When VIX positioning risk is low relative to the VIX / SPX hedge cost ratio, we propose underweighting VIX calls and overweighting SPX puts (one might even consider tactically going short VIX calls & long SPX puts), and vice-versa if the signal were flipped. As an example, such a signal would have led investors to be well-positioned for both the grind lower of 2022 and the vol shock in Aug24; see Exhibit 51. To systematize this process, we z-score both signal inputs and take the difference. The subsequent returns of a VIX call vs SPX put tail hedge strategy are skewed positive when the signal is extreme (i.e., VIX imbalances high vs VIX / SPX hedge cost) and the reverse when the signal is low (Exhibit 52).

Exhibit 50: Extreme negative VIX / SPX betas are more frequent when positioning risk in VIX options is elevated

Empirical distribution of 3m realized VIX / SPX beta, conditional on the level 3m prior of the customer VIX option change in value after a +10pt vol shock

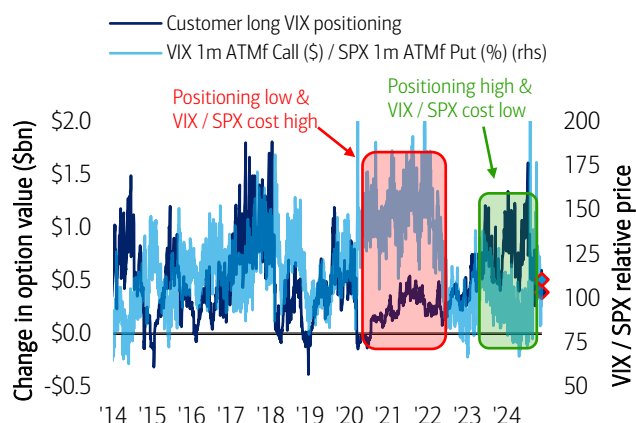


Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. The realized beta is computed using daily point changes and percent returns for VIX and SPX respectively.

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Exhibit 51: Using VIX imbalances & the relative cost of VIX / SPX hedges as a signal would have positioned investors well into the grind lower of '22 and the vol shock in Aug24 ...

Cust VIX option change in value after a +10pt shift in the 1m VIX future compared with the ratio of VIX 1m ATMf call to SPX 1m ATMf put prices

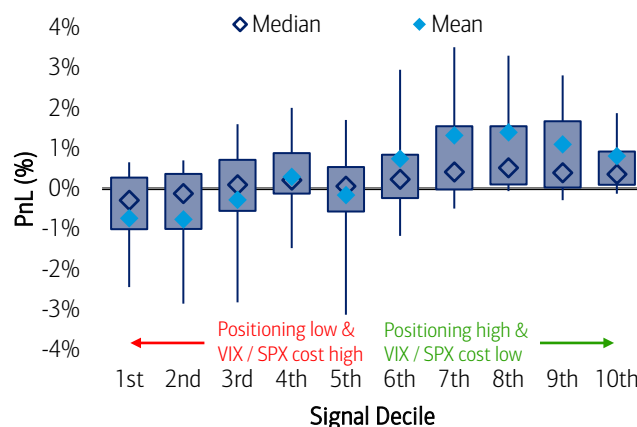


Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. The Customer long VIX positioning computes the change in value of the "customer book of options" for a +10pt shift in the 1m VIX future (holding implied vols & option positions constant).

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Exhibit 52: ... applying this signal to a systematic long VIX call / short SPX put tail hedge, we find that performance would have skewed positive when the signal is high and the reverse when the signal is low

Hypothetical backtested distribution of 3m returns of systematic long 3m VIX call & short 3m SPX put tail hedge vs VIX imbalance signal 3m prior*



Source: BofA Global Research, CBOE. Data from Jan-14 to 21-Nov-24. *The VIX imbalance signal is constructed as the difference in z-scores of (A) and (B). (A) = the customer change in option value after a +10pt shift in the 1m VIX future. (B) = VIX 3m ATMf call / SPX 3m ATMf put ratio. Box edges are the 25th & 75th %iles, while the whiskers are 10th & 90th %iles. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Europe, still cheap & unloved; Don't get caught short

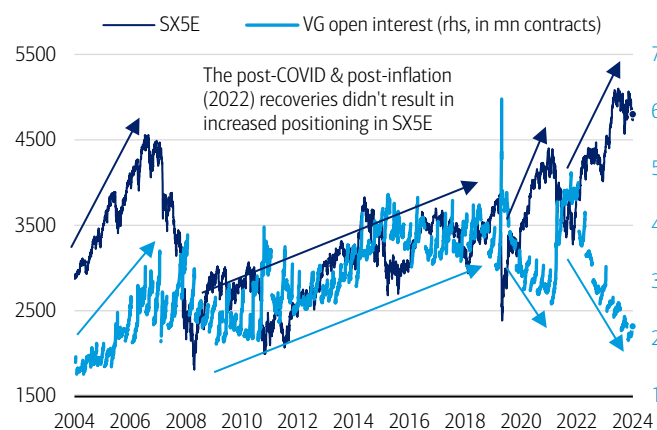
Upside & downside risks are also higher for EU equities, but a lack of positioning intensity in Europe makes it less likely that we witness bigger downside shocks than the US, in our view (aside EU-specific catalysts). This has a few implications:

- 1) Establish EU upside hedges in case of a domestic upside surprise.
- 2) If the consensus is that the EU is a short versus the US, consider shorting vol, not delta. Harvesting EU vol premium likely entails less mark-to-market pain as there aren't huge crowds needing to exit EU stocks and many are likely to remain or will go increasingly underweight.
- 3) While it still has a long way to go, the SXXP is an increasingly liquid, well diversified EU equity derivatives instrument and an attractive vol funder.

European equities are at an interesting juncture. On one hand, domestic growth headwinds, political uncertainty and trade tariff risks loom over an equity market that's not far from all-time highs (Exhibit 53). On the other hand, there is a risk that animal spirits awakening with a generational shift in US public policy also lifts globally cheap European equities (see Exhibit 54), arguably less than their US counterparts.

Exhibit 53: The EU equity market is not far off highs, but derivatives positioning is at its lowest since the GFC

SX5E and SX5E futures (VG) open interest in the front two contracts (excluding roll week)



Source: BofA Global Research. Data: 11-Nov-04 to 22-Nov-24.

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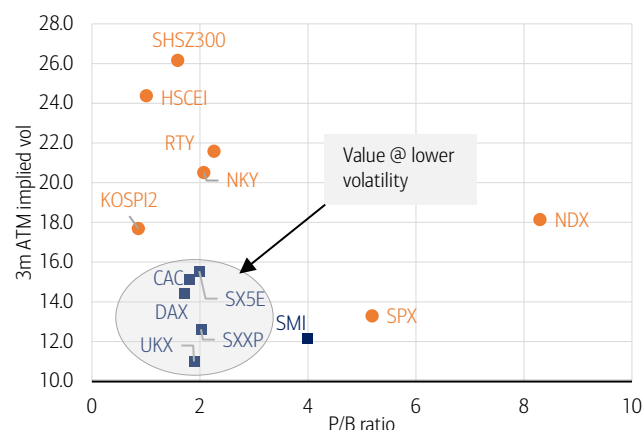
We believe investors are likely positioned for the former, less favourable outcome. This is guided by a combination of weaker macroeconomic fundamentals in Europe, a trade war overhang that may weigh on the region whether or not tariffs are implemented, and a relative preference for US equities. Indeed Exhibit 53 shows that the open interest in SX5E (VG) futures, Europe's most liquid equity derivatives instrument, is near the lowest it has been since the GFC.

Instead, positioning has grown in US equity longs relative to Europe, as net investor positioning in SPX (ES) versus SX5E (VG) futures shows in Exhibit 55. Relatively stretched positioning is echoed in the recent record divergence of Total Return Future (TRF) spreads between the two regions too, suggesting greater demand for US equity (synthetic) longs than in Europe (Exhibit 56).

Given this setup, we encourage global investors consider upside hedges in Europe in case of a surprise upside catalyst, such as a quick end to the Russia-Ukraine war, or an unlikely, but possible fiscal boost in Germany (see for example: [Upside hedges for a positive surprise in unloved Europe](#)).

Exhibit 54: European equities largely remain a play on cheap stocks globally, with relatively lower option costs

3m ATM IV (y-axis) and P/B ratio of global equity indices.

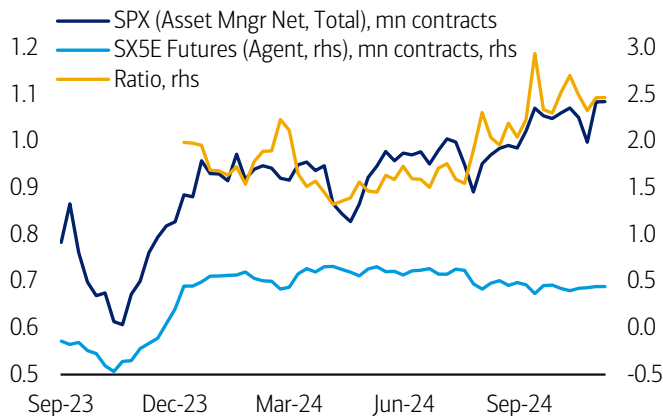


Source: BofA Global Research. Data as of 22-Nov-24 (Note: We proxy the implied vol of SHSZ300 using that of ASHR US).

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Exhibit 55: Interest has grown instead in US equities, which can be seen from diverging net investor positioning in SPX vs SX5E futures
CFTC Asset Manager Net positioning in SPX Futures, & Eurex Agents net SX5E futures exposure (in contracts to compare positioning ex performance)

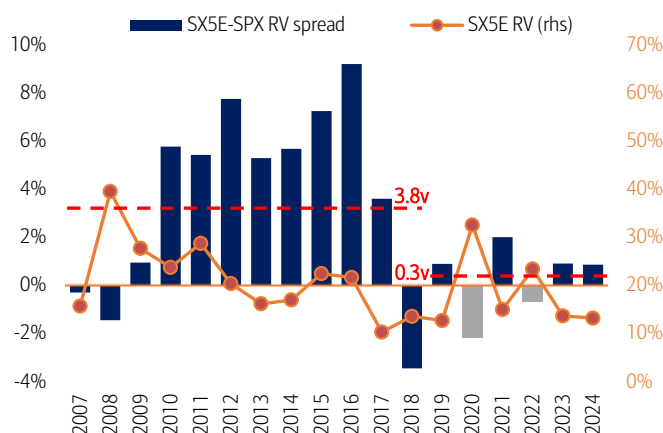


Source: BofA Global Research, Bloomberg, Deutsche Börse. Data from 01-Sep-23 to 19-Nov-24.
Eurex defines participant type Agent as those who're trading on behalf of customers.

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But where does this leave us vis-a-vis EU equity vol? Here, historical perspective relative to US equity vol is instructive. In Exhibit 57, we see the distribution of the 1y vol spread between SX5E and the SPX by calendar year. After being elevated through the majority of the last decade while Europe navigated its sovereign debt crisis before entering the QE era, this spread has been muted in recent years. It averaged just +0.3v since 2019 vs +3.8v in the 12 years prior. In fact, it is the SPX that has been more volatile in periods of elevated SX5E volatility in the past 5 years (viz, 2020 and 2022). Importantly, relatively muted SX5E volatility can also be evidenced from the behaviour of implied volatility in that time, as Exhibit 58 shows. The largest shocks to short-dated variance in the SX5E since 2019 have been overshadowed by those in the SPX, by over 1.5v on average.

Exhibit 57: The SX5E vol spread to the SPX has been muted in recent years; indeed, the SPX was more volatile in high vol years ('20 & '22)
SX5E-SPX 1y vol spread by year and 1y SX5E (RV) – grey bars show that in high SX5E vol years recently (RV > 20%), the SPX was more volatile

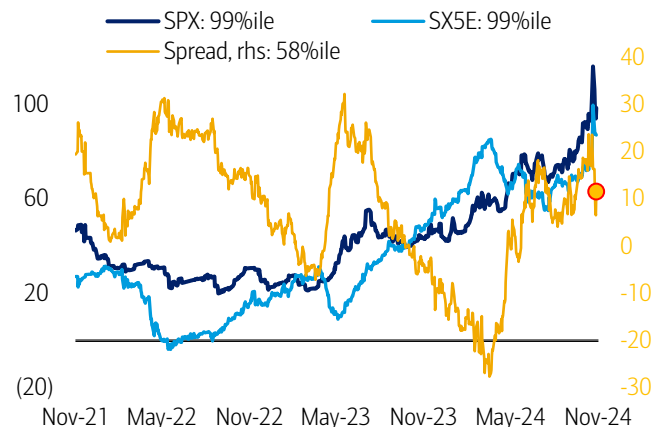


Source: BofA Global Research, Bloomberg. Data: 2-Jan-07 to 22-Nov-24.

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Exhibit 56: This divergence is also evident from a widening differential between Total Return Future (TRF) spreads

Rolling 1Y Total Return Future (TRF) spread for SPX and SX5E & their spread. A widening difference suggests more demand for synthetic SPX longs.

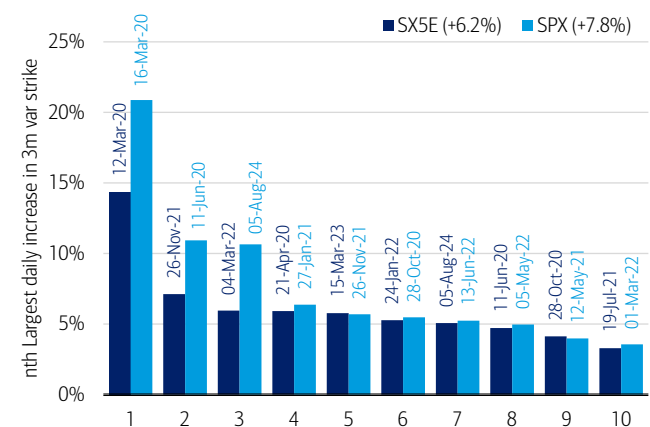


Source: BofA Global Research, Bloomberg. Data from 01-Sep-23 to 22-Nov-24.

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Exhibit 58: Moreover, the largest upside shocks to SX5E implied volatility have been outpaced by those in SPX (by >1.5v on average)

Largest 1d gains in 3m var for SX5E and SPX since Nov-19 (average in parenthesis).



Source: BofA Global Research, Bloomberg. Data: 14-Nov-19 to 22-Nov-24.

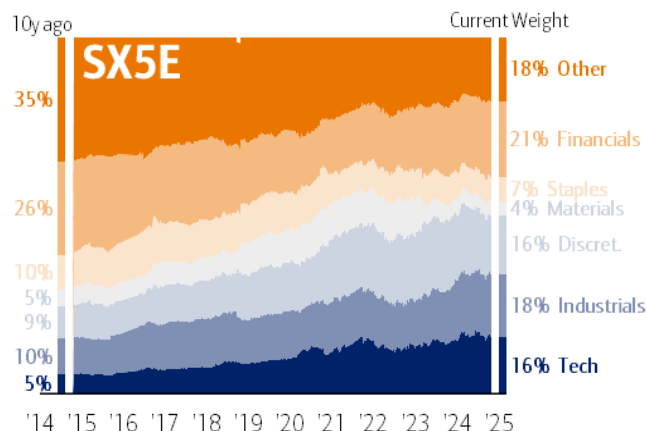
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This is consistent with investor positioning being less stretched in the EU, as otherwise, we would have experienced more volatility from 'crowds' exiting positions in crises. Of course, EU-centric risks (like the Russian invasion of Ukraine in '22 or snap French elections in '24) have still impacted SX5E vol more than the SPX. But given that the US

remains the engine of DM growth and likely more stretched in positioning terms, we see greater upside risk to SPX vol in case of a global shock, and certainly in case of a US-led crisis. We would encourage investors to consider thinking of broad European indices for harvesting vol premium or as a funding (short) leg in long-short volatility trades.

Exhibit 59: SX5E index members' sector composition has changed significantly over the years, resulting in a pro-cyclical tilt...

SX5E Index weights by GICS sectors

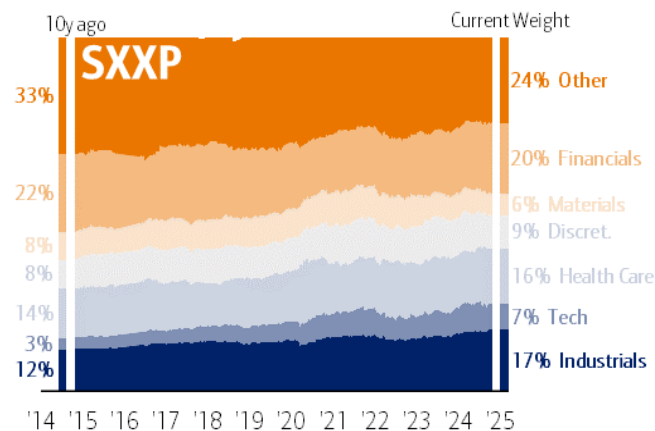


Source: BofA Global Research, Bloomberg. Data from 25-Oct-14 till 24-Oct-24. Weights use past constituents, categorized by GICS sectors. Sectors are ordered by their 10y (signed) difference in index weights.

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Exhibit 60: ... whilst SXXP sector weights have been more stable, with diversified exposure from sectors including (lower vol) Healthcare

SXXP Index weights by GICS sectors



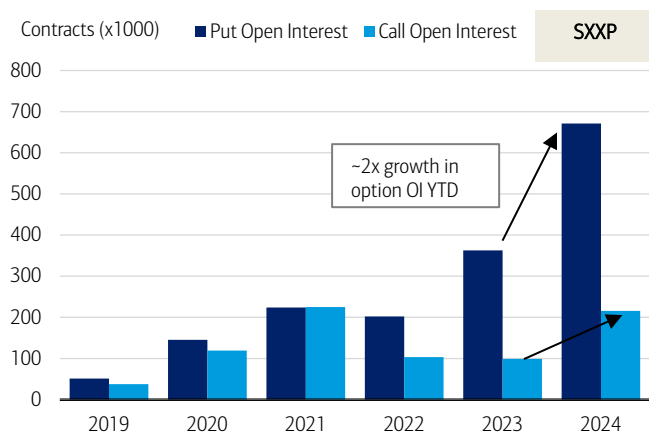
Source: BofA Global Research, Bloomberg. Data from 25-Oct-14 till 24-Oct-24. Weights use past constituents, categorized by GICS sectors. Sectors are ordered by their 10y (signed) difference in index weights.

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Aside the SX5E, which is the most popular equity index for derivatives in Europe, we also highlight the STOXX600 (SXXP), which has had a more stable and diversified exposure to sectors than the SX5E (see Exhibit 59 and Exhibit 60). While the SXXP has lower options liquidity than the SX5E, option open interest has nearly doubled in the past year (Exhibit 61). We regard the SXXP as a suitable short leg, liquidity permitting, for harvesting vol premium (Exhibit 62), macro-directional opportunities or volatility relative value, including dispersion.

Exhibit 61: SXXP option open interest, while still a fraction of that in the SX5E, has grown by over 2x YTD

Put and call OI averaged by year in SXXP index options, in 1000s of contracts

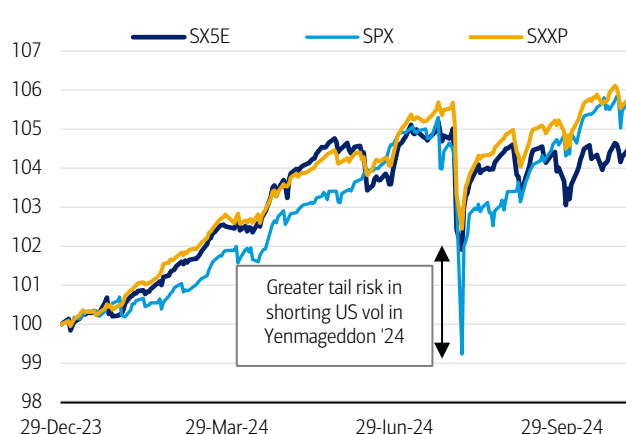


Source: BofA Global Research, Bloomberg. Data: Jan-19 to Nov-24 (2024 avg through 21-Nov-24)

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Exhibit 62: While we don't recommend shorting var outright, it's notable that short EU var would have experienced less tail risk in 2024

Hypothetical backtest: -1m var on SX5E, SPX & SXXP (\$0.25vega per \$100)



Source: BofA Global Research. Data: 29-Dec-23 to 7-Nov-24. This performance is backtested & doesn't represent actual performance of any account or fund. Back-tested performance depicts theoretical (not actual) performance of a particular strategy over the period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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How to trade our global themes

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How to trade a boom without having to time the bust

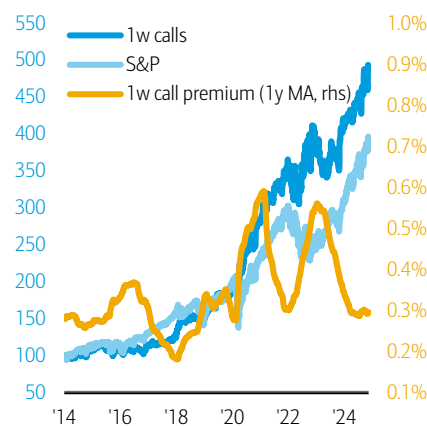
What: Introducing the “A-trade”, which systematically buys partially delta-hedged SPX 1w 30D/5D 1x2 call ratios (+1x 30D call, -2x 5D calls) to asymmetrically hedge the risk of a US equity boom without having to time the inevitable bust.

Why: (i) Short-dated S&P calls have directionally underpriced US equity upside; (ii) selling away the extreme right tail via call ratios tends to improve the trade's cost-benefit; and (iii) adding a partial delta-hedge enhances defensive characteristics in times of weakness while retaining upside participation. The result is a strategy that would have generated a Sharpe ratio near 1 and Calmar ratio near 0.8 over the past decade, remained robust across bear markets, and helped those underweight risk to keep up with the S&P if used via equity replacement.

The upside in US equities defied everyone's expectations in 2024. As noted in our Outlook, our view is that the pain trade remains higher, especially if the tech bubble serves as an appropriate analogy. However, the road higher can often be bumpy, and timing the top is challenging when progressing through a boom-bust cycle. As a result, we aim to find asymmetric options strategies that can capture upside while being robust to intermittent market weakness and thus reduce the need to time the top.

Exhibit 63: Despite no structural decline in premium, long SPX 1w 30D calls would've outperformed the S&P by 2.4% p.a. since '14

Hypothetical P&L of SPX 1w 30D long calls and S&P (total returns) vs 1w 30D call premiums

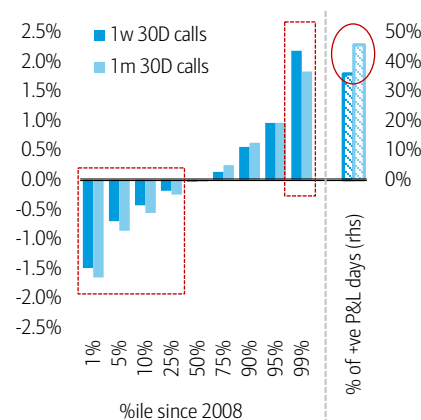


Source: BofA Global Research. Data from 1-Jan-14 to 15-Nov-24. 1w call P&L scaled to S&P vol. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 64: While 1m calls would've had more positive P&L days, the 1w call's longer right tail would've driven its outperformance

Distribution & hit rate of hypothetical 1-day P&Ls for S&P 1w & 1m long 30D calls

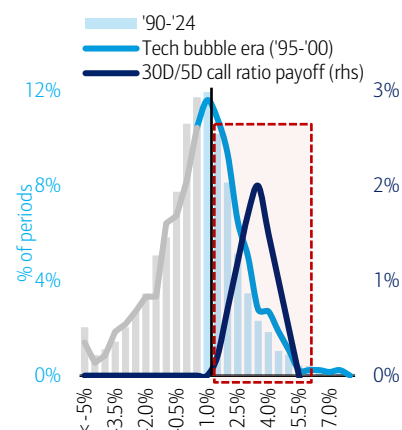


Source: BofA Global Research. Data from 4-Jan-08 to 15-Nov-24. Scaled to ann. vol of 10%. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 65: Over half of 1w S&P rallies fall within the typical positive payoff range of 1w 30D/5D call ratios

Distribution of 5d SPX returns vs SPX 30D/5D 1x2 call ratio payoff (color bars = positive return)



Source: BofA Global Research. Data from 1-Jan-90 to 15-Nov-24. Strikes of call ratio are based on median strikes from 4-Jan-08 to 15-Nov-24.

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Why 1-week calls? Over the past decade, long SPX 1w 30D (30-delta) call options would have outperformed the S&P by more than 2.4% p.a., even though there has been no structural cheapening of call premia over this period (Exhibit 63). Moreover, despite conventional wisdom that longer-dated options are “cheaper” than shorter-dated ones, 1w 30D calls would have outperformed their 1m counterparts by ~60% since 2008, even though their relative premia have not changed materially over time. 1m calls even had



more frequent positive P&L days, but the right skew in the P&L distribution of 1w calls was sufficiently favorable to drive the outperformance (Exhibit 64).

Why call ratios? Call ratios cost less than outright calls but also cap upside. The historical distribution of S&P 1w returns reveals that over 55% of positive returns fall in the +0.75% to +3% range – roughly the typical positive payoff region for a 30D/5D 1x2 call ratio structure (Exhibit 65). Moreover, the lower cost would have allowed call ratios to outperform outright calls close to 95% of the time since 2008, lopsided enough to dominate the rare occurrences when the upside cap is undesirable. By being net short options, call ratios could also be well-suited to a rising vol regime as typically seen in a bubble, with spread-based structures being more cost-efficient than outright ones. Furthermore, over the last decade, the 5-delta call vol risk premium (VRP) has remained supported while 30-delta call vol now trades close to fair (Exhibit 66).

Why (partially) delta-hedge? Applying a partial (daily) delta-hedging scheme to the trade aims to improve performance in drawdowns by creating a more straddle-like payout profile while still retaining an upside tilt (Exhibit 67). As shown in Exhibit 68, the result is an upside trade that can participate in steady rallies (e.g., 2021, 2024) while also being defensive in bear markets (e.g., Covid shock, 2022).

Exhibit 66: The vol risk premium (VRP) for SPX 1w 5D calls remains supported while the VRP for 1w 30D calls has collapsed to zero

1y rolling median of S&P 1w 30D & 5D vol risk premium (VRP*)

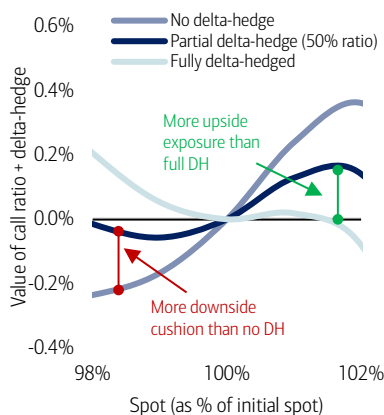


Source: BofA Global Research. Data from 4-Jan-08 to 8-Nov-24.
*Vol risk premium is calculated as implied vol minus subsequent gamma weighted realized vol.

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Exhibit 67: Partially delta-hedging call ratios injects some defensiveness while retaining upside participation

Hypothetical profile across spot scenarios for 1w 30D/5D call ratio with various delta-hedge ratios

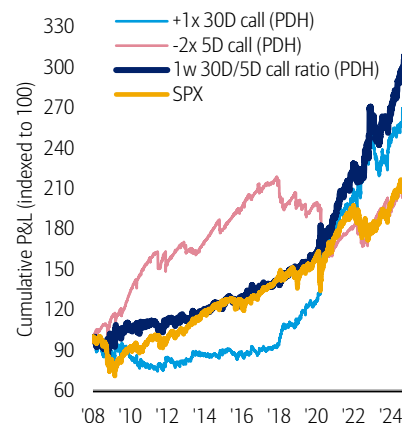


Source: BofA Global Research. Hypothetical value assuming no change to implied vol and other parameters.

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Exhibit 68: Partially delta-hedged call ratios offer call-like upside with reduced cost and a defensive profile in selloffs

Hypothetical backtest of S&P 1w 30D/5D 1x2 call ratios with partial delta hedge (PDH)



Source: BofA Global Research. Data from 4-Jan-08 to 15-Nov-24. Scaled to ann. vol of 10% for call ratio. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Keeping up with the S&P, with less risk: Systematically buying partially delta-hedged SPX 1w 30D/5D 1x2 call ratios would have had a greater annual return than the S&P 500 (total return) over the past decade with 13 percentage points less of a max drawdown, yielding 1.4x the Sharpe and 2.2x the Calmar ratio of delta-one exposure to US equities (Exhibit 69 & Exhibit 70). Strikingly, the defensive properties of the strategy would have allowed it to be relatively robust in periods of S&P weakness like the convex Covid shock (call ratio's max drawdown in Feb-20 to Mar-20 period was -14% vs S&P's -34%) or the slower 2022 drawdown (where it ended the year up vs S&P's -18.6%; Exhibit 70), highlighting its value as an upside hedge that reduces the risk of timing the top.

These results point to equity replacement as a natural use case in portfolios that may be underweight risk (particularly tech) and at risk of S&P underperformance. To demonstrate this, we illustrate the hypothetical performance of a portfolio with 70% notional of equal-weight S&P (SPW) and 30% of call ratios. As seen in Exhibit 69 & Exhibit 70, such a portfolio would have outperformed the S&P over the past decade on a risk-adjusted basis and achieved higher Sharpe (0.77 vs 0.67) and Calmar (0.43 vs 0.37) ratio.

Risks to the strategy involve periods of mild drawdowns or sideways price action, or rallies that are not strong enough to counteract vol moving lower (as in 1H23).

Exhibit 69: Our partially delta-hedged SPX call ratio strategy would have generated a Sharpe ratio near 1 and Calmar near 0.8 over the past decade...

Hypothetical backtested performance of (i) partially delta-hedged call ratios, (ii) equity replacement portfolio (70% SPW / 30% call ratios)

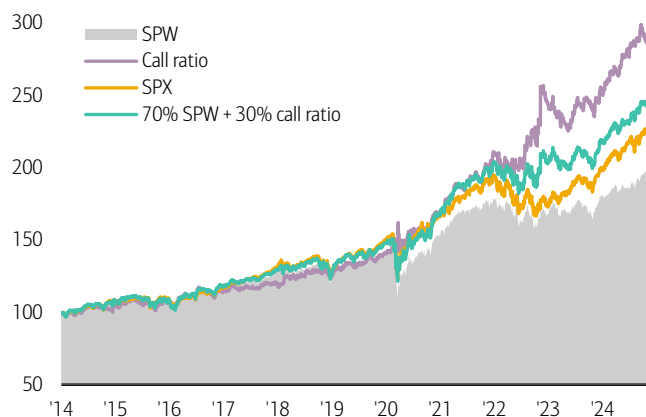
Since 2014	Ann. return	Ann. Vol	Max Drawdown	Sharpe ratio	Calmar ratio
Call ratio	17.0%	17.4%	-20.6%	0.98	0.82
SPW	10.6%	18.0%	-39.0%	0.53	0.27
SPX	12.6%	17.4%	-33.8%	0.67	0.37
70% SPW + 30% call ratio	12.8%	15.8%	-29.6%	0.77	0.43

Source: BofA Global Research. Data from 1-Jan-14 to 15-Nov-24. SPW = equal-weight S&P, SPX = S&P 500 total returns. *Call ratio illustratively has 7x leverage. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 70: ...while keeping up with the S&P when used in both outright and equity replacement of equal weight S&P (SPW) contexts, on a risk-adjusted basis

Hypothetical backtested performance of (i) partially delta-hedged call ratios, (ii) equity replacement portfolio (70% SPW / 30% call ratios)



Source: BofA Global Research. Data from 1-Jan-14 to 15-Nov-24. SPW = equal-weight S&P, SPX = S&P 500 total returns. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Convexity sans carry: Fighting fragility by taming the skew

What: We introduce the *SkeU-trade* - a positively convex yet carry-neutral SPX option strategy that stands to gain in both upside and downside fragility shocks. We pair this with a 'grind lower' hedge for a defensive overlay to risk assets.

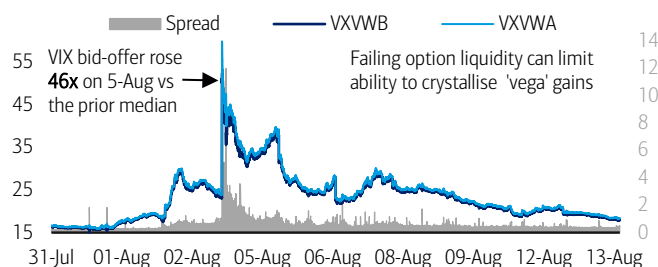
Why: Anticipating implied vol changes by making a "skew-adjustment" when delta-hedging a long put option can both lock-in gains in case of a sharp selloff and be better positioned for subsequent dip-buying. We use this technique to boost the carry of a dynamically-sized, delta-hedged SPX put ratio, enhancing its reactivity to fragile up-moves without capitulating on gains in left-tail scenarios. We also show how pairing this strategy with a grind-lower hedge would have provided an effective defensive overlay for the SPX, boosting risk-adjusted returns by a quarter whilst significantly reducing tail risk.

Risk: Equities grinding, not crashing lower is a risk for the SkeU-trade, which is why we recommend pairing it with a 'grind lower' hedge.

The Aug'24 shock reminded investors not only that volatility convexity isn't dead, but that failing trading liquidity remains a feature of even usually liquid markets (like SPX, VIX derivatives, Exhibit 71). It further reinforced the adage that buying equity dips remains a strong source of alpha, highlighting the need for hedges to be resilient to sharp moves in both directions.

For defensive strategies, such shocks illustrate the benefits of delta-hedging long option positions, which 'lock in' gains when realized volatility increases without needing to trade in and out of potentially less liquid options (Exhibit 72). However, this isn't a panacea either as delta-hedging long options alone using a traditional (Black-Scholes) approach can be punitively expensive in the long run, on account of the vol risk premium (Exhibit 73). Is there a way to tilt the delta profile of a long delta-hedged option to reduce the cost of carry and yet retain the convexity?

Exhibit 71: The Aug24 shock was a stark reminder that failing trading liquidity in critical moments remains a feature of usually liquid markets
VXVWA/B: vol bid and offer using S&P500 options used in VIX calculation

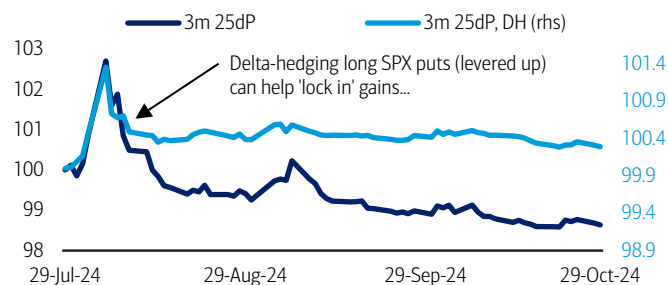


Source: BofA Global Research. The vol bid-offer spread (in grey) of the VIX using S&P500 option quotes used in the VIX calculation at 30s intervals from 31-Jul-24 to 13-Aug-24.

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Exhibit 72: A well known way to lock in hedge gains without needing to trade out of/roll option longs is to delta-hedge them...

Long SPX 3m 25d puts, naked & delta-hedged daily (DH) at the close.

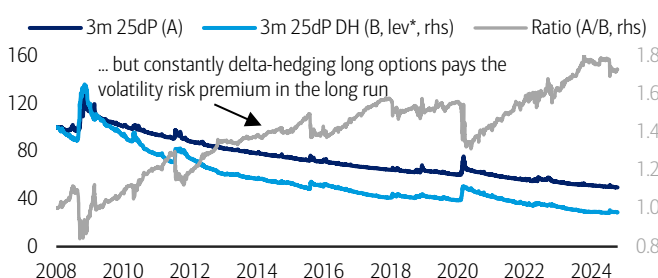


Source: BofA Global Research. Data: 29-Jul-24 to 29-Oct-24. (Assuming equal notional on both strategies for simplicity here, hence the lower vol delta-hedged put strategy is shown on the RHS)

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Exhibit 73: ... but doing this in the long run is prohibitively expensive on account of a positive volatility risk premium; can we do better?

Long SPX 3m 25d puts, naked & delta-hedged daily at the close, & their ratio.



Source: BofA Global Research. Data: 18-Jan-08 to 29-Oct-24. *The long DH put strategy is levered up (by 5.9x) to match the volatility of the long unhedged put strategy.

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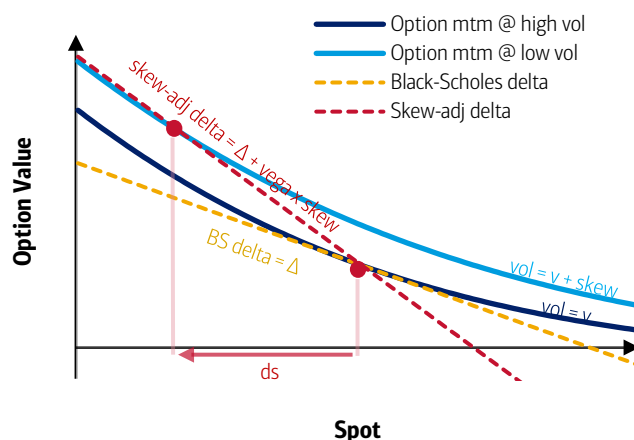
Anticipating vol → symmetric convexity & lower carry:

When delta-hedging an option, investors seek to neutralise the option's directional risk to the underlying. In a 'Black-Scholes' world where there is no volatility skew, delta-hedging assumes that any given move in the underlying is equally likely to be up or down. In reality however, risk assets like equities experience sharper down than up-moves, often coupled with rising implied volatility when equities fall, and vice versa. As a result, Black-Scholes option delta typically *underestimates* the true delta of an option. While this 'shadow' delta is hard to pin down a-priori, one approach is to anticipate it by observing implied volatility skew and calculating a skew-adjusted delta (see Exhibit 74).

The steeper the volatility skew and/or the closer the underlying is to the option strike, the greater the magnitude of the put option's skew-adjusted delta compared to its Black-Scholes equivalent. This translates into being long more futures as the put's delta-hedge. This will lead to *initial* underperformance versus using Black-Scholes delta in a drawdown (Exhibit 78) but can often lead to significant outperformance in subsequent recovery (Exhibit 79). Done systematically, however, we see that delta-hedging long 3m 5d puts in SPX, SX5E and NKY (Exhibit 75, Exhibit 76 & Exhibit 77, respectively) would have offered a superior long-run carry profile while retaining positive convexity in major crises.

Exhibit 74: Downward moves in spot are often accompanied by a rise in implied volatility. Adjusting the option delta to account for this allows for more symmetrical exposure to option convexity

Illustration of the value of a put option (with respect to spot) at a lower vol level and at a higher vol level, along with the Black-Scholes delta and the skew-adjusted delta.

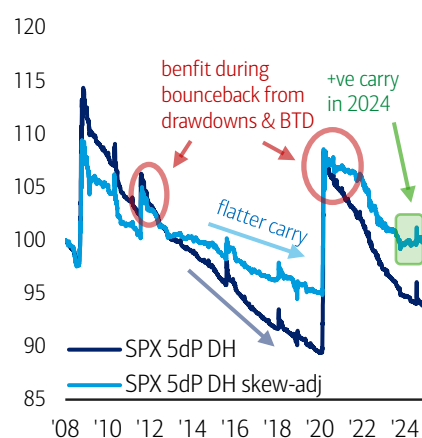


Source: BofA Global Research. Skew-adjusted delta = Black-Scholes delta + vega x skew. Both option price curves are individually shown with constant vol with respect to spot. The overall illustration depicts the option implied vol jumping to high vol (as a result of skew) for given a down-move ds in spot.

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Exhibit 75: Using a skew-adjusted delta for hedging long SPX puts would have offered flatter carry, outperformance in recoveries and reduced sensitivity to market direction

Hypothetical backtest of SPX 3m 5d puts (entered monthly, held to expiry) delta-hedged with Black-Scholes delta & a skew-adjusted delta.

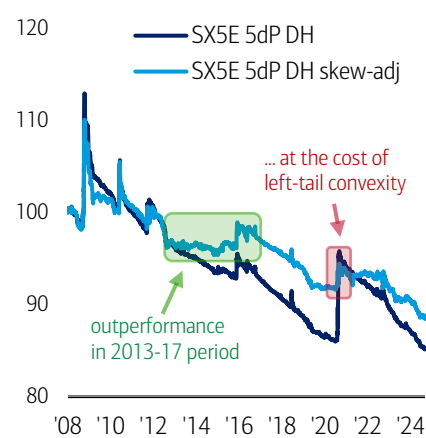


Source: BofA Global Research. Data from 18-Jan-08 to 11-Nov-24. Skew-adjusted delta = Black-Scholes delta + vega x skew, where skew is measured between 97.5% and 102.5% of the option's strike. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 76: In SX5E, long puts delta-hedged with a skew-adjustment would have provided positive carry from '13 to '17, but generated smaller gains during COVID

Hypothetical backtest of SX5E 3m 5d puts (entered monthly, held to expiry) delta-hedged with Black-Scholes delta & a skew-adjusted delta.

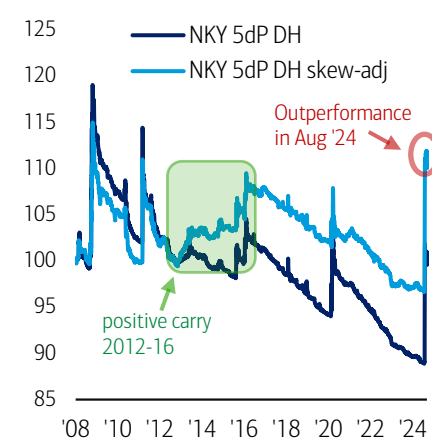


Source: BofA Global Research. Data from 18-Jan-08 to 11-Nov-24. Skew-adjusted delta = Black-Scholes delta + vega x skew, where skew is measured between 97.5% and 102.5% of the option's strike. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 77: For NKY puts, the skew-adjustment would have amplified gains made in the Aug-24 fragility shock, due to the violent bounceback from the high vega region

Hypothetical backtest of NKY 3m 5d puts (entered monthly, held to expiry) delta-hedged with Black-Scholes delta & a skew-adjusted delta.



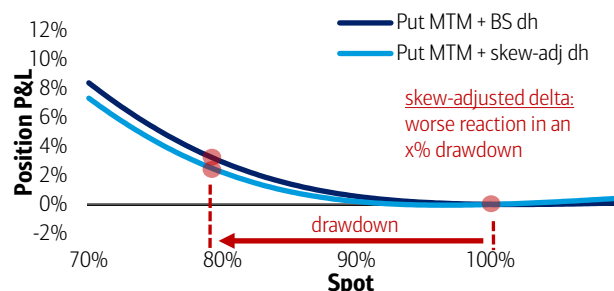
Source: BofA Global Research. Data from 18-Jan-08 to 11-Nov-24. Skew-adjusted delta = Black-Scholes delta + vega x skew, where skew is measured between 97.5% and 102.5% of the option's strike. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 78: Delta-hedging a long put with a skew-adjustment may initially underperform vs. using Black-Scholes delta in a drawdown...

The theoretical p&l of two SPX 3m 5-delta puts, one using a Black-Scholes delta and the other using a skew-adjusted delta, in an instantaneous drawdown in the underlying



Source: BofA Global Research. Data as of 25-Sep-24.

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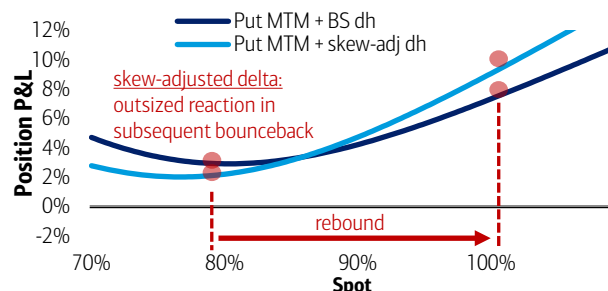
How is this different to just adding more delta? During strong market rallies or periods where 'buy-the-dip' behaviour is rife, it is possible to see long skew-adjusted delta-hedged (DH) puts exhibit flat or even positive carry (like in 2024 YTD; Exhibit 75). Indeed, one might question how using this adjustment is any different to simply adding a constant delta (overhedge) to the put in acknowledgement of equities' positive risk premium. Exhibit 80, which compares the performance of the skew-adjusted DH put to a Black-Scholes DH put with a constant 1.7 delta added (the constant is chosen such that the average delta of both structures are equal), illustrates the relative benefits of the skew-adjusted version over time. Indeed, we see the strongest outperformance after market routs when one might see volatile upmoves and/or dip buying. This outperformance comes at the cost of smaller gains in deep, sustained drawdowns, such as during Mar'20 and the GFC.

Application to a fragility hedge – the SkeU-trade: Adjusting for skew when delta-hedging an OTM put should usually lead to a more symmetric convex profile, rather than outsized 'left-tail' performance. Hence, investors may use our long skew-adj 5d put (from Exhibit 75) as a component of a systematic fragility hedge to better capture both down and up-fragility moves (the latter being particularly pertinent when buying dips remains a strong source of alpha). In Exhibit 81, we introduce the SkeU-trade: a systematic 3m -25d+5d delta-hedged put ratio strategy which applies the skew-adjustment to its long 5d leg and sizes the long put at inception to make the structure zero premium. This results in 4.9x 5d puts bought for each 25d put sold on average. The SkeU-trade's backtest profile is similar to that of our W-trade which buys fewer 5d puts (3x) per 25d put sold (see our 2020 Outlook: Deeper into the abyss). We note that the SkeU-trade would have provided superior carry to the W-trade historically and reacted strongly in fragile up-moves that often occur during or after a multi-day drawdown, thus achieving a symmetric U-shaped profile to market shocks (Exhibit 82).

As part of a diversified hedging portfolio: Despite being less of a left-tail hedge than the W-trade and despite being less *reliable* versus contractual hedges (Exhibit 82 & Exhibit 84), the SkeU-trade would have provided an effective defensive overlay for SPX when paired with a 'grind-lower' hedging strategy (shown in Exhibit 83). Indeed, their combination, when sized for cVaR minimization of the overlay, would have boosted SPX risk-adjusted returns by a quarter whilst significantly reducing tail risk (Exhibit 83).

Exhibit 79: ... but in a rebound, the skew-adjustment can help significantly outperform vs. using Black-Scholes delta

The theoretical p&l of two SPX 3m 5-delta puts, one using a Black-Scholes delta and the other using a skew-adjusted delta, in a subsequent rebound after delta-hedging

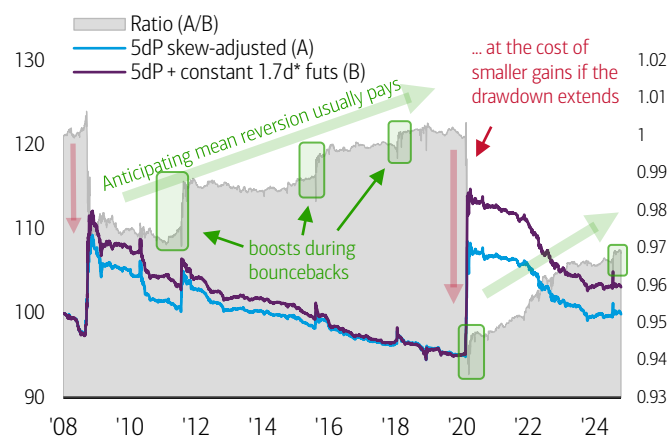


Source: BofA Global Research. Data as of 25-Sep-24.

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Exhibit 80: So how is this any different to just adding a fixed overhedge to your long put? You are taking a risk on mean reversion on equity dips, which is rewarded in the long run.

SPX 3m 5d delta-hedged puts using a skew-adjusted delta (A) vs using a Black-Scholes delta plus a constant 1.7 delta of futures (B) with the ratio (A/B) on the right-hand axis

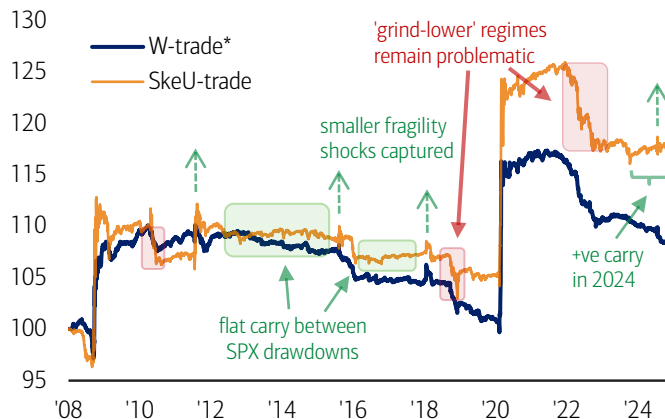


Source: BofA Global Research. Data: 18-Jan-08 to 11-Nov-24. *The difference in delta on average between the skew-adjusted delta used and the Black-Scholes delta used is 1.7, so to make both backtests carry the same delta on average we have added 1.7d of futures to the Black-Scholes delta-hedged version. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 81: Enter the SkeU-trade: Applying the skew-adjustment to the long leg of a put ratio strategy would have provided a similar profile to our fragility hedge, the W-trade, but with superior carry...

Backtested performance of the W-trade* and SkeU-trade** strategies using SPX options

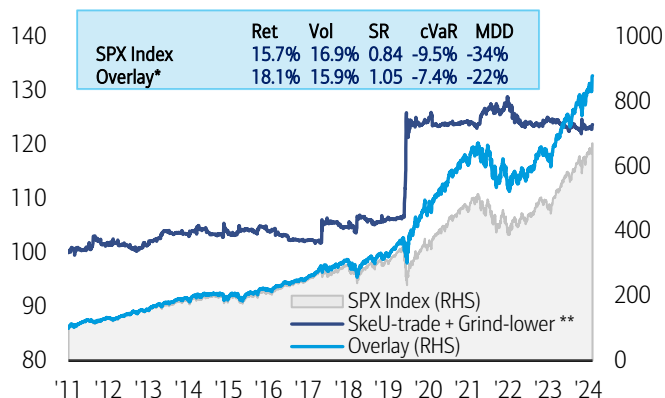


Source: BofA Global Research. Data: 18-Jan-08 to 11-Nov-24. *For an introduction to the W-trade, please read: *Global Equity Derivatives 2020 Outlook. Deeper into the abyss*. **The SkeU-trade strategy sells 1 x 3m 25d DH puts and buys N x 3m 5d DH puts, where the long leg uses a skew-adjustment for its delta hedge and sizing N is chosen such that the whole structure costs zero premium at inception (the average value of N within the backtest is 4.9). Each structure is entered monthly and held to expiry. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 83: When paired with a 'grind-lower' hedge, the SkeU-trade would have provided an effective SPX overlay, boosting Sharpe Ratio by 25% whilst significantly reducing tail risk

Backtested performance of the combination of the SkeU-trade (shown in Exhibit 81) with a grind-lower strategy** and the overlay of this hedge-combo onto the SPX Index*

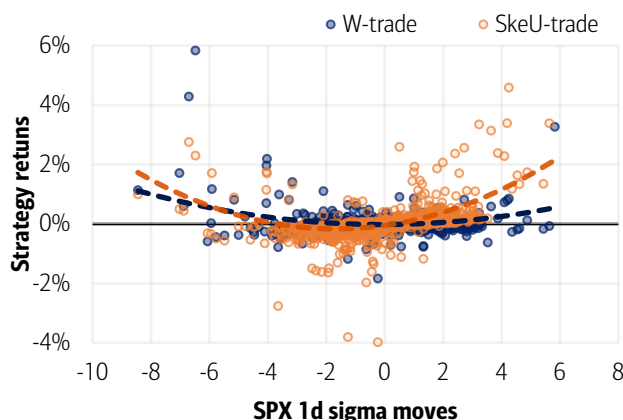


Source: BofA Global Research. Data: 5-Oct-11 to 11-Nov-24. The weights of two strategies in the hedge combo (SkeU-trade & BAEICPCO) are chosen to make the average of the top %-ile of daily returns the same for both strategies. *The notional of the hedge combo is chosen using 5% cVaR minimization of the combo onto the S&P, given a maximum tracking error of 5%. **The grind-lower strategy used in our hedge combo is BAEICPCO Index (contact us for more information). This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 82: ... and though both strategies are clearly convex in the tails, the W-trade would have paid out more in fragile down-moves while the SkeU-trade has a more symmetric U-shaped profile

The daily returns of the W-trade and the SkeU-trade (both shown in Exhibit 81) backtested strategies (y-axis) relative to daily sigma moves* in the SPX, with respective quadratic regression curves

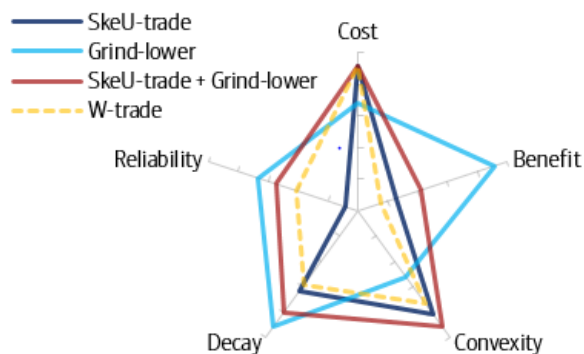


Source: BofA Global Research. Data: 18-Jan-08 to 11-Nov-24. *Daily sigma moves are measured as the 1-day return divided by the 6m trailing standard deviation of returns. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Exhibit 84: Relative to the W-trade, the SkeU-trade gives up Reliability for superior Benefit, Convexity & Decay. Pairing it with a grind lower strategy can yield a robust defensive outcome

Backtested performance of the SkeU-trade, the Grind-lower hedge, the hedge combo from Exhibit 83 and the W-trade evaluated on the five dimensions of our hedging framework (for explanation of these hedging metrics, read here: [The Leading Edge of your Hedge](#)) relative to a broad universe of SPX option hedges



Source: BofA Global Research. Data: 5-Oct-11 to 11-Nov-24. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

BofA GLOBAL RESEARCH

30x NKY fragility hedge: Dynamically sized 1x3 put ratios

What: NKY SkeU 1x3* PR. Systematically trade dynamically sized delta-hedged 3m NKY 1x3* put ratios to protect against fragility in Japan. Since 2008, fragility shocks have led to gains that were 30x the average cost per annum.

Why: Alongside US stocks, we see larger than normal fragility risk in Japan; especially to the downside. The dynamics that pushed yen carry trade positioning to extremes in August, remain in place. The longer US rates stay elevated, the larger the risks carry trade positions getting re-built in 2025. This could translate to increasingly asymmetric Nikkei risks with stocks “grinding” higher as long as the currency remains well behaved. While a repeat of the Aug-24 “YenMageddon” seems unlikely, a sharp yen rally could easily catalyze another sudden unwind of long stock positions.

Risk: Risks would be stocks grinding lower on muted vol & skew moves

Enhancing our W-trade and focus on downside fragility

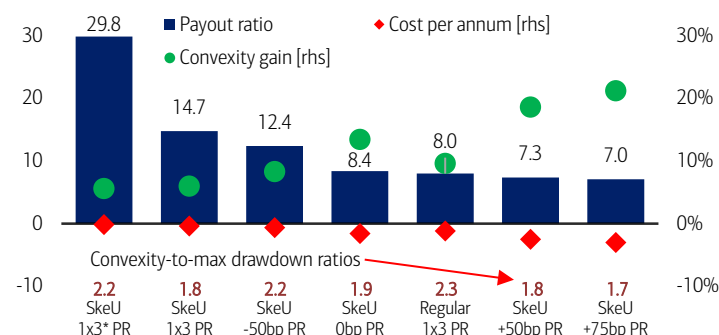
We introduced the W-trade in our [2020 Outlook](#) as our preferred fragility hedge. The trade includes shorting 3m 1x3 put ratios (sell 1x 25-delta put to buy 3x 5-delta puts and hedge delta daily). Tailoring this further to the Nikkei's prevailing asymmetric risks while limiting carry costs led us to the following enhancements:

1. **Focus only on the put ratio portion** of the W-trade (the original W-trade also has a 1x2 call ratio leg)
2. **Over-hedging delta** based on historically observed skew spikes amid fragility shocks (as per method in the previous section)
3. **Dynamically adjust put ratio sizing** to target the same cost* over time. Intuitively, this means buying >3x 5-delta puts when skew is flat and buying <3x when skew is steep (*= cost target at -77bps, corresponding to the average cost of a 1x3 put ratio)

Over-hedging delta on the 1x3 put ratios cut carry cost by up to 85% without giving up too much convexity (Exhibit 85 & Exhibit 86). Indeed, the average gain-to-cost ratio for the dynamically sized 1x3* put ratio improved from 8.0x to 29.8x.

Exhibit 85: NKY SkeU 1x3* PR would on average have gained 30x the carry cost per annum during quiet times. Its gain-to max drawdown ratio of 2.2 also ranked near the top of all strategies, second only to 2.3

Hypothetical back-tested performance statistics of various structures

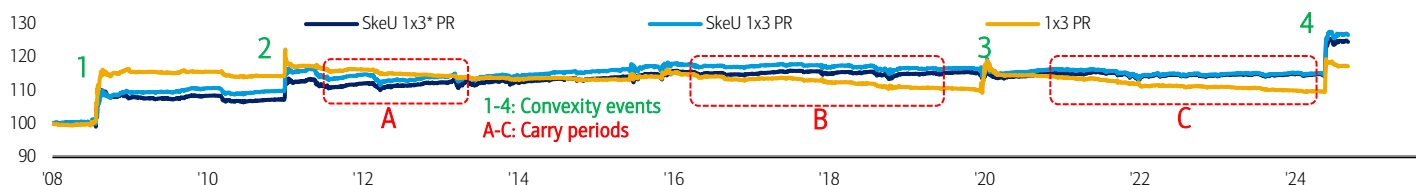


Source: BofA Global Research, Bloomberg. Data from 17-Mar-08 to 18-Nov-24. “SkeU” denotes strategies that apply the skew-adjusted delta hedging explained in the Convexity sans carry: Fighting fragility by taming the skew. By varying the long 5-delta put leg, the trades above apply different sizing to 3-month 5/25-delta put ratios. Either using a fixed ratio (e.g. 1x3) or a dynamic ratio by targeting a net put ratio cost (e.g. -50bps). * = Dynamic sizing that targets the long-term avg cost (-77bps) of a 1x3 put ratio. Each structure is entered monthly, held to expiry, and delta hedged. Carry periods are defined as Mar '11 to Jun '13, Jan '17 to Feb '20 and Jan '21 to Jul '24; Convexity events are the 2008 GFC, 2011 earthquake, 2020 Covid and 2024 “YenMageddon”.

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Exhibit 86: Skew-induced over-hedging and a dynamically sized put leg can keep carry costs close to flat. While the 1x3 put ratio structure from our original W-trade spiked enough to recoup 8 years of carry costs, SkeU 1x3* PR would have recouped nearly 30 years of costs (29.8x payout ratio)

The 1x3 Nikkei put ratio leg from our original W-trade as well as the two enhanced 1x3 put ratios on the Nikkei. All structures are delta hedged daily



Source: BofA Global Research, Bloomberg. Data from 17-Mar-08 to 18-Nov-24. * = See text below Exhibit 85 above. Transaction costs have been added; please ask us for details. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Equity/rates hybrid to hedge the return of bond vigilantes

What: Buy SPX Mar25 95% put contingent on 10y SOFR > ATMf+15bps for 0.53% (65% disc. vs vanilla put, +17 correl bid, ref. 5917, 3.87%). *Risk limited to premium paid.*

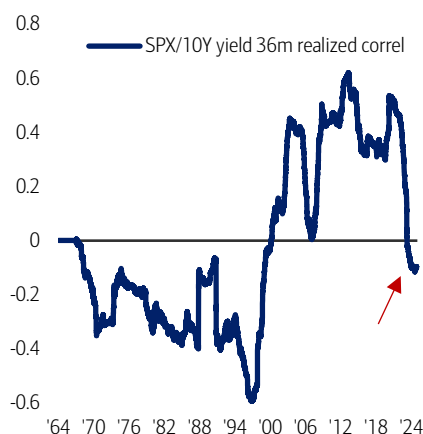
Why: US stocks briefly ignored the spike in rates post-election, but long-end rates unanchoring (on inflation or fiscal concerns) is a major risk to the bull market. Sell positive implied correlation and get a 65% discount on S&P puts.

Since inflation accelerated higher in 2021, US equities have been more vulnerable to rising interest rates than any other time in the last 25yrs (Exhibit 87). But for a brief moment after the US election, rates didn't seem to matter. Indeed, despite the spike in rates, even the more rate-sensitive stocks (e.g., small caps) reacted more positively than after the 2016 election and than the options market was implying.

However, as the post-election euphoria eases, we expect rates to once again be a key driver of equity returns. If rates stay elevated or rise modestly from here, we think one can follow the playbook of the last 3yrs, where rate-sensitive stocks like small caps underperform even if growth is strong (see our SPX over RTY outperformance trade). But if inflation returns in earnest or bond vigilantes call time on profligate policy, and the long end of the Treasury curve unanchors, we worry the entire equity market is at risk.

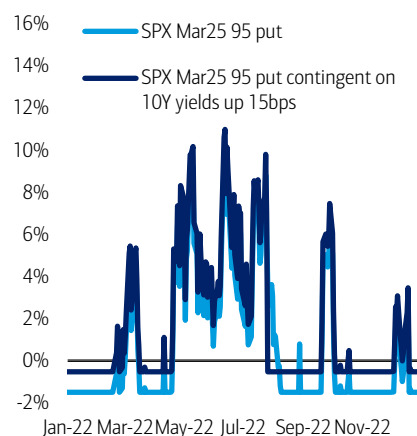
Equity/rates implied correlation is back in positive territory, supported by the election reaction and strong US data since the summer. As a result, one can get particularly attractive discounts on S&P puts if made contingent on higher rates. For example, S&P Mar25 95% puts contingent on 10Y rates up 15bps or more by expiry sell +17 implied correlation and cost 65% less than the vanilla S&P puts. The trade would have performed well in the 2022 rates-driven bear market (Exhibit 88), and 10Y yields are already above levels which catalyzed meaningful equity selloffs over the last 15 months (Exhibit 89).

Exhibit 87: Inflation risks made equities vulnerable to higher rates in the last 3yrs
3yr rolling correl between SPX & 10Y Tsy yields



Source: BofA Global Research. Data as of 20-Nov-24. Correl of daily equity returns and daily changes in 10Y Treasury yields.
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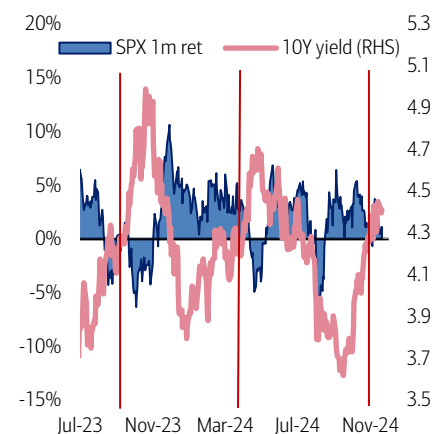
Exhibit 88: S&P puts contingent on higher 10Y rates would have worked well in 2022
Hypothetical expiry P&L at current costs



Source: BofA Global Research. Data as of 20-Nov-24. Current prices, past returns.

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Exhibit 89: US 10Y yields are already above levels that hurt stocks in the last 15m
10Y Tsy yields and S&P 500 1m returns



Source: BofA Global Research. Data as of 20-Nov-24.

BofA GLOBAL RESEARCH



Are you big enough? Hedge US large caps leading the rally

What: Buy SPX over RTY 103% Jun25 outperformance call contingent on RTY up for 0.59% (82% discount to vanilla outperf., +75 correl bid).
Risk limited to premium paid.

Why: Small caps jumped on the US election results, but high rates & the AI boom likely to keep large-cap Tech the market leader. Owing to 98th %ile RTY vol vs SPX vol, the contingent trade costs just 18% as much as the vanilla & would have paid off 67% as often in the last 10y.

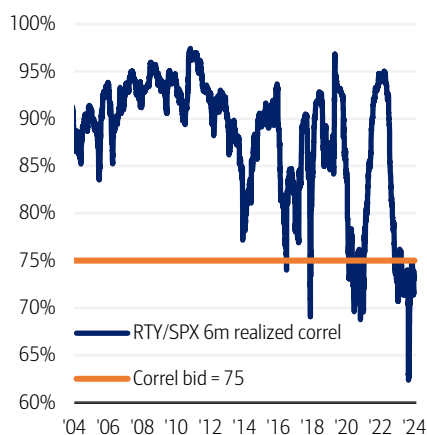
The election raised hopes of a long-awaited rotation into laggard cyclical and small caps (e.g. RTY +7.7%, SPX 3.7% from Nov 5th through 11th), which our equity strategists think is more likely than usual to continue given positioning (see [Strategy Snippet](#)). But much of the RTY outperformance has retraced as rates found new local highs, a dynamic we suggested hedging through RTY down / 10Y rates up dual digitals (see also our S&P puts contingent on higher 10Y yields to hedge inflation and bond vigilante risks).

Investors who rotated into small caps (or who are hesitant to get all the way up to equal-weight on large-cap US equities and/or US Tech) may want to hedge the risk of underperforming if large-cap Tech remains the market leader. Solid economic growth, an accommodative Fed, and the potential for the AI boom to become more bubble-like makes this risk worth hedging, in our view (see also our “A-trade”, designed to trade a boom without having to time the bust).

We particularly like SPX over RTY Jun25 103% outperformance calls *contingent on RTY up at expiry*. The entry point benefits from a relatively high implied correlation (+75 bid, in line with recent realized; Exhibit 90) and, particularly, from extremely high RTY 6m vol vs SPX vol (10yr 98th %ile; Exhibit 91). As a result, the contingent trade costs just 18% as much as the vanilla outperformance call, even though it would have paid off 67% as often in the last 10yrs (assuming current costs; Exhibit 92). In other words, RTY was up two thirds of the time that the S&P outperformed by 3% or more, and high rates and the ongoing AI boom may only strengthen this large-over-small dynamic from here.

Exhibit 90: The trade's implied correlation of +75 is high in absolute terms and in line with recent realized

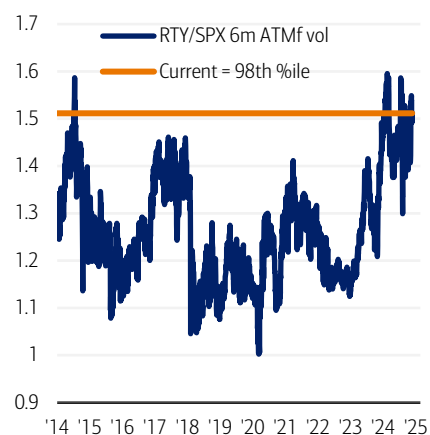
RTY/SPX 6m realized correl & trade's correl bid



Source: BofA Global Research. Data as of 20-Nov-24.
 Correlation of daily returns.

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Exhibit 91: The RTY/SPX 6m vol ratio near 10yr highs helps create a historically attractive discount vs the vanilla outperf. call
 RTY/SPX 6m ATMf vol ratio

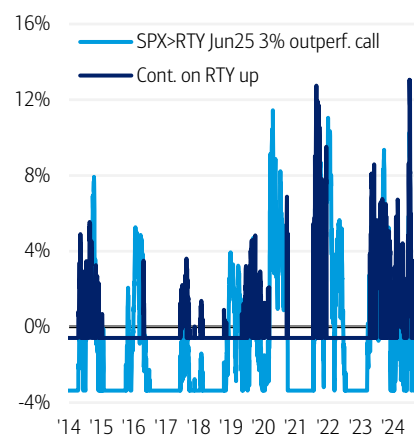


Source: BofA Global Research. Data as of 20-Nov-24.

BofA GLOBAL RESEARCH

Exhibit 92: The contingent trade costs just 18% as much as the vanilla & would have paid off 67% as often in the last 10yrs

Hypothetical expiry P&L assuming current costs



Source: BofA Global Research. Data as of 20-Nov-24. Current costs, prior returns.

BofA GLOBAL RESEARCH

Riders of the bubble: Historic entry point for NDX upvar

What: Buy (i) NDX Dec25 ATM upvar for 17v indicatively, or (ii) NDX Dec25 80% upvar for 20.15v indicatively (t & t-1 convention, Dec25 var ref. 25.1v).

Why: An AI stock bubble is likely to see vol move higher along with spot, as in the tech bubble. NDX upvar is well-suited to hedge this risk, in our view. Dislocations in deep OTM put skew have created a historically attractive entry point for upside conditioned variance relative to unconditional variance. At current prices, the ATM upvar & 80% upvar would have had average payoffs of 3.4v and 3.3v respectively since '86, while having strong performance in the tech bubble period. Even in the relatively low vol ChatGPT era (since Nov '22), the average ATM upvar payoff would have been essentially flat with a >50% breakeven rate.

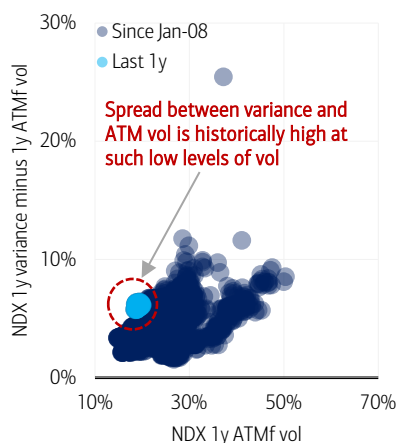
Risks: Losses would involve the variance strike paid.

As noted in our Outlook, drawing an analogy between today's AI boom and the tech bubble suggests that vol has further room to rise (Exhibit 11). Indeed, vol rising with spot has been one of the signatures of asset bubbles historically (Exhibit 9 & Exhibit 10). For such a scenario, we find NDX Dec25 upvar interesting as a vol-based hedge.

In terms of entry point, the upside conditionality offers an attractive discount to unconditional variance. Why? The spread between 1y variance and 1y ATMf vol for NDX has never been higher for these levels of vol (Exhibit 93), and much of this spread is being driven by elevated deep OTM put vol (not call vol) post-Covid (Exhibit 94). Based on current indicative pricing, ATM upvar would have generated a mean payoff of 3.4v (vs 1.2v for the unconditional) since 1986, with a >50% breakeven rate (Exhibit 95). Even the less stringent (but more costly) 80% upvar would have had an average payoff of 3.3v with >50% breakeven rate over this period. Looking at the relatively low vol ChatGPT era (since Nov '22), the ATM upvar would have still generated an average payoff of 0.06v with a >50% breakeven rate. More importantly, both iterations would have performed very well in the tech bubble (Exhibit 95). The key risk to this trade, apart from the spot conditionality, is a low vol environment (like in the 2013-19 period).

Exhibit 93: The 1y var-ATMf vol spread for Nasdaq has never been higher for these vol levels since 2008...

NDX 1y var-ATMf vol spread vs NDX 1y ATMf vol

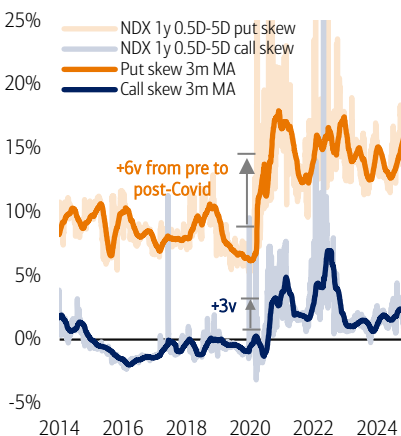


Source: BofA Global Research. Data from 1-Jan-08 to 15-Nov-24.

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Exhibit 94: ...with dislocated deep OTM put vol driving this spread and creating an attractive entry point for upvar

NDX 1y 0.5D-5D call skew and put skew

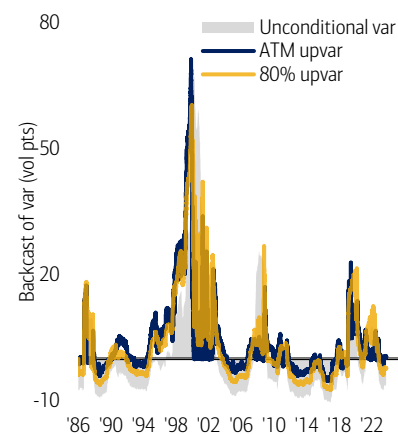


Source: BofA Global Research. Data from 1-Jan-08 to 15-Nov-24. 0.5D-5D skew is calculated as 0.5D vol minus 5D vol (for call and put side respectively).

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Exhibit 95: ATM and 80% upvar on NDX have performed well vs unconditional var, particularly in the tech bubble

Backcast of NDX Dec25 ATM and 80% upvar



Source: BofA Global Research. Data from 1-Jan-86 to 15-Nov-24. Assumes current prices and historical returns/variance.

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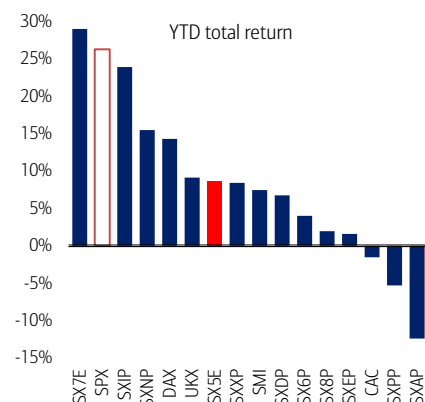
Upside hedges for a positive surprise in unloved Europe

What: Buy (i) SX5E Mar25 calls outright, or (ii) vs short SX6P (Utilities) calls for a small credit. (iii) Buy a SX5E vs SX6P Mar25 ATM outperformance call for 1.3%, conditioned on SX6P up at expiry (correl bid: 30%, 73% disc. Vs vanilla outperf.).

Why: Europe's bear case has been well made (tariff threats, domestic growth headwinds, disinflation risk, political uncertainty, etc). However, EU stocks are still up YTD (Exhibit 96). We find it prudent to have cheap upside hedges on, in the region in case of a positive surprise as few expect this or are positioned for it. We find most value in owning SX5E calls (Exhibit 97), which is financials, industrials and tech-heavy (i.e., cyclically-geared). For upside-tilted RV, we like a 0 premium SX5E vs SX6P call switch that benefits from relatively elevated SX6P (defensive sector) vol (see Exhibit 97). At current pricing, Exhibit 99 shows that the call switch would have generated 1.1% on average and been profitable >80% of the time in the past 10y. The risk-limited outperf. call, which benefits from positive SX5E-SX6P correl (Exhibit 100), would have generated a 4:1 payout on average.

Risks: Limited to premium paid for trades (i) and (iii). Utilities rising significantly more than SX5E (unlikely in a positive growth shock) is a risk for (ii).

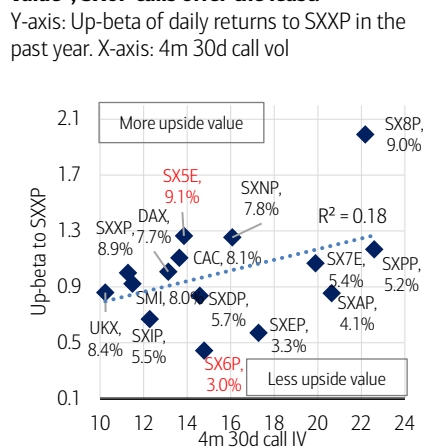
Exhibit 96: The SX5E is in the middle of the pack in terms of YTD returns in Europe...
YTD total return of EU sectors, indices & SPX



Source: BofA Global Research. Data as of 21-Nov-24.

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Exhibit 97: SX5E calls offer most upside value*; SX6P calls offer the least.
Y-axis: Up-beta of daily returns to SXXP in the past year. X-axis: 4m 30d call vol

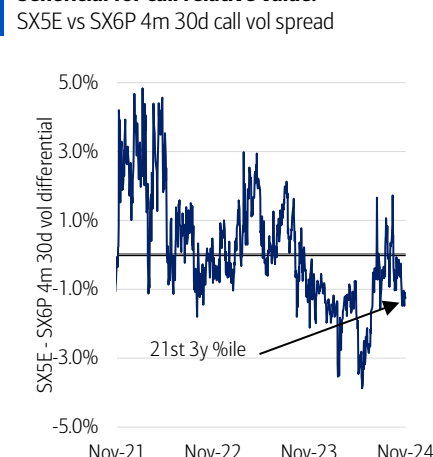


Source: BofA Global Research. Data: 21-Nov-23 to 21-Nov-24.

*Upside value = Up-beta to SXXP / 4m call 30d call IV

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Exhibit 98: SX5E vol is low compared to SX6P, beneficial for call relative value.
SX5E vs SX6P 4m 30d call vol spread

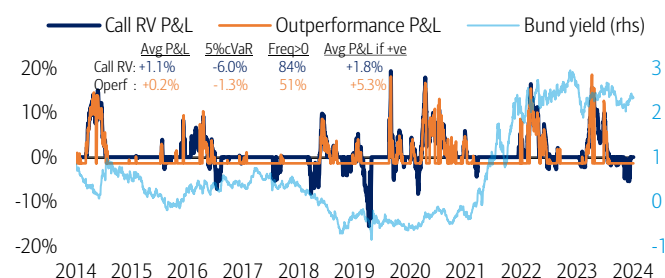


Source: BofA Global Research. Data: 21-Nov-21 to 21-Nov-24.

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Exhibit 99: The SX5E-SX6P call switch would have been more profitable than the outperformance call, but has more tail risk

Backcast (+) of (i) SX5E vs SX6P 4m 30d call switch & (ii) 4m ATM outperformance call on SX5E over SX6P, contingent on SX6P up at expiry.



Source: BofA Global Research. Data: 21-Nov-14 to 21-Nov-24.

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Exhibit 100: SX5E is positively correlated to SX6P, helping to cheapen the price of outperformance options.

4m realised correlation of daily returns of SX5E vs SX6P



Source: BofA Global Research. Data: 21-Nov-14 to 21-Nov-24.

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†: This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

How to trade European Dollar Darlings and Tariff Targets

What: Buy a Jun25 ATM (113% strike) qUSD call on return dispersion on a basket of low (vs history) vol stocks that are most positively & negatively geared to USD for 1.92%, indicatively.

Why: We anticipate the coming months to be rife with US trade policy discussions, and the USD could be the lightning rod for policy directions. Swings in the currency are to be expected and a call on return dispersion on low(er) vol, USD-sensitive stocks can offer a way to position for currency volatility.

Risks: Losses will be limited to the premium paid

The weeks following the 2024 US elections have been marked with strong rotations at the regional, sector, factor and also at the stock level. Given the experience during President-elect Trump's last tenure, we'd expect ongoing rhetoric on US exceptionalism to remain an important driver in financial markets globally in the coming months. A prominent effect already has been the large USD move (DXY +3.5% since elections) and volatility (3-week realised vol of 8.9% vs a YTD median of 5% for DXY). European stocks with large positive sensitivity to the USD face a double-edged sword – while these companies' revenue streams may benefit from USD strength, some of these very names may become targets within a punitive US trade tariff regime. On the flip side, companies with large negative USD exposure may suffer on continued FX strength, or conversely may rebound if some of the FX gains are given up in a scenario where discussions around restrictive trade policies cool off. In all these scenarios, we see value in positioning for dispersion within a selection of positively and negatively geared (to USD) names. Exhibit 101 shows that stocks with such a profile already have higher implied vols than those with low (magnitude) USD beta, however, we see opportunities to select names from the highlighted zones where implied vol is low vs its own history. Exhibit 102 shows such a selection of names, which have historically performed well at current pricing (Exhibit 103) and most particularly in periods of large moves in the US Dollar (Exhibit 104).

Exhibit 102: A custom selection of names with large (magnitude) beta to USD and low implied vol vs history prices up well for...

European names with large (positive or negative) beta to DXY, and low Implied vol vs history

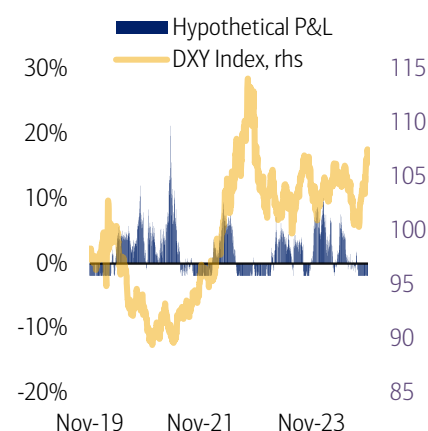
		6M ATM F	
	DXY Beta	IV	IV 3y%ile
VNA GY	- 1.43	28.1%	28%
ORA FP	- 0.94	15.3%	15%
ENGI FP	- 0.69	17.8%	18%
RIO LN	- 0.68	23.8%	24%
MT NA	- 0.67	29.5%	29%
ITX SQ	- 0.63	22.3%	22%
PGHN SE	0.29	23.7%	24%
ENI IM	0.31	18.8%	19%
ALC SE	0.41	21.6%	22%
SLHN SE	0.48	17.9%	18%
RR/ LN	0.53	33.0%	33%
SHELL NA	0.73	19.5%	19%

Source: BofA Global Research, Bloomberg. Data from 20-Nov-21 till 20-Nov-24. 6m beta, as in Exhibit 101. We average beta and IV %ile ranks within the liquid European universe to select top 6 positively and negatively geared (to USD) European names. Betas may also include unrelated macro sensitivities

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Exhibit 103: ... a call option on return dispersion. Historically this has performed well in periods of large USD swings...

Hypothetical back-cast (†) of a Jun25 ATM call on return dispersion within the selected names

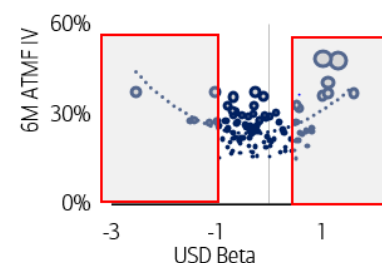


Source: BofA Global Research, Bloomberg. Data from 21-Nov-19 till 25-Nov-24.

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Exhibit 101: European stocks with the largest (magnitude) USD betas also have larger implied volatilities

SXXP Top 100 liquid names implied vol vs 6M beta of 2-day returns to DXY Index

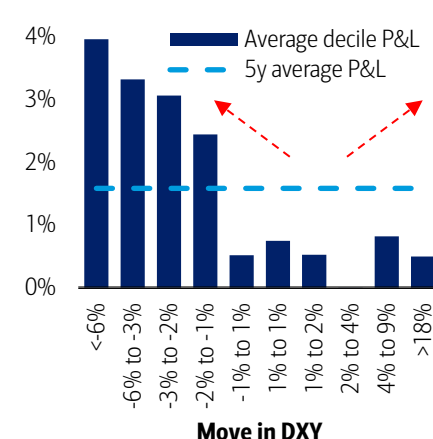


Source: BofA Global Research, Bloomberg. Data from 21-Nov-21 till 20-Nov-24. Bubble size is proportional to the 3y percentile rank of 6M ATM F IV. USD beta is the 6m beta of 2-day returns from a 2-factor regression to SXXP and DXY index

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Exhibit 104:and this bi-directional exposure of the return basket can be seen in its performance vs large USD returns

Average trade P&L within each decile of DXY returns in the last 5 years



Source: BofA Global Research, Bloomberg. Data from 27-Nov-19 till 25-Nov-24

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†: This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.



Global FTSE stocks pay USD dividends: buy cheap div futures for an FX uplift

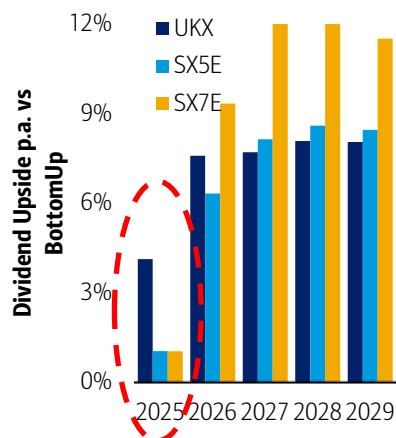
What: Buy Dec25 FTSE dividends as a Dollar (vs GBP) strength beneficiary (ref: 299.5)

Why: FTSE dividends have lagged the underlying equity index, resulting in 1Y FTSE implied dividend yields dropping to below the 10th percentile of the last 2 years (Exhibit 106). This has been during a period that the FTSE index itself has underperformed the SX5E by nearly 9%, spotlighting FTSE dividend futures as a potential value opportunity in Europe. Indeed, the annualized upside potential to bottom-up estimates for Dec25 is significantly higher (4.1%) for FTSE vs SX5E and EU-Banks (SX7E) dividends of the same tenor (Exhibit 105). Moreover, FTSE dividends should be a beneficiary of USD strength as ~43% of FTSE dividends are announced by globally exposed FTSE companies in USD (Exhibit 107). This currency translation effect from USD to GBP between announcement and payment dates has historically resulted in a negative beta of the constant tenor div futures to FX moves, when regressed vs the FTSE index and GBPUSD. Liquidity on FTSE dividend futures is limited (~5% of SX5E dividend futures' open interest) but not negligible, and the product can thematically gain interest in case of continued Dollar strength.

Risks: Losses will be 1-for-1 on long futures.

Exhibit 105: FTSE Dec25 dividend futures stand out vs SX5E and SX7E for having a large upside to estimates, despite the short tenor

Upside (p.a.) to Bottom-Up estimates

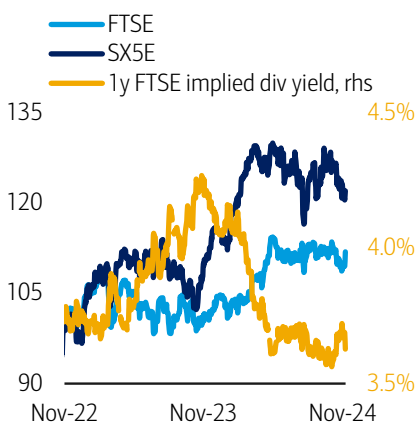


Source: BofA Global Research, Bloomberg. Data as of 22-Nov-24

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Exhibit 106: FTSE has lagged EU equities while FTSE divs underperformed the index itself, making divs look cheap

FTSE, SX5E spot vs FTSE implied 1y* div yields

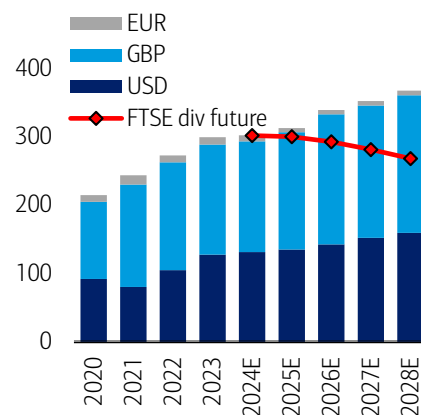


Source: BofA Global Research, Bloomberg. Data from 23-Nov-22 to 22-Nov-24. *We compute constant tenor 1y dividends

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Exhibit 107: c. 43% of Dec25 FTSE divs will likely be announced in USD, resulting in short cable sensitivity

FTSE's bottom-up dividend points announced by currency



Source: BofA Global Research, Bloomberg. Data as of 21-Nov-24. We use the currency plot of bottom-up estimates as of the end of the prior year to proxy the split in paid dividends. For 2024 to 2028, we use current analyst bottom-up estimates

BofA GLOBAL RESEARCH

Taiwan risk: Low-cost tail hedge for 22x max payout

What: Buy 19-Dec-25 85%/65% Best of Put spreads on NKY/KOSPI2/SPX for 0.90% (indic. quanto USD, 56% discount to avg vanilla)

Why: For a proper tail event, investors tend to target >20x max payout ratios. This tail hedge offers a 22x max payout ratio (Exhibit 108 & Exhibit 109). It does, however, require a correlated shock, but we believe correlation is the key variable to take advantage of when it comes to Taiwan. Not only would a large geopolitical event stress markets globally, this particular issue would also reverberate via Nvidia and other tech companies. That is, with Taiwan and TSMC down, Nvidia and others would likely take a very large hit, pushing major indices, including the S&P, much lower.

Low probability but massive impact...

While a low probability event, many investors still grapple with the tail risk of a Taiwan conflict. Not only would a conflict have a direct impact on Tech giants such as Nvidia via Taiwan's TSMC, but it could potentially involve "Russia-style sanctions" on China, reverberating across asset markets globally.

...And a tail difficult to completely ignore...

With TSMC close to being a US\$1 trillion company and Taiwan rallying ~40% since Q4 last year, many global portfolios – EM-focused ones in particular – have a meaningful direct exposure to Taiwan, making it harder to ignore tail risk hedging considerations.

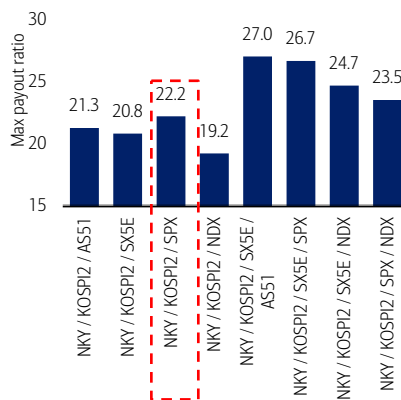
...But challenging to hedge: In a worst-case scenarios there may not be any settlement at key exchanges (e.g. in Taiwan). Be careful to avoid having a hedge that cannot be monetized during the event it is designed for.

...so one should probably avoid exchanges in Taiwan, mainland China and even Hong Kong: Consequently, investors wanting to hedge Taiwan risks should focus on liquid proxy assets that are likely to react to a large conflict between China and the US.

Best of put (BoP) spreads are **long volatility, correlation and convexity**; all parameters that tend to spike during large selloffs. At 90bps cost per annum (~13 months) investors can make up to 2,000bps, or 22x the premium spent. This would be achieved if the best performing index falls by at least 35%. It sounds like a lot, but that corresponds to S&P levels seen 1.5 years ago (S&P hit 3,856 on March 13th, 2023). Best of put option structures are suitable for these types of scenarios. Not only do they benefit from correlation moving towards one in global risk-off events (indiscriminate selling), they also provide a cheaper than normal downside hedge. Particularly important when timing is uncertain as one may have to carry the hedges for an extended period time. While BoPs are exotic options ("light" exotic), they may still suit non-derivatives specialists due to

Exhibit 108: Among baskets of 3, the BoP spread on NKY / KOSPI2 / SPX offers the highest max payout ratio of 22.2x. Adding SX5E would increase the max payout to 26.7x, but is less close to semi-conductors

Max payout ratios of various index components of 85%/65% BoP spreads (payout as of expiry)



Source: BofA Global Research. Data: as of 20 Nov 2024.

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Exhibit 109: Depending on the components of the 8 baskets, investors can achieve max payout ratios ranging from 19 to 27x

Cost of 85%/65% BoP spreads and max payout ratios across eight different basket combinations of underlying indices: N=NKY, K=KOSPI2, S=SPX, E=SX5E, ND=NDX, A=AS51

					85/65 BoP spread	
Underlying asset					Cost (% of spot)	Max payout ratio
A1	A2	A3	A4			
1	N	K	A	--	0.94%	21
2	N	K	E	--	0.96%	21
3	N	K	S	--	0.90%	22
4	N	K	ND	--	1.04%	19
5	N	K	E	A	0.74%	27
6	N	K	E	S	0.75%	27
7	N	K	E	ND	0.81%	25
8	N	K	S	ND	0.85%	24

Source: BofA Global Research. Data: as of 20 Nov 2024.

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Risk: Limited to premium paid



their buy-and-hold nature. They require minimal maintenance, no need to roll them mid-way through the trade and no delta hedging. A Best-of-Put option pays out based on the best performing asset. For instance, if the Nikkei, the Kospi and S&P 500 are down 30%, 35% and 40%, respectively, the option would pay out based on the Nikkei's 30% drop.

AI = Inevitable electricity demand = Nuclear only option

What: Our global screen of nuclear and Uranium-focused companies set to benefit from a massive increase in electricity demand.

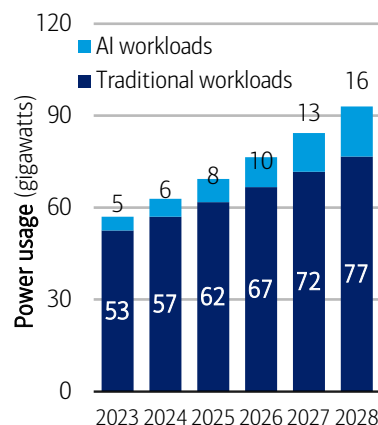
Why: One of the most certain aspects of AI is an explosion in electricity demand. The International Energy Agency (IEA) expects that data centres could be using a total of 1,000 terawatt hours annually by 2026. This is more than any country except for China, US and India (based on 2023 Electricity consumption). Large tech companies look set to fuel demand for Small Modular Reactors (SMR) as the power source for projects they have in development (see [Introduction to SMR's](#)).

Our global nuclear screen includes companies that are seen as leaders in SMR development or are engaged in related operations such as uranium mining (Exhibit 111). Much nuclear innovation comes from Asia and we believe exposure to Japan, Korea and China is key. Some alternatives are more Western-focused. The NLR ETF, for instance, only has 9% in Asia vs 56% in ours (NLR up 29% YTD vs +64% for global).

Some of these stocks have already rallied substantially, however, and risks range from regulatory to negative public opinion. As such, we believe investors should use options instead of delta-1 exposure.

Exhibit 110: AI workloads' power usage may grow at a 25-33% CAGR through 2028

Global data center power usage by application



Source: Schneider Electric, The AI Disruption: Challenges and Guidance for Data Center Design. AI workloads: Uses GPU/TPU heavily; Traditional workloads: Uses CPU primarily. See [Industrials/Multi-Industry](#).

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Exhibit 111: Our global nuclear screen is market cap weighted and has a high Asia exposure (56.8%) that captures surging regional demand & technological leadership

Global nuclear screen ordered by market cap

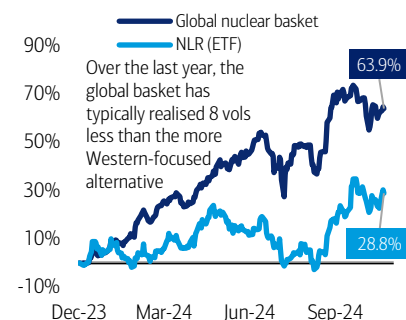
Ticker	Name	Country	YTD*	Weight
6501 JP	Hitachi Ltd	Japan	82%	26.4%
CEG US	Constellation	US	118%	18.3%
7011 JP	MHI	Japan	171%	11.9%
PEG US	PEG	US	55%	10.7%
1816 HK	CGN Power	China	33%	6.3%
601985 C1	CNNC	China	25%	6.2%
CCO CN	Cameco Corp	Canada	45%	6.0%
BWXT US	BWX Tech	US	72%	2.9%
KAP LI	Kazatomprom	Kazakhstan	5%	2.5%
034020 KS	Doosan Enerbility	Korea	34%	2.2%
SMR US	NuScale Power	US	794%	1.6%
NXE CN	NexGen Energy	Canada	30%	1.1%
UEC US	Uranium Energy	US	25%	0.8%
DMML CN	Denison Mines	Canada	43%	0.5%
052690 KS	KEPCO	Korea	12%	0.4%
1164 HK	CGN Mining	HK	1%	0.4%
YCA LN	Yellow Cake	UK	-13%	0.4%
PON AU	Paladin Energy	Australia	-18%	0.4%
EFR CN	Energy Fuels	US	5%	0.3%
NNE US	NANO Nuclear	US	548%	0.2%
BOE AU	Boss Energy	Australia	-31%	0.2%
DYL AU	Deep Yellow	Australia	8%	0.2%
FCU CN	Fission Uranium	Canada	-16%	0.1%

Source: BofA Global Research. Data from 29-Dec-2023 through 20-Nov-2024. *YTD: year-to-date return. This is a screen and not a recommended list either individually or as a group of stocks. Investors should consider the fundamentals of the companies and their own individual circumstances / objectives before making any investment decisions.

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Exhibit 112: Our global nuclear screen would have outperformed the more Western-focused ETF YTD. Additionally, on lower volatility

Global nuclear basket performance vs the NLR



Source: BofA Global Research. Data from 29-Dec-2023 through 20-Nov-2024. This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Our analysts forecast US\$135/lb potential for Uranium through 2026, see [Bullish uranium](#). What is more, APAC lead this transformation, with China's ambitious 150+



reactor pipeline and Japan's accelerating restarts providing strong demand visibility. The emergence of AI and data centre power requirements adds a compelling secular growth driver (Exhibit 110), as Tech giants increasingly seek reliable, carbon-free baseload power, see [Introduction to Small Modular Nuclear Reactors \(SMR's\)](#).

Our screen includes companies whose focus on nuclear reactors or uranium mining is large relative to other focus areas. To assess this, we screen for companies globally by: **(i)** Selecting stocks whose company description includes "Nuclear", "Uranium" or "reactor" (358 globally), **(ii)** Selecting stocks that meet minimum requirements of US\$0.5 Bn market capitalization and US\$3 Mn in 60-day average daily turnover (61), **(iii)** Removing too volatile stocks to ensure options tradability, limiting 1-month realised vol to 150% (58) and **(iv)** Select stocks that are deemed closest to the theme by requiring that (a) they derive >50% of its total revenue from a nuclear power sector, (b) score >50 on the United Nations Sustainable Development Goal (SDG) 7 "Affordable and Clean Energy", or (c) Are leaders in SMR development and/or are key domestic suppliers of reactors or are main domestic producers of reactors (23).

Rising stock fragility boosts EU vol dispersion opportunity

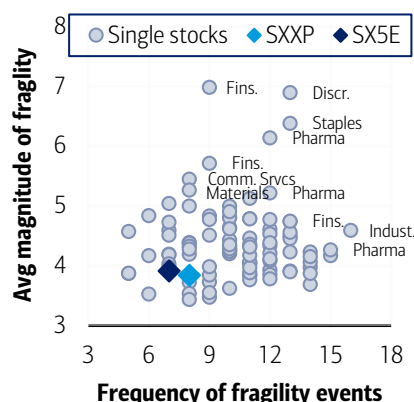
What: Long Dec25 VolSwap dispersion on custom names (Exhibit 115) vs the ESTX50 at 9.5 vol pts (indic.).

Why: A highly fragile environment and greater idiosyncratic risk for EU stocks creates a favourable backdrop for vol dispersion strategies

Risks: arise if the realised stock vs index vol spread falls below initial implied or in a correlated market shock.

Exhibit 113: Fragility is frequent and pronounced across European stocks too.

No. and avg magnitude of fragility events* across SXXP, SX5E and top 100 SXXP's constituents.

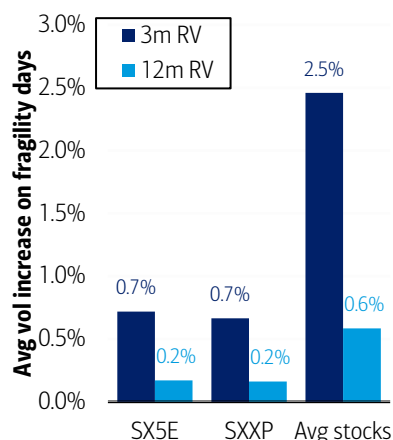


Source: BofA Global Research, Bloomberg. Data 21-Nov-22 to 21-Nov-24. *See footnote of Exhibit 114.

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Exhibit 114: Realized vol tends to surge on days marked by fragility & more so on stocks.

Average realized vol increase on fragility days (vs previous day) for SXXP, SX5E and top 100 SXXP's constituents.



Source: BofA Global Research, Bloomberg. Data 21-Nov-22 to 21-Nov-24. *A fragility event is here defined as a sigma move (a stock's 1d return divided by its 21d trailing volatility) with magnitude > 3.

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Record-high stock fragility as a driver for dispersion: 2024 is shaping up to be a year marked by near-record levels of single stock fragility, both in terms of frequency and magnitude (Exhibit 22 & Exhibit 23). This trend is far from confined to US Tech; it is clearly visible across European equities too and extends across multiple sectors within the region (Exhibit 113). While fragility (often linked to the 4th moment of the return distribution, i.e., kurtosis) and volatility (variance) are two separate concepts, the former can significantly drive the latter as the magnitude (squared) of extreme returns inflate variance too. Indeed, Exhibit 114 shows that historically, realised vol has surged on days marked by fragility, and more so for single stocks than indices. In our view, a continued highly fragile environment creates a favourable backdrop for stock vs index vol dispersion strategies.

Our favourite vol dispersion basket vs ESTX50: Across a liquid universe of EU names, we screen for the top 20 candidates that are most under-pricing their historical and recent vol (i.e., median 12m RV and latest 6m RV, respectively) out to Dec25. In our view, this tenor is near enough to benefit from (short-dated) stock fragility and option liquidity but also far enough out to take advantage of a steep ESTX50 vol term structure. The basket (Exhibit 115) is well diversified across sectors, with weights assigned to each name proportional to their historical and recent RV versus their current (mid) volswap level. At current pricing, our custom dispersion basket vs ESTX50 would have offered frequently positive performance (2.1% on average, when >0), and positive carry lately (Exhibit 116).

Exhibit 115: Our volswap dispersion basket vs ESTX50 is offered at only 9.5v...

Basket weights, with Dec25 volswap offer for stocks, *bid for SX5E. We use Vega-neutral sizing.

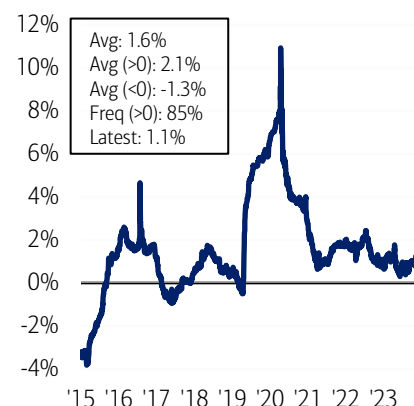
Sector	Weight	Volswap offer
STLAM IM Discretionary	9.1%	33.3%
SHELL NA Energy	6.8%	22.1%
SAN SQ Financials	5.4%	30.4%
ENI IM Energy	5.4%	22.5%
UCG IM Financials	6.6%	33.6%
IFX GY IT	4.8%	38.6%
ABI BB Staples	4.8%	23.9%
SREN SW Financials	5.7%	22.9%
PGHN SW Financials	5.9%	26.2%
AIR FP Industrials	5.1%	28.3%
NESN SW Staples	5.0%	17.2%
BBVA SQ Financials	5.1%	31.5%
DHL GY Industrials	4.1%	25.4%
SGO FP Industrials	3.9%	28.2%
ITX SQ Discretionary	3.8%	24.6%
UBSG SW Financials	4.6%	29.5%
G IM Financials	2.9%	22.9%
MC FP Discretionary	3.6%	31.3%
INGA NA Financials	4.0%	27.4%
IBE SQ Utilities	3.4%	20.5%
SX5E*	-100%	17.9%

Source: Source: BofA Global Research. Data as of 21-Nov-24

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Exhibit 116: ... and delivered frequently positive P&L, with positive carry lately.

Back-cast (†) for 12m (~Dec25) stock minus index RV spread, minus current implied spread (9.5v).



Source: BofA Global Research, Bloomberg. Data from 21-Nov-14 to 21-Nov-24. †: This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

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Leveraging EU Delta-1 dislocations for carry or hedging

Record SX5E TRFs imply stretched equity demand that's hard to sustain in 2025

What: Short delta-hedged Dec25 SX5E Total Return Futures (TRFs) *in the new year* to benefit from a de-escalation of factors causing the recent spike.

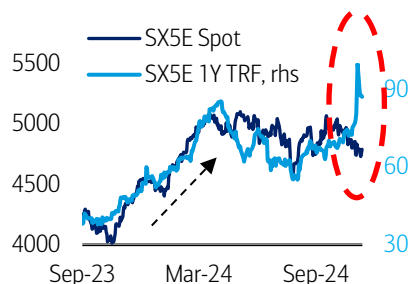
Why: Watch for opportunities in the new year to sell (potentially still) elevated TRF spreads in anticipation of extended positioning dissipating given a relatively sobering EU economic outlook. We see attractive (5:1) risk-reward for generating carry that may benefit further from outright levels declining.

Risks: Spikes in the TRF level can cause MtM losses, as can a disproportionate drop in SX5E price return futures (the delta-hedge for the short TRF position).

Elevated SX5E TRF spreads (Exhibit 117) and an explosive if short-lived recent spike reflect (i) long positioning (and anticipation of more longs) via synthetic products into this year's rally (Exhibit 118, Exhibit 119), (ii) year-end balance sheet pressures, and (iii) potential news flow on Banks' G-SIB scoring. We think this demand is inconsistent with a relatively weak economic backdrop. Moreover, a recent disconnect between TRFs and SX5E returns as well as investor positioning suggests this is a significant carry opportunity via short delta-hedged SX5E Dec25 TRFs. We estimate the reward/risk of entering the trade at current levels and holding for ~3 months is ~5:1 (=avg 3M P&L vs worst 1-day MtM loss, Exhibit 120). The trade benefits from high carry (given high TRF spreads) but also potentially positive P&L if spreads decline. Crucially, we caution against establishing a position before year-end as ongoing balance sheet pressures could widen spreads further before they correct lower in the new year. The position can be hedged with long Mar25 price return (VG) futures. TRF sellers could benefit from accrued funding likely being larger than accrued dividends in Q1 but need to be aware of implied dividend and rates risk in the VG contract. For example, a scenario where short-term rates are 50bp lower (than anticipated) could impact VGH5 futures (but not TRFs) by up to 11bps.

Exhibit 118: This year's rally in equities likely drove demand for long synthetic exposure and the subsequent rise in TRF spreads, but this relationship broke in Nov24

SX5E 1Y constant tenor TRF spreads and the SX5E index level

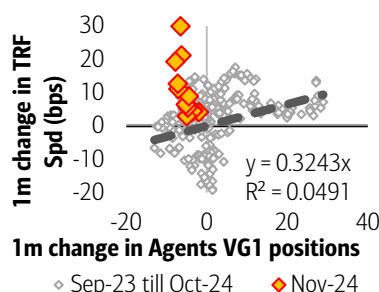


Source: BofA Global Research, Bloomberg. Data from 11-Sep-23 to 22-Nov-24.

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Exhibit 119: Widening TRF spreads can be indicative of long positioning in equities, similar to Agents increasing positions in SX5E futures. Nov24 has been an outlier though

1m changes in Eurex Agent Net positioning (bn EUR) vs changes in 1Y SX5E TRF spreads

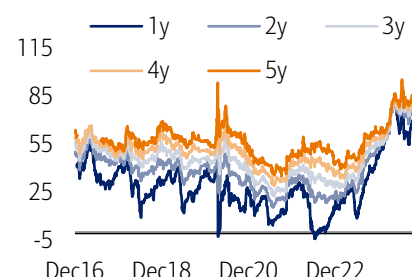


Source: BofA Global Research, Bloomberg, Deutsche Börse. Data from 11-Sep-23 to 18-Nov-24. Eurex defines participant type Agent as those who're trading on behalf of customers

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Exhibit 117: SX5E Total Return Future (TRF) spreads spiked to historical highs, and despite reversing some of the Nov24 move, still remain within the top 8-year percentile

1y – 5y constant tenor ESTX50 TRF spreads

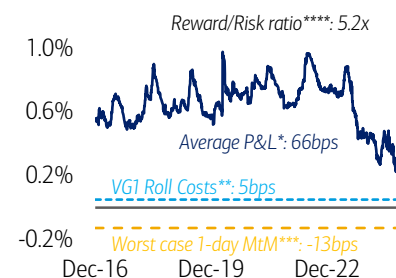


Source: BofA Global Research, Bloomberg. Data from 02-Dec-16 to 20-Nov-24.

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Exhibit 120: Short SX5E Dec25 TRFs sold at current levels would perform well historically, with a 5x reward/(MtM)risk ratio

Hypothetical back-cast (†) of short 1Y TRFs closed 3m later



Source: BofA Global Research, Bloomberg. Data from 2-Dec-16 to 19-Nov-24. *We look at the P&L of a Dec25 SX5E TRF sold at current levels in early Jan'25 and closed at historical ~9M constant tenor TRF levels at Mar25 expiry. We consider the "Traded Basis" P&L – please note short TRF also has Accrued Dividends and Funding P&L **The VGH5 future delta hedge will need to be rolled if the trade is carried beyond Mar25 expiry. VG futures are also exposed to rates and div changes ****1-day P&L if 1Y TRF spreads reach their highest level historically, on day 1 of the trade. *****Average P&L of a trade held for 3M, divided by the worst 1-day MtM loss. BofA GLOBAL RESEARCH

†: This performance is backtested and does not represent the actual performance of any account or fund. Backtested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.



Stretched div yields make SX7E div calendars a better hedge for EZ Banks

What: Investors with long positions in EU-Banks (SX7E) should consider hedging with -Dec27/+Dec26 SX7E dividend futures spreads

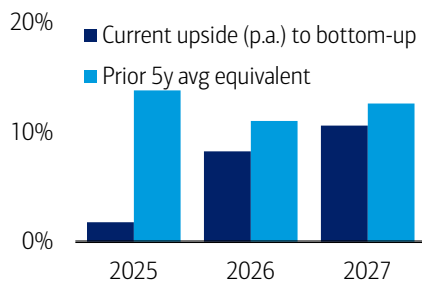
Why: The EU-Banks index (SX7E) is among the best performers vs European sector peers this year, and implied dividend yields are high.

Risks: A long/short div futures spread would make losses if the Dec26 contract proportionally underperforms Dec27 contract. Considering this as a hedge, Dec26 underperformance in a sell-off is a risk.

The EU-Banks index (SX7E) is among the best performers vs European sector peers this year (+20.4% vs 4.5% for SXXP), coincident with a significant steepening in German yield curves. Our latest [European Fund Managers Survey](#) suggests an investor overweight on Banks persists despite growth concerns in the region. While economic data surprises may still have the potential to give Banks a further boost and result in upside, we caution against maintaining unhedged positions given a likely European underperformance in the next US presidential tenure. We've discussed ([GEVI: 10-Sep-24](#)) how relative value dividend futures calendar trades have exhibited a defensive profile that has carried better this year than traditional puts or put spreads (Exhibit 123). A long-short calendar spread with a larger position in long Dec26 futures than short Dec27 futures may offer an attractive balance between convexity and carry (Exhibit 124 illustrates this for Dec25 and Dec26). Moreover, historically elevated implied dividend yields (Exhibit 121) suggest dividends may have relatively less upside potential than the underlying index itself, and hence a smaller "upside beta" than "downside beta". This is further supported by the lower upside potential vs bottom-up estimates in SX7E dividend futures currently, compared to what's typically been seen in past years at this time of the year (Exhibit 122).

Exhibit 122: While there remains upside to the dividend futures vs bottom-up estimates, this upside is lower than what's been seen in past years at this point in the cycle

Annualized upside to div futures vs bottom-up, currently and at equivalent points in past cycles

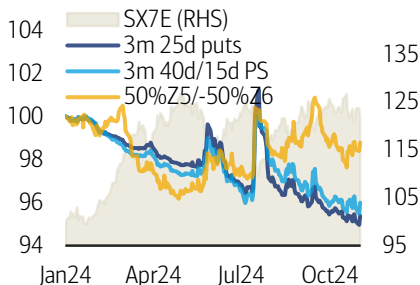


Source: BofA Global Research, Bloomberg. Data from 15-Nov-19 till 15-Nov-24

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Exhibit 123: The spread between DBEZ5 and DBEZ6 would have provided an effective alternative to SX7E option hedges this year

YTD backtested† performance of long SX7E 3m 25d puts, 3m 40d/15d put spreads and a +DBEZ5-DBEZ6 futures spread

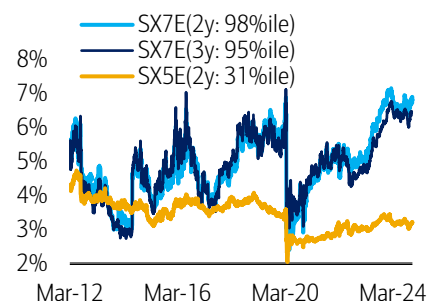


Source: BofA Global Research. Data: 17-Jan-24 to 19-Nov-24.
*(de-levered to 0.5x, i.e., +0.5x DBEZ5 -0.5x DBEZ6, rebalanced monthly to match SX7E put hedge volatility). † Please see Backtesting disclaimer below

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Exhibit 121: EU-Banks (SX7E) Implied dividend yields have reached close to decade highs, even as the sector has outperformed peers alongside steepening yield curves

Constant tenor SX7E and SX5E implied div yields

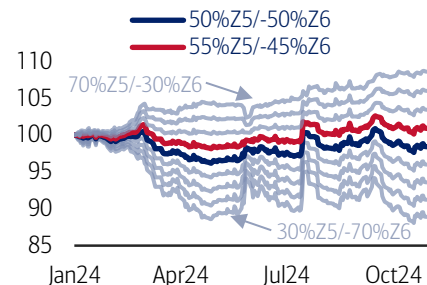


Source: BofA Global research, Bloomberg. Data from 2-Mar-12 to 22-Nov-24

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Exhibit 124: Adjusting the sizing between the Dec25 and Dec26 dividend futures can allow for a more neutral cost of carry

YTD backtested† performance of long DBEZ5 & short DBEZ6 futures (rebalanced monthly) whilst varying the respective weights in 5% increments



Source: BofA Global Research. Data: 17-Jan-24 to 19-Nov-24.
We de-lever the baseline futures spread to 0.5x and ensure gross leverage sums to 1 for the variations. † Please see Backtesting disclaimer below

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† This performance is backtested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein



China > Japan: Hedging a reversal of the ABC trade

What: XUA Jun25 110% 130% call spreads for 2.94% (indic. 22% delta) Partially funded by selling a NKY Jun25 110% call conditional on US 2yr rates lower, for 1.02% (47% of vanilla, indicative only)

Why: China in the medium term is uncertain, hence asymmetric exposure remains key. Instead of committing to China upside, a hedge offers skin in the game. This hedge can make you 20% in case China rallies but only lose 1.9% in case of a sell-off (given that the Nikkei does not rally more than 10% in a falling US rates scenario). Capture the ABC (=Anything But China) reversal and avoid overpaying by recognising that lower US rates allow China to ease more aggressively while a stronger yen may limit Nikkei upside

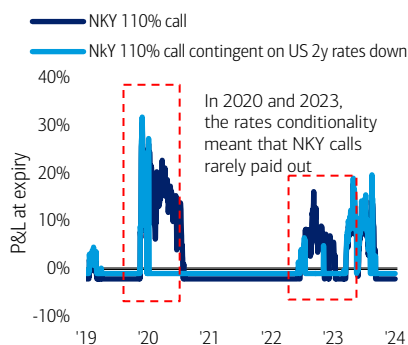
Reversal of the ABC trade translates to China upside and Japan headwinds:

The prevailing China skepticism, limited positioning, and lower valuations could drive Chinese stocks higher next year, even in scenarios when global stocks may struggle. Lower US rates could also cap Nikkei upside while allowing China to continue easing policy.

We are not necessarily negative on Japan, but a potential environment of lower US rates next year – and thereby a stronger yen (Exhibit 126) – may pose stronger Nikkei headwinds compared to many other markets. We expect another rate hike by the BoJ in January (link: [11-Nov BoJ Watch](#)) and the market is pricing 2 more US rate cuts in the next 7 months (by Jun-25).

Exhibit 125: Outside of 2024*, the short Nikkei call would only have lost money 15% of the time if conditioning it on US 2-year rates falling

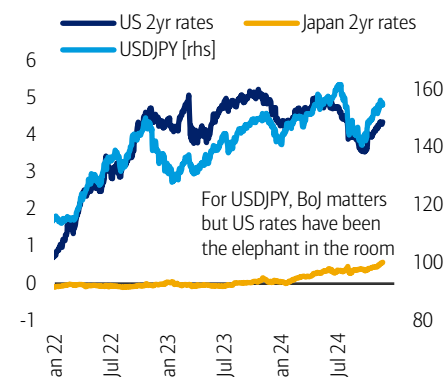
P&L at expiry using current cost of trade on historical NKY moves and 2yr USD OIS swap rate



Source: BofA Global Research and Bloomberg. Data from 25-Nov-19 through 25-Nov-24. * With rates peaking in 2023, it may not be surprising that the hybrid = vanilla for much of 2024. However, falling US rates remain a headwind for Nikkei.
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Exhibit 126: USD/JPY rallied more than 9% from the Sep lows as US rates rose from 3.27% to 4.05%. While Japanese rates matter, the larger factor will likely remain what happens to the US rates market

US and Japanese 2-year rates versus USD/JPY



Source: BofA Global Research. Data from 2-Jan-2022 to 21-Nov-2024.

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Cheapen rich XUA upside by selling rich NKY vol: With XUA (=futures on FTSE China A50) volatility still elevated, we like Nikkei call vol as a funding leg as the carry-like dynamics in Japan argues for a relatively well-behaved market on the upside. That is, a market grinding higher on low vol as long as USD/JPY doesn't sell-off. In fact, Nikkei vol ranks among the richest globally.

Trump 2.0 did not cause much China stress & tariff talks may spur more easing

The initial reaction to the Trump election win was only slightly negative for China. FTSE China A50 fell by 1.2% on Nov 6th while rallying 3.4% the day after (election results started coming out during the morning hours in Asia on Nov 6th). This illustrates that a lot of the tariff risks associated with Trump are, at least to a reasonable extent, likely priced into Chinese stocks already. And in fact, as our China Strategy team argues (see [US election impact on China](#)), any adverse tariff announcements could be used to justify more aggressive stimulus in China.

Risk: Negative payout if the Nikkei rallies >10% at the same time as US rates decline. The China leg would involve the premium paid

Lower US rates → US\$ not rallying → Allowing China to keep easing

Many commentators across “the Street” had been arguing for more aggressive China stimulus for some time. Eventually, China had its policy pivot in late September. Knowing that China wants to project strength via its currency, the timing of its policy easing just after the Fed started cutting rates may suggest that China feels more comfortable to ease when US rates are declining; ideally in conjunction with a softer US\$.

Lower US rates → Stronger Yen → Limiting NKY upside

We are not calling for a repeat of the August “YenMageddon” sell-off, but BoJ rate hikes as well as falling US rates could pose headwinds for the Nikkei. As Exhibit 126 illustrates, US rates are the elephant in the room when it comes to USD/JPY. Should US rates remain elevated next year, the NKY call will not pay out (Exhibit 125) and investors would walk away with a cheap upside call spread on China A50.



More trades on our Global Radar

Tariffs may only add to US over EU outperf.; hybrids can cheapen & limit risk

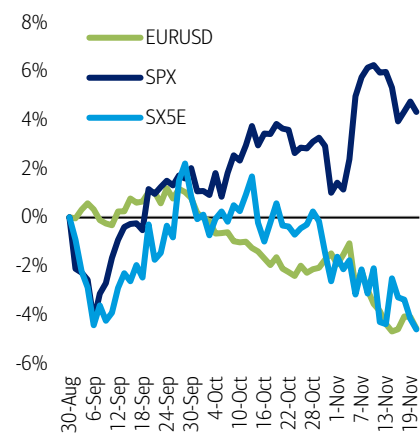
Trade: Buy SPX over SX5E Mar25 ATMf outperf. call cont. on EURUSD down for 2.21% indic. (43% disc. vs vanilla outperf.).

Why: US resiliency vs. fragile European growth may only intensify under a Trump administration and potentially higher tariffs (see [15-Nov Global Econ Letter](#)), supporting the case for US equities to continue outperforming European stocks (Exhibit 127). But the trade seems increasingly consensus, and policy and positioning risks are real. Hence, we prefer risk-limited expressions, with a EURUSD down contingency that aligns with the tariff threat. The contingent option costs 57% as much while paying off 69% as often since 2008 (Exhibit 128).

Risks: Limited to the premium paid.

Exhibit 127: The outperformance of SPX over SX5E has accelerated since the US election

SPX and SX5E cumulative returns since Sep-24

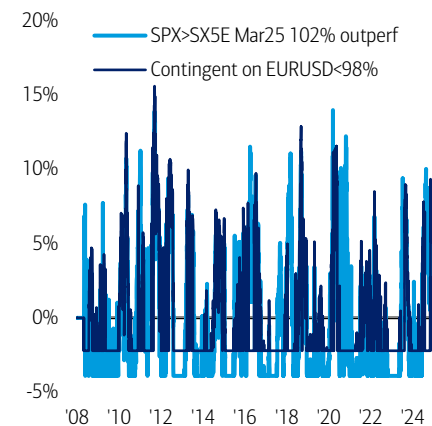


Source: BofA Global Research.

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Exhibit 128: The contingent option costs 57% as much while paying off 69% as often since '08

Hypothetical expiry P&L at current costs



Source: BofA Global Research. Current prices, past returns.

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Summary of Trades Opened (27-Nov-24)

Price data for open level reflects the price on open date and does not necessarily reflect the price at which the trade could be executed at the date of this report. Our trades are structured to be executed on the open date and are not necessarily appropriate to execute as formulated beyond that date.

Exhibit 129: Summary of trades opened in this report

For a full list of our open trades and trades closed over the last 12 months, please see [Global Equity Volatility Insights: Fade vol crush to own fragile momentum](#).

Trade Description	Open Date	Open Level	Expected Trade Term	Rationale	Risks
Buy SPX over RTY 103% Jun25 outperf. call cont. on RTY up	26-Nov-24	0.59%	20-Jun-25	High rates & AI boom likely to keep large-cap Tech mkt leader. Owing to 98th %ile RTY vol vs SPX vol, the contingent trade costs just 18% as much as vanilla & would have paid off 67% as often in the last 10y.	Limited to premium paid
Buy SPX Mar25 95% put contingent on 10y SOFR > ATMf+15bps	26-Nov-24	0.53%	21-Mar-25	Long-end rates unanchoring (on inflation or fiscal concerns) is a major risk to the bull market. Sell positive implied correlation and get a 65% discount on S&P puts.	Limited to premium paid
Buy SPX over SX5E Mar25 ATM outperf call cont. on EURUSD down	26-Nov-24	2.21%	21-Mar-25	US election results support case for continued US over EU outperf., but policy and positioning risks are real. The contingent option costs 57% as much as vanilla & would have paid off 69% as often since '08.	Limited to premium paid
Buy NDX Dec25 ATM upvar for 17v indicatively (t & t-1 convention, Dec25 var ref. 25.1v)	26-Nov-24	17v	19-Dec-25	Dislocations in deep OTM put skew have created a historically attractive entry point for upside conditioned variance relative to unconditional variance, especially if a higher vol bubble regime is likely.	Losses would involve the variance strike paid
Buy NDX Dec25 80% upvar for 20.15v indicatively (t & t-1 convention, Dec25 var ref. 25.1v)	26-Nov-24	20.15v	19-Dec-25	Dislocations in deep OTM put skew have created a historically attractive entry point for upside conditioned variance relative to unconditional variance, especially if a higher vol bubble regime is likely.	Losses would involve the variance strike paid
Buy SX5E Mar25 calls vs short SX6P (Utilities) calls for a small credit.	26-Nov-24	8 bps credit	21-Mar-25	We find it prudent to have cheap upside hedges on in Europe in case of a positive surprise as few expect this or are positioned for it. SX5E vol is low compared to SX6P, beneficial for call relative value.	Utilities rising significantly more than SX5E (unlikely in a positive growth shock)
Buy a SX5E vs SX6P Mar25 ATM outperformance call conditioned on SX6P up at expiry	26-Nov-24	1.3%	21-Mar-25	We find it prudent to have cheap upside hedges on in Europe in case of a positive surprise as few expect this or are positioned for it. SX5E is positively correlated to SX6P, helping to cheapen the price of outperformance options.	Limited to the premium paid
Buy a Jun25 ATMf (115% strike) call on return dispersion on a basket of low (vs history) vol stocks that are most positively & negatively geared to USD	26-Nov-24	2%	20-Jun-25	We anticipate the coming months to be rife with US trade policy discussions. Swings in the USD are to be expected and a call on return dispersion on low(er) vol, USD-sensitive stocks can offer a way to position for currency volatility.	Limited to the premium paid
Buy Dec25 FTSE div futures as a Dollar (vs Sterling) strength beneficiary	26-Nov-24	299.5	19-Dec-25	FTSE Dec25 dividend futures stand out vs SX5E and SX7E for having a large discount. FTSE dividends should be a beneficiary of USD strength, as ~43% of its dividends are announced by globally-exposed FTSE companies in USD.	Losses will be 1-for-1 on long futures.
Long Dec25 VolSwap dispersion on custom names vs the ESTX50	26-Nov-24	9.5v	19-Dec-25	A highly fragile environment and greater idiosyncratic risk for EU stocks creates a favourable backdrop for vol dispersion strategies	Risks arise if the realised stock vs index vol spread falls below initial implied or in a correlated market shock.
Systematically trade dynamically sized delta-hedged 3m NKY 1x3* put ratios	26-Nov-24	-77bps	14-Feb-25	30x NKY fragility hedge: Dynamically sized 1x3 put ratios	Stocks grinding lower on muted vol & skew moves
Buy 19-Dec-25 85%/65% Best of Put spreads on NKY/KOSPI2/SPX for 0.90% (indic. quanto USD, 56% discount to avg vanilla)	26-Nov-24	0.90%	19-Dec-25	Taiwan risk: Low-cost tail hedge for 22x max payout	Limited to premium paid
XUA Jun25 110% 130% call spreads for 2.94% (indic. 22% delta)	26-Nov-24	2.94%/1.02%	13-Jun-25	China > Japan: Hedging a reversal of the ABC trade	Negative payout if the Nikkei rallies >10% at the same time as US rates decline. The China leg would involve the premium paid
Partially funded by selling a NKY Jun25 110% call conditional on US 2yr rates lower, for 1.02% (47% of vanilla, corr bid @0.1, indicative only)					

Source: Bloomberg, BofA Global Research. Prices reflective of most recently available data which may be delayed in some cases. "Trade Value" represents current valuation of trades initiated on the "Open Date".

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List of abbreviations

Exhibit 130: List of abbreviations

Abbreviations used in this report

ODTE: Zero-days-to-expiry	JPY: Japanese Yen
ABC: Anything But China	LDI: Liability-Driven Investments
Adj: Adjusted	LTCM: Long-Term Capital Management
AI: Artificial Intelligence	MA: Moving Average
ATM: At-the-money	Mag7: Magnificent 7 stocks are Microsoft, Apple, Amazon, Nvidia, Alphabet, Meta and Tesla
ATMf: At-the-money-forward	MHI: Mitsubishi Heavy Industries Ltd
AUM: Assets Under Management	MtM: Mark-to-market
Avg: Average	NDX: Nasdaq 100
BoE: Bank of England	NFP: Non-farm payrolls
BoJ: Bank of Japan	NIRP: Negative interest rate policy
BTd: Buy-the-dip	NKY: Nikkei 225 Index
CAPE: Cyclically-adjusted price-to-earnings ratio	NTM: next twelve months
CAGR: Compound Annual Growth Rate	OCC: Options Clearing Corporation
CB: Central Bank	OI: Open Interest
Comm. Svcs / Comms: Communication Services	OIS: Overnight Indexed Swap
Correl / corr: Correlation	OTM: Out-of-the-money
Covid: Covid-19 virus	Outperf: Outperformance
CPI: Consumer Price Index	OW: Overwriting
CPU: Central Processing Unit	P.a.: Per Annum
CS: Call spread	P/B ratio: Price-to-book ratio
CSS: Critical Stress Signal	P/E ratio: Price-to-earnings ratio
CTA: Commodity Trading Advisors (trend followers)	PDH: partial delta-hedge
cVar: Conditional Value-at-Risk	PnL / P&L: Profit and loss
DBEZ5: EURO STOXX Banks Index Dividend Futures Dec25	PR: Put Ratio
DBEZ6: EURO STOXX Banks Index Dividend Futures Dec26	PS: put spread
DH: Delta hedged	QoQ: Quarter-over-quarter
Disc: Discount	R.V.: Realized Volatility
Discr / Cons Disc: Consumer Discretionary	Refi: Refinancing
Div: Dividend	RTY: Russell 2000
DR: Diversification ratio	S&P: S&P 500
DXY: US Dollar Index Spot Rate	SMR: Small Modular Reactor
EM: Emerging Markets	SOFR: Secured overnight financing rate
EOD: End-of-day	SPW: Equal-weight S&P 500
EPS: Earnings-per-share	SPX: S&P 500 Index
ESTX50: Euro STOXX 50 index	SVB: Silicon Valley Bank
ETF: Exchange Traded Fund	SX5E: Euro Stoxx 50 Index
ETN: Exchange Traded Note	SX6P: STOXX Europe 600 Utilities index
EU: European Union	SX7E: STOXX Europe 600 Banks index
EZ: Eurozone	SXEP: STOXX Europe 600 Oil & Gas index
FCI-G: Fed's index measuring financial conditions' impact on GDP growth	SXXP: STOXX Europe 600 index
Fed: Federal Reserve Bank	Tech: Technology
FFR: Federal Funds Rate	TPU: Tensor Processing Unit
Fins: Financials	TR: Total Return
FOMC: Federal Open Market Committee	TRF: Total Return Future
FOMO: Fear of Missing Out	TRP: Tail liquidity risk premium
Fwd: Forward	TSMC: Taiwan Semiconductor Manufacturing Company
FX: Foreign Exchange	Tsy: Treasury
G-SIB: Global Systemically Important Banks	UKX: FTSE 100 Index
GDP: Gross Domestic Product	UNSDG: United Nations Sustainable Development Goal
GFC: Global Financial Crisis	US: United States
GFSI: Global Financial Stress Indicator	UST: US Treasury bonds
GICS: Global Industry Classification Standard	Var: Variance
GPU: Graphic Processing Unit	VG1: Generic 1st Euro Stoxx 50 Future
HF: Hedge fund	VIX: CBOE Volatility Index
HH Index: Herfindahl-Hirschman Index	Vol: Volatility
HSI: Hang Seng Index	VRP: Vol Risk Premium
HY: High yield	WIRP: Bloomberg World Interest Rate Probability analysis
I.V.: Implied Volatility	YCC: Yield curve control
IEA: International Energy Agency	YoY: Year-over-year
IG: Investment grade	YTD: Year-to-date
Indust/Indus: Industrials	YTW: Yield-to-worst

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